Polaris Vega User Guide for ST, VT, and XT Models

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Read Me First!

This guide provides detailed information about the Polaris Vega® ST, VT, and XT Systems. Read this section before continuing with the rest of the guide.

Warnings



In all NDI documentation, warnings are marked by this symbol. Follow the information in the accompanying paragraph to avoid personal injury.

1. Do not use the Polaris Vega System in the presence of flammable materials, such as anaesthetics, solvents, cleaning agents, and endogenous gases. Flammable materials may ignite and cause life-threatening injury.
2. Do not connect the Polaris Vega System to a host computer or network that is not IEC 60950 and/or IEC 60601 approved. If you connect the system to a non-approved host computer or network, you may increase leakage currents beyond safe limits and cause personal injury.
3. Do not transport or store the Position Sensor outside the recommended storage conditions, as this may cause the system to go out of calibration. Reliance on data provided by an out of calibration Position Sensor may lead to inaccurate conclusions and may cause personal injury. A calibration procedure must be performed before using the Position Sensor after it has been transported or stored outside the recommended storage temperature range.
4. Do not protect or shield either the Position Sensor or tools with methods not approved by NDI. Non-approved methods, such as drapes or covers, will interrupt the optical path and degrade the performance of the system. Reliance on data provided by a Position Sensor with an interrupted optical path may lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.
5. The Polaris Vega System requires special precautions regarding EMC. It must be installed and put into service in accordance with the EMC information detailed in “Electromagnetic Compatibility” on page 84. Failure to do so may result in personal injury.
6. Radio frequency communications equipment, including portable and mobile devices, may affect the Polaris Vega System and result in personal injury.
7. Do not use the Polaris Vega System either adjacent to, or stacked with, other equipment as this may result in increased electromagnetic emissions and/or decreased electromagnetic immunity. If the Polaris Vega System is used adjacent to, or stacked with other systems, check that both systems are operating normally. Failure to do so may result in personal injury.
8. Use of accessories and cables other than those specified or provided by NDI could result in increased electromagnetic emissions or decreased electromagnetic immunity of this equipment and result in improper operation.
9. Do not incorporate non-NDI components with the Polaris Vega System. The accuracy of results produced by applications that incorporate non-NDI components with the Polaris Vega System is unknown. Reliance on these results may result in personal injury.
10. All user maintenance must be done by appropriately trained personnel and while equipment is not in use. Individual components of the Polaris Vega System contain no user-serviceable parts. Maintenance by untrained personnel may present an electric shock hazard.
11. Do not attempt to bypass the grounding prong on the power cord by using a three-prong to two-prong adapter. The system must be properly grounded to ensure safe operation. Failure to do so presents an electric shock hazard.

12. Do not immerse any part of the Polaris Vega System or allow fluid to enter the equipment. If fluids enter any part of the system, they may damage it and present a risk of personal injury.
13. Do not sterilize the Polaris Vega Position Sensor as this may cause irreversible damage to its components. Reliance on data provided by a damaged Position Sensor may lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.
14. Do not use the Position Sensor without inspecting it for cleanliness and damage before use. The Position Sensor should also be monitored during use. Reliance on data provided by an unclean or damaged Position Sensor may lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.
15. Do not use the Polaris Vega System for absolute measurements; the system is designed for relative measurements only. Treating measurements as absolute may result in an incorrect interpretation of results. Incorrect interpretations may result in personal injury.
16. Do not rely on unqualified 3D results for stray markers. There are no built-in checks to determine if the 3D results for stray markers represent real markers, phantom markers or IR interference, so the host application must identify and qualify the reported 3D results for stray markers. Reliance on unqualified 3D data may lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.
17. When using the API command `BX2` or reply option `0x0800` with the API commands `BX` or `TX`, you must take appropriate action to detect the following events: the tool or marker is out of volume, the bump sensor has been tripped, the system is outside of the optimal operating temperature range, or other fault and error conditions exist. You must determine whether these events are detrimental to your application. If one or more of the events listed occurs, reply option `0800` enables the system to return data that may lead to inaccurate conclusions and may cause personal injury. For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide*.
18. Do not use a unique geometry tool whose design does not conform to the Polaris Vega System's unique geometry constraints. These unique geometry constraints ensure that the system can correctly identify a tool and can distinguish between multiple unique geometry tools. When two indistinguishable tools are used, the first tool that is detected will be tracked. If that tool moves out of the measurement volume, the second tool will be tracked. If this is repeated, the tracking data will appear to jump between the two tools. Reliance on data produced by an incorrectly identified tool or two indistinguishable tools can lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.
19. Do not use a tool with a tip without first verifying the tip offset. Any application that uses a tool with a tip must provide a means to determine the location of the tip. Reliance on data produced by a tool with an inaccurate tip offset may lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.
20. Do not use markers without inspecting them for cleanliness and damage both before and during use. Reliance on data produced by unclean or damaged markers may lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.
21. Do not obstruct the normal flow of air around the Position Sensor (for example, draping or bagging the Position Sensor). Doing so will affect the Position Sensor's operational environment, possibly beyond its recommended thresholds. Reliance on data provided by a Position Sensor that is outside of recommended thresholds may lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.
22. Do not use the Polaris Vega System in an x-ray, proton radiation, or MRI environment without first determining the performance, including accuracy, of the Position Sensor in these environments. NDI has not validated the Position Sensor in these environments. Failure to do so may result in personal injury.

23. Do not look directly into the laser-emitting aperture. The Class 2 laser module on the Position Sensor emits radiation that is visible and may be harmful to the human eye. Direct viewing of the laser diode emission at close range may cause eye damage.
24. Do not use controls or adjustments or perform procedures other than specified in this guide. Doing so may result in hazardous light exposure.
25. Ensure that people with restricted movement or reflexes (for example, patients undergoing medical procedures) do not look directly into the laser-emitting aperture. Patients undergoing medical procedures may be restricted in the availability of adverse-effects reflexes (turning away eyes and/or head, closing eyes) due to pharmaceutical influences and/or mechanical restraints. The Class 2 laser module on the Position Sensor emits radiation that is visible and may be harmful to the human eye. Direct viewing of the laser diode emission at close range may cause eye damage.
26. Position the system components so that they can be easily disconnected from the external power. Failure to do so may result in an electric shock hazard and possible personal injury.
27. The Polaris Vega System has been evaluated as a potential component of a Medical Electrical System intended for use in health care facilities outside of the patient environment. Failure to test in the final end-use configuration may result in personal injury.

Cautions

Caution! **In all NDI documentation, cautions are marked with the word "Caution!". Follow the information in the accompanying paragraph to avoid damage to equipment.**

1. To ship the Polaris Vega System, repack it in the original containers with all protective packaging. The provided packaging is designed to prevent damage to the equipment.
2. Always place the Position Sensor on a rigid support system. If not supported, the Position Sensor may fall, which may affect the calibration and damage the Position Sensor.
3. Use only 70% isopropanol solution and a soft lint-free cloth to remove handling smudges from the enclosure and illuminator covers. Other fluids may cause damage to the illuminator filters. Do not use any paper products for cleaning. Paper products may cause scratches on the illuminator filters.
4. Do not handle passive markers with bare hands as this will leave residue from skin that affects the marker's reflectivity. Take care not to drop or scuff the markers, as this also affects the reflectivity of the markers.
5. Do not block the vent on the System Control Unit. Blocking this may cause the System Control Unit to overheat and may result in equipment damage.

Disclaimers

1. Read the entire *Polaris Vega User Guide for ST, VT, and XT Models* before attempting to operate the Polaris Vega System.
2. This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:
   a) this device may not cause harmful interference, and
   b) this device must accept any interference received, including interference that may cause undesired operation.

For more information, see “Radio Frequency Emissions” on page 88.

3. The user must determine the suitability of the Polaris Vega System for their own application.
4. This equipment has been investigated with regard to safety from electrical shock and fire hazard. The inspection authority has not investigated other physiological effects.
5. The Vega Position Sensor requires a thermal stabilization period in order to provide reliable measurements. When the Position Sensor is powered on, the power light will flash to indicate that the system is warming up. When the light stops flashing, the system is ready for use.
6. Northern Digital Inc. has not investigated the implications of incorporating the Vega Position Sensor with an automatic position device, or any other automated closed loop systems. Using the Polaris Vega System in such an application is solely the responsibility of the user.
7. The Polaris Vega System emits IR light that may interfere with IR-controlled devices, such as operating room tables. It is recommended that you test the Polaris Vega System if you intend to use it in an environment where other IR-controlled devices are in use.

Contact Information

If you have any questions regarding the content of this guide or the operation of this product, please contact us:

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Updates

NDI is committed to continuous improvements in the quality and versatility of its software and hardware. To obtain the best results with your NDI system, register with the NDI Customer Support site and check it regularly for update information: https://support.ndigital.com.

1 Polaris Vega System Overview

The Polaris Vega System is an optical measurement solution that uses near-infrared (IR) light to calculate the 3D positions of markers attached to tools. A typical Polaris Vega System includes a Position Sensor, a PoE midspan injector, cables, and one or more tools. NDI provides a suite of utilities, NDI ToolBox, for you to configure, upgrade, troubleshoot, and test your Polaris system. The Polaris system also requires a host computer to run NDI ToolBox to collect, display, and manipulate information.

The Position Sensor is the main component of the Polaris Vega System. Its primary function is to track the position and orientation of markers attached to tools and to translate that information into measurement data for integration into tool navigation and visualization applications. Each Position Sensor can track multiple tools at the same time within a pre-calibrated measurement volume.



The Polaris Vega System has been evaluated as a potential component of a Medical Electrical System intended for use in health care facilities outside of the patient environment. Failure to test in the final end-use configuration may result in personal injury.

The Polaris Vega System interacts with each component as follows:

1. Depending on the type of tool, the Position Sensor collects infrared (IR) light in one of the following ways:
    - For passive tools, the Position Sensor emits IR light from its illuminators, flooding the surrounding area. The IR light reflects off passive markers back to the Position Sensor.
    - For active wireless tools, the Position Sensor emits an IR “chirp” signal. A receiver on the tool detects the signal, and the controller on the tool activates its IRED markers to emit IR light. The Position Sensor detects the emitted light.
    - For active wired tools, the System Control Unit triggers active markers attached to the connected tools. The markers emit IR light, which the Position Sensor detects.
2. Using the reflected or emitted light, the Position Sensor measures the positions of the markers and calculates the transformations (the positions and orientations) of the tools the markers are attached to.
3. The Position Sensor sends the transformation data and status information to the host computer for collection, display, and further manipulation.

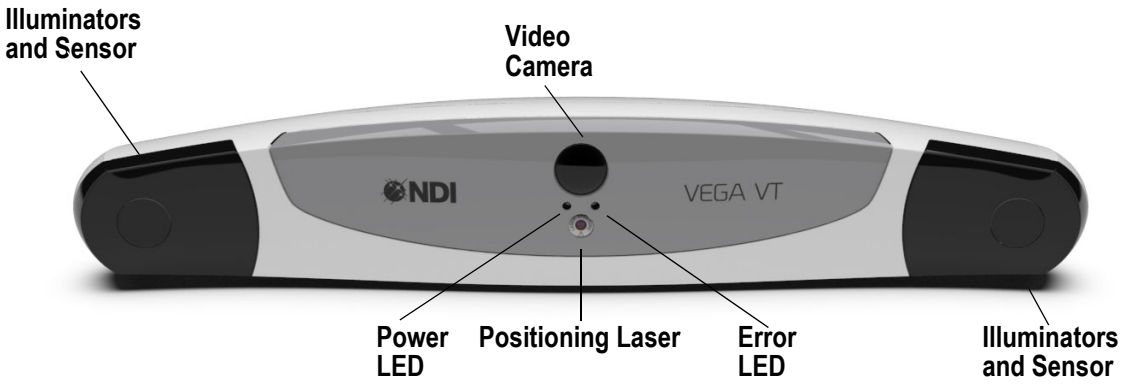
For more information on tools, see “Polaris Vega System Tools” on page 36. For more information on how the Position Sensor detects markers, see “Marker Detection and Tool Tracking” on page 33.

1.1 Position Sensor

The Position Sensor can be branded to custom specifications. For more information on this option, contact NDI. See “Contact Information” on page viii.

Front View

Figure 1-1 Vega VT Position Sensor - Front View



Illuminators Two arrays of infrared light-emitting diodes (IREDs) provide IR light to illuminate the passive markers on passive tools or trigger the active markers on active wireless tools.

Sensors Two sensors, each with a lens and an image sensor, collect IR light that is reflected from passive markers or emitted by active markers.

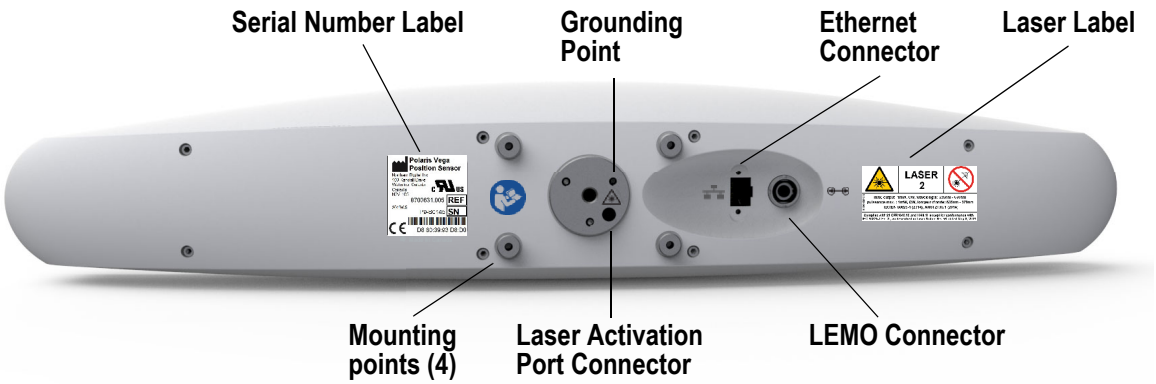
Video Camera A video camera on the Polaris Vega VT Position Sensor captures a live video stream of the Vega measurement volume.

Positioning Laser The optional positioning laser indicates the general centre of the characterized measurement volume within a tolerance. For more information, see “Positioning Laser” on page 46.

Indicator LEDs The power and error LEDs combine to indicate the status of the Position Sensor. For more information about the statuses, see Table 1-1. For more information about troubleshooting the Error LED, see “Troubleshooting” on page 89.

Rear View

Figure 1-2 Vega ST Position Sensor - Rear View



Mounting points Four M4 x 0.7 mm pitch x 10 mm deep threaded holes. For more information, see “Mounting the Position Sensor” on page 15.

Ethernet Power and I/O Interface Provides Ethernet communication with the host computer and PoE to the Position Sensor. The Ethernet port is compatible with the IEEE 802.3at Type 2 PoE standard.

LEMO Connector Provides serial communication with the host for Polaris Vega ST Position Sensors. If this option is not included, the connector port is covered and unusable.

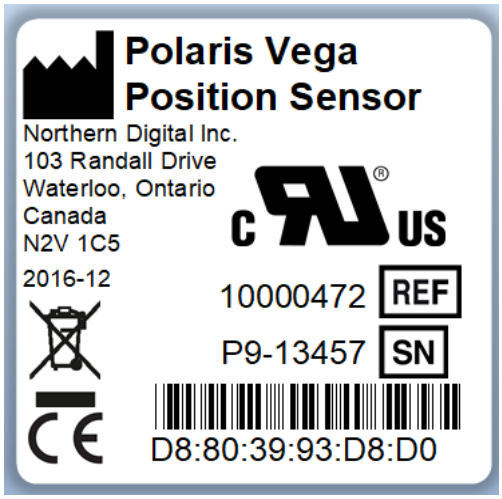
Grounding Point An M3x0.5, 6mm tapped hole for a dedicated ground connection to be fitted to the Position Sensor, if the system end-use application use requires one.

Laser Label (included with the optional positioning laser) Shows the classification, output, wavelength, standards, and a warning.

Laser Activation Port A 3.5 mm jack socket to connect and activate the laser using an external, customer-supplied switch. For information on activating the laser through the user parameters, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

Serial Number Label Shows the catalogue number, model, serial number, and manufacture date of the Position Sensor.

Figure 1-3 Position Sensor Serial Number Label



Features and Capabilities

For a detailed description of the different Position Sensors, see “Vega Models” on page 10.

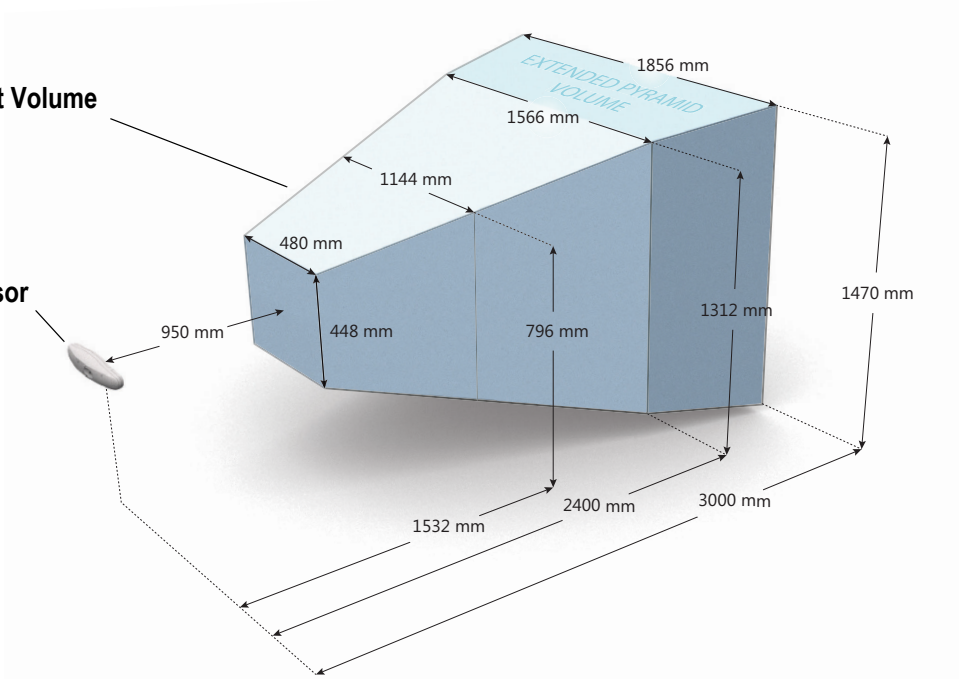
1.2 System Control Unit

The optional System Control Unit provides the interface between the Position Sensor and active wired tools. For more information, contact NDI.

1.3 Characterized Measurement Volume

The characterized measurement volume, as shown in Figure 1-4, is the volume within the detection region of the Position Sensor where accuracy is within specified limits. NDI cannot guarantee measurement accuracy outside this region. For the Polaris Vega System accuracy specification, see “Polaris Vega System Performance” on page 101.

Figure 1-4 Polaris Vega Measurement Volume



Note **The back section of the measurement volume, 2400 mm to 3000 mm from the Position Sensor, is only available in the optional extended pyramid volume.**

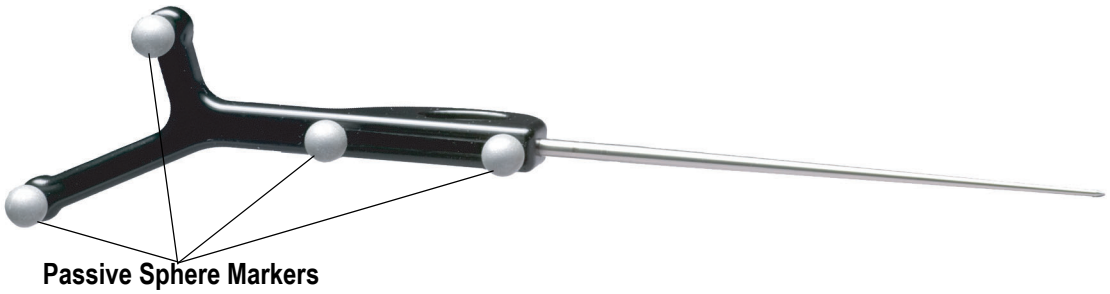
1.4 Tools

A tool is a rigid structure on which three or more markers are fixed so that there is no relative movement between them. The Polaris Vega System tracks the positions and orientations of tools and can also report the positions of individual markers.

The Polaris Vega System can track passive, active wireless, and active wired tools. To track active wired tools, you must purchase the optional System Control Unit separately. Passive tools reflect IR light from the Position Sensor illuminators. Active wireless tools emit IR light in response to pulses of IR light from the Position Sensor illuminators. Active wired tools emit IR light when triggered by the System Control Unit. Each tool must have a tool definition file that describes the tool to the Position Sensor. For more information on tools and tool definition files, see “Polaris Vega System Tools” on page 36. For information on tool design and construction, see the *Polaris Tool Design Guide*.

Figure 1-5 shows an example of a probe that incorporates four NDI Passive Sphere™ markers.

Figure 1-5 Passive Tool with Sphere Markers



For more information on the tools available from NDI for use with the Polaris Vega System, contact NDI. See “Contact Information” on page viii.

1.5 Host Computer Requirements and Communication

The system requires a customer-supplied host computer. The host computer must be approved to IEC 60950 or IEC 60601.

To use an NDI-provided host application, such as NDI ToolBox, NDI recommends the computer meets the following minimum requirements:

- 1 GHz or faster 64-bit (X64) processor
- 2 GB RAM (64-bit)
- 1 GB available hard disk space (64-bit)
- Screen resolution of 1024 x 768 (1280 x 1024 recommended)
- Gigabit network interface

For a list of the operating systems each NDI application was tested with, consult the readme.htm file installed with the application.

Communication Options for the Host System

The Polaris Vega System supports a gigabit Ethernet connection between the Position Sensor and host computer. PoE supplies power to the Position Sensor.

You can connect the Position Sensor directly to the host computer via Ethernet and a PoE midspan injector or you can use a customer-supplied network switch or router, as shown in Figure 1-6 and Figure 1-7. To connect multiple Position Sensors to the host computer, you must use a network switch or router.

Figure 1-6 Connections for Polaris Vega Systems Without a System Control Unit

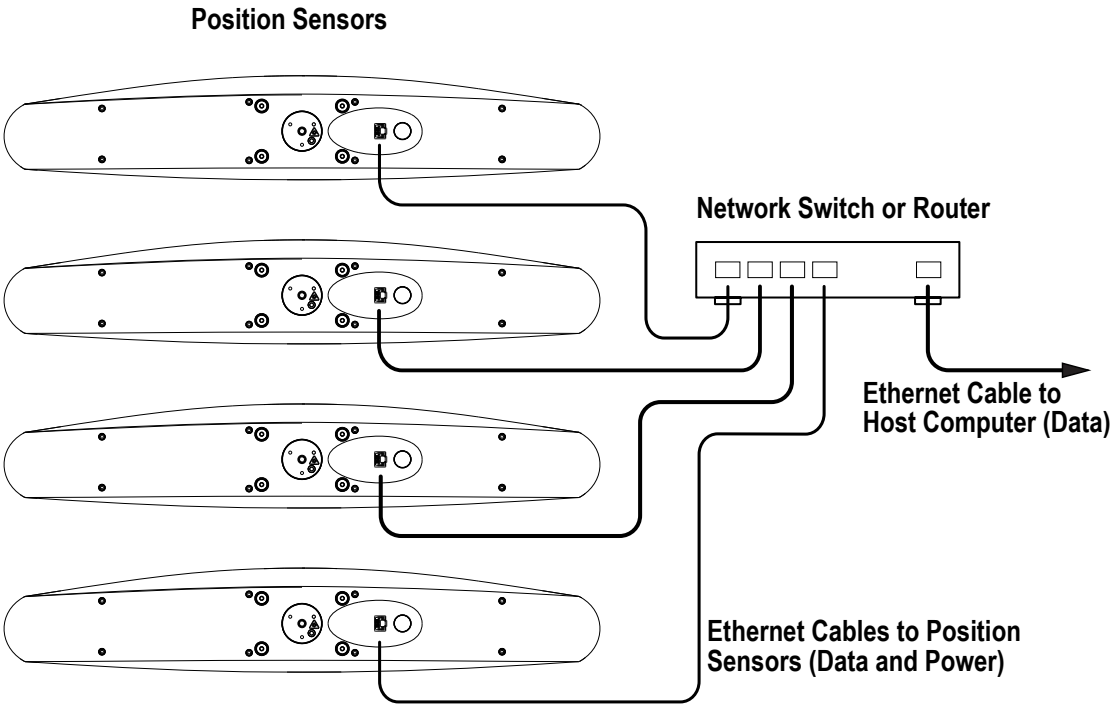
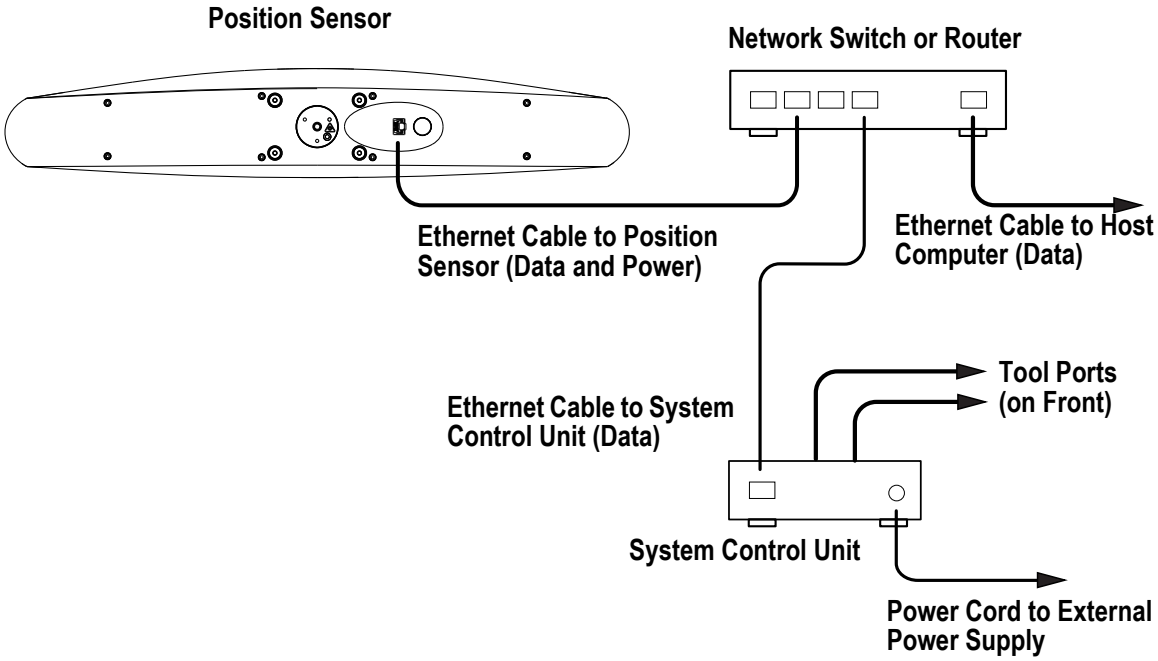


Figure 1-7 Connections for Polaris Vega Systems With a System Control Unit



NDI recommends using a controlled network to ensure the measurement application and communications between the Vega Position Sensor and host computer receive an appropriate quality and level of service.

Note **The customer-supplied Ethernet switch or router must be approved to the IEC 60950 or IEC 60601 standard and must meet the following minimum specification: IEEE802.3at compliant PoE type 2 device.**

You should use Cat 5e or higher shielded Ethernet cables in any Vega configuration.

When connecting multiple Position Sensors to the same network switch or router, ensure that each port on the switch or router will supply enough power to the connected Position Sensor. If the port does not supply sufficient power for tracking, the Position Sensor raises an alert.

If you install multiple Vega Position Sensors on the same subnet, or want to synchronize other devices on the same network with the Position Sensor, Precision Time Protocol (PTP) synchronizes tracking and aligns the timestamps from other connected devices on the same subnet.

Note **To synchronize multiple Position Sensors on the same network, the Position Sensors must have the same frame frequency to align the frame numbers.**

The Frame Component payload of the API command `BX2` includes the timestamp for a frame. For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide.*

1.6 Status LEDs and Audio Codes

For information about the System Control Unit LEDs, see the data sheet for your System Control Unit.

Position Sensor LEDs

The front of the Position Sensor includes two LEDs that combine to indicate the status of the Position Sensor, as described in Table 1-1.

Table 1-1 Position Sensor Indicator LEDs

Power LED (Green)	Error LED (Amber)	Position Sensor Status
Off	Off	No power.
Flashing (4 Hz)	Off	The system is booting up.
Flashing (2 Hz)	Any state	The Position Sensor is warming up. When the Position Sensor is ready for use, the power LED stops flashing and remains on.
On	Off	The Position Sensor is ready for use with no faults or error conditions.
Flashing (1 Hz with 80% Duty Cycle)	Off	For Vega VT systems only. The video camera is streaming video. Note: Any error condition takes precedence over this streaming indication.

Table 1-1 Position Sensor Indicator LEDs (Continued)

Power LED (Green)	Error LED (Amber)	Position Sensor Status
On	On	Minor recoverable error condition. For example, the bump sensor detected a bump. If you do not correct the condition, the system is still operational, but might not receive tracking data unless the application overrides the “bad data” filter. This is not a fault. Do not return the system to NDI. Instead, determine the condition. To determine what the condition is, use the alerts user parameters, as described in the *Polaris Vega and Lyra Application Programming Interface Guide*.
On	Flashing	Major recoverable fault which prevents operation. For example, incompatible firmware. This is a repairable fault. Do not return the system to NDI. To determine what the fault is, use the alerts user parameters, as described in the *Polaris Vega and Lyra Application Programming Interface Guide*.
Off	On	Non-recoverable fault. Return the Position Sensor to NDI for service.

Ethernet LEDs

The Ethernet LEDs indicate the status of the Ethernet connection with the host computer, as described in Table 1-2.

Table 1-2 Ethernet Status LEDs

LED		Status
Speed Indication	Single Flash	10 Mbps
	Double flash	100 Mbps
	Triple flash	1000 Mbps
Link/Activity LED	Off	No link
	On	Link
	Flash	Activity

Position Sensor Audio Codes

The Position Sensor emits two beeps each time it completes a power-up cycle or if you use the API command `RESET`.

Tip: You can turn off the beep feature using NDI ToolBox software or by setting the value of the user parameter `Param.System Beeper` to **0**. You can also use the API command `BEEP` to cause the Position Sensor to emit beeps. For more information on user parameters and API commands, see the *Polaris Vega and Lyra Application Programming Interface Guide.*

1.7 NDI Software

As a registered customer, you can download the following NDI software from the NDI Customer Support site at https://support.ndigital.com:

- **NDI ToolBox:** A collection of utilities to configure, upgrade, troubleshoot, and test the Polaris Vega System. Command line functionality also allows you to complete a ToolBox task, such as upgrading firmware, from within your application software. For more information, see the *NDI ToolBox Online Help.*
- **NDI Combined API Sample (CAPI):** A sample program, source code, and documentation. This program provides an example of how to write programs to operate the Polaris Vega System.
- **NDI Cygna-6D:** A utility for characterizing and creating tool definition files for tools used with the Polaris Vega System.
- **Polaris Accuracy Assessment:** Software to use with the Accuracy Assessment Kit for the Polaris family of Position Sensors.

1.8 System Parameters and API Commands

User parameters store values for aspects of the Polaris Vega System. Some parameters store values for the full system configuration, such as the combined firmware revision; others store values for a particular hardware device in the system, such as the illuminator rate on the Position Sensor.

You can use API commands to view read-only parameters or to change the values for configurable parameters. To retrieve a user parameter value, use the API command `GET`. To set the value of a user parameter, use the API command `SET`. For more information on user parameters, see the *Polaris Vega and Lyra Application Programming Interface Guide.*

2 Vega Models

2.1 Vega ST Position Sensor

The Vega ST Position Sensor provides high volumetric accuracy at different tracking frequencies and supports a number of optional features, as described in Table 2-1.

Table 2-1 Vega ST Position Sensor Features and Capabilities

Feature / Capability	
Frame Frequency	• 60 Hz fixed
Average Latency*	• < 17ms (typical)
Supported Tool Types	• Passive Spheres • Radix™ Lenses • Active wireless tools • Active wired tools For more information on tools and markers, see "Polaris Vega System Tools" on page 36.
Communication Interface	• Ethernet
Other Configuration Options	• Positioning Laser • Extended Pyramid Volume • Gamma Radiation Hardening • Serial Communications

*Latency is scene and context dependent. Individual frames, especially during the lock-on stage (beginning of tracking), may exhibit longer latency.

Figure 2-1 Vega ST Position Sensor with Positioning Laser



2.2 Vega VT Position Sensor

The Vega VT Position Sensor is similar to the Vega ST but is equipped with a video camera and supports a different set of features.

Table 2-2 Vega VT Position Sensor Features and Capabilities

Feature / Capability	
Frame Frequency	• 60 Hz fixed
Average Latency*	• < 17ms (typical)

Table 2-2 Vega VT Position Sensor (Continued)Features and Capabilities

Feature / Capability	
Supported Tool Types	• Passive Spheres • Radix Lenses • Active wireless tools • Active wired tools For more information on tools and markers, see "Polaris Vega System Tools" on page 36.
Communication Interface	• Ethernet
Other Configuration Options	• Positioning Laser • Extended Pyramid Volume

*Latency is scene and context dependent. Individual frames, especially during the lock-on stage (beginning of tracking), may exhibit longer latency.

Figure 2-2 Vega VT Position Sensor



2.3 Vega XT Position Sensor

The Vega XT Position Sensor provides a higher tracking frequency and lower latency than the Vega ST and VT Position Sensors.

Table 2-3 Vega XT Position Sensor Features and Capabilities

Feature / Capability	
Frame Frequency	• Configurable up to 400 Hz
Average Latency*	• < 3 ms (typical) at 400 Hz Frame Frequency
Supported Tool Types	• Passive Spheres • Radix Lenses • Active wireless tools • Active wired tools For more information on tools and markers, see "Polaris Vega System Tools" on page 36.
Communication Interface	• Ethernet
Other Configuration Options	• Positioning Laser • Extended Pyramid Volume

*Latency is scene and context dependent. Individual frames, especially during the lock-on stage (beginning of tracking), may exhibit longer latency.

Figure 2-3 Vega XT Position Sensor with Positioning Laser



3 Setting Up the Polaris Vega System

3.1 Unpacking the System

The Polaris Vega System usually ships to end users with a Position Sensor and cables. If your system includes the optional System Control Unit, it is also shipped as part of the system. OEM partners may receive a different configuration.

Handle all system components with care.

Note **Keep the packaging in good condition and use it if you must return the system to NDI for repair. For instructions on returning the system to NDI, see "Return Procedure" on page 93.**

Remove the Network Warning Label

When you first remove the Vega Position Sensor from the packaging, review the warning label over the Ethernet port, then remove the label using the finger tab on the right. Figure 3-1 shows a sample label.

Figure 3-1 Network Warning Label



Do not connect the Polaris Vega System to a host computer or network that is not IEC 60950 and/or IEC 60601 approved. If you connect the system to a non-approved host computer or network, you may increase leakage currents beyond safe limits and cause personal injury.

3.2 Operating Environment

Warnings

To avoid the risk of personal injury, read the following warnings before using the Polaris Vega System.

1. Do not use the Polaris Vega System in the presence of flammable materials, such as anaesthetics, solvents, cleaning agents, and endogenous gases. Flammable materials may ignite and cause life-threatening injury.
2. Do not protect or shield either the Position Sensor or tools with methods not approved by NDI. Non-approved methods, such as drapes or covers, will interrupt the optical path and degrade the performance of the system. Reliance on data provided by a Position Sensor with an interrupted optical path may lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.
3. The Polaris Vega System requires special precautions regarding EMC. It must be installed and put into service in accordance with the EMC information detailed in “Electromagnetic Compatibility” on page 84. Failure to do so may result in personal injury.
4. Do not use the Polaris Vega System either adjacent to, or stacked with, other equipment as this may result in increased electromagnetic emissions and/or decreased electromagnetic immunity. If the Polaris Vega System is used adjacent to, or stacked with other systems, check that both systems are operating normally. Failure to do so may result in personal injury.
5. Radio frequency communications equipment, including portable and mobile devices, may affect the Polaris Vega System and result in personal injury.
6. Do not immerse any part of the Polaris Vega System or allow fluid to enter the equipment. If fluids enter any part of the system, they may damage it and present a risk of personal injury.
7. Do not use the Polaris Vega System in an x-ray, proton radiation, or MRI environment without first determining the performance, including accuracy, of the Position Sensor in these environments. NDI has not validated the Position Sensor in these environments. Failure to do so may result in personal injury.
8. Position the system components so that they can be easily disconnected from the external power. Failure to do so may result in an electric shock hazard and possible personal injury.
9. Do not obstruct the normal flow of air around the Position Sensor (for example, draping or bagging the Position Sensor). Doing so will affect the Position Sensor's operational environment, possibly beyond its recommended thresholds. Reliance on data provided by a Position Sensor that is outside of recommended thresholds may lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.

Operating Conditions

To operate correctly, the system must be set up in an environment that meets the following criteria:

- There must be a clear line of sight between the Position Sensor and the tools to be tracked. The tools must be inside the characterized measurement volume. For the dimensions of the characterized measurement volume, see Figure 5-3 or Figure 5-4.
- Ensure that the environment you place the Position Sensor in meets the operating conditions outlined in Table 13-1.
- Make sure that sources of background IR light in the 800 nm to 1100 nm range (e.g. sunlight, some operating room lights) are minimized. The Position Sensor is sensitive to IR light. Since the Position Sensor functions by detecting IR light reflected from, or emitted by, markers, other sources of IR light can interfere with the Polaris Vega System.
- In a setup where two Position Sensors face each other, the exposure timing of one of them should be shifted so the two do not illuminate the measurement volume at the same time. For more information, see the `Param.Exposure.Time Slot.Passive` information in the *Polaris Vega and Lyra Application Programming Interface Guide*.

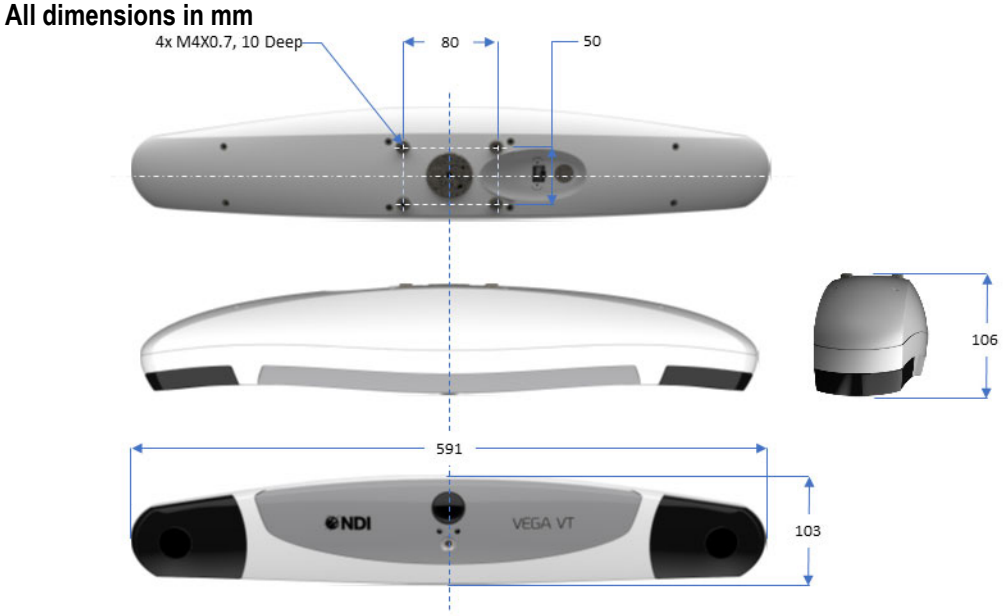
- Make sure that there are no large reflective surfaces within the field of view. See “Field of View and Characterized Measurement Volume” on page 30.
- Make sure that the tools do not have flat reflective surfaces. Certain tool shapes and surfaces can cause reflections that may interfere with the Polaris Vega System. For more information, see the *Polaris Tool Design Guide.*
- Before using the system, make sure the power LED on the Position Sensor is solid green and that the error LED is off. This indicates that the system is ready for use without any major or minor faults.

3.3 Mounting the Position Sensor

Caution! **Always place the Position Sensor on a rigid support system. If not supported, the Position Sensor may fall, which may affect the calibration and damage the Position Sensor.**

Mount the Position Sensor using the four included M4 x 0.7 mm pitch x 10 mm deep threaded holes. Figure 3-2 shows the Position Sensor dimensions and mounting arrangement.

Figure 3-2 Position Sensor Mounting Details



Note **Before you design a custom enclosure or any attachments for the Position Sensor, contact NDI for assistance. See “Contact Information” on page viii.**

NDI provides Figure 3-2 for information purposes only. Dimensions are subject to change without notice.

3.4 Mounting the System Control Unit

For dimensions and mounting options for the System Control Unit, see the data sheet for your System Control Unit.

3.5 Connecting the Hardware



Warnings

To avoid the risk of personal injury, read the following warnings before using the Polaris Vega System.

1. Use of accessories and cables other than those specified or provided by NDI could result in increased electromagnetic emissions or decreased electromagnetic immunity of this equipment and result in improper operation.
2. Do not connect the Polaris Vega System to a host computer or network that is not IEC 60950 and/or IEC 60601 approved. If you connect the system to a non-approved host computer or network, you may increase leakage currents beyond safe limits and cause personal injury.
3. Do not attempt to bypass the grounding prong on the power cord by using a three-prong to two-prong adapter. The system must be properly grounded to ensure safe operation. Failure to do so presents an electric shock hazard.
4. Do not incorporate non-NDI components with the Polaris Vega System. The accuracy of results produced by applications that incorporate non-NDI components with the Polaris Vega System is unknown. Reliance on these results may result in personal injury.
5. Do not use the Position Sensor without inspecting it for cleanliness and damage before use. The Position Sensor should also be monitored during use. Reliance on data provided by an unclean or damaged Position Sensor may lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.

Connect the Polaris Components to the Host Computer

You can connect the Position Sensor directly to the host computer via Ethernet and a PoE midspan injector or you can use a customer-supplied network switch or router. To connect multiple Position Sensors to the host computer, or if your system includes a System Control Unit, you must use a network switch or router. For configuration options and diagrams, see “Communication Options for the Host System” on page 5.

Note **The Vega ST Position Sensor is available with an optional LEMO connector to support serial communications between the Position Sensor and host computer. This option is only recommended for customers that are migrating from Polaris Spectra to Polaris Vega. It is not recommended for new designs. To set up serial connections, see “Setting up for Serial Communications” on page 67.**

1. Connect the Ethernet cable to the port on the Position Sensor.
2. Perform one of the following actions:
   - For configurations without a switch or router, connect the Ethernet cable to the PoE.
   - For configurations using a switch with embedded PoE, connect the Ethernet cable to the switch.

- For configurations using a router or switch without embedded PoE, connect the router or switch to a PoE midspan injector and then connect the Position Sensor to the PoE midspan injector.

3. Perform one of the following actions:
   - For configurations without a switch or router, connect the PoE midspan injector to the host computer.
   - For configurations with a switch or router, connect the switch or router to the host computer.
4. If your system includes a System Control Unit, perform the following actions:
   a) Connect the switch or router to the host computer and System Control Unit.
   b) Connect power to the DC power in on the System Control Unit.
   c) Connect any tools to the System Control Unit.
5. Make sure all the cables are connected firmly and placed where they will not be stressed, stepped on, or bent.

After you apply power via PoE, the system automatically starts booting up, for approximately 30 seconds, then starts warming up. After booting up, the system can communicate with the host computer but, depending on ambient conditions, might need several minutes to warm up fully before it can track accurately. After the system warms up fully, the Power LED on the front panel of the Position Sensor is solid green. If your system includes the optional System Control Unit, the Power and Status LEDs on the System Control Unit are also solid green.

The system is designed to be power fail-safe; when you remove power, the system turns off.

Note **Before you remove power to the Position Sensor, ensure that the firmware is not in the process of being updated. If you remove power to the Position Sensor during a firmware update, the update might fail. For more information, see "Updating the Firmware" on page 74.**

Note **NDI does not recommend integrating the Vega Position Sensor into a 10Base-T network.**

The internal Position Sensor-to-Video Camera network is 192.168.184.xxx. This means the Position Sensor will not work correctly if it is connected to a 192.168.184.xxx network.

Check the Bump Sensor and Internal Battery Status

The Position Sensor contains a bump sensor that detects when the Position Sensor has suffered an impact that may affect its calibration. The Position Sensor also contains an internal battery that powers the system's real-time clock and powers the bump sensor when the system is powered off.

After unpacking the unit, NDI recommends that you check the bump status to determine if the unit experienced a potentially damaging impact during shipping and handling and check the battery status.

For more information on checking the bump sensor and battery, see "Bump Sensor and Internal Battery" on page 49.

3.6 Installing NDI ToolBox

NDI ToolBox is a collection of utilities to configure, upgrade, troubleshoot, and test the Polaris Vega System.

The ToolBox video client utility uses GStreamer to handle the video stream from the Vega video camera. You can install GStreamer as part of the ToolBox installation or install your own version of GStreamer separately from the ToolBox installation.

You can download the latest software and documentation, visit the NDI Customer Support site at https://support.ndigital.com. For instructions on using ToolBox, see the *NDI ToolBox Online Help.*

Install on Windows

1. Navigate to where you saved the NDI ToolBox installer.
2. Double-click **install.exe**.
3. Follow the on-screen instructions.

Install on Linux

You may require execution permissions to install ToolBox. The default installation location is **<user_name>/NDIToolBox** or **~/NDIToolBox**.

1. Navigate to where you saved the NDI ToolBox installer.
2. Run the following commands:

```
sudo chmod +x install.sh
./install.sh
```

3. Follow the on-screen instructions.

Tip: To silently install ToolBox, run the installer from the command line with the '-q" option. You can also specify a different directory than the default. For example, `./installer.sh -q -dir /opt/ToolBox`. If you omit the `-dir` option, the installation uses the default install location.

Install on macOS

To manage NDI software on a Mac platform, you require administrator account privileges.

1. Navigate to where you saved the NDI ToolBox installer.
2. Double-click the **install.dmg** file.
3. Double-click **NDI Installer** and enter your administrator password.
4. Follow the on-screen instructions.

Note **The NDI ToolBox download includes a Java virtual machine (VM). The Java VM included in the NDI ToolBox download is fully compatible with NDI ToolBox. Other versions of Java VM may cause NDI ToolBox to exhibit unusual or unpredictable behaviour.**

3.7 Connecting to a System

ToolBox uses the multicast DNS (mDNS) protocol to automatically detect devices (Position Sensors or System Control Units) on the network and display them in the connection window. Windows 10 and macOS include the mDNS functionality. If the host computer is running Windows 7 or 8, you can install Apple's Bonjour software to provide the mDNS functionality. On Linux computers, you can enable Avahi.

When you start the ToolBox software, it may automatically connect to a system, depending on the available connection options. For more information about connection options, see the *NDI ToolBox Online Help.* After ToolBox is running, you can connect to a different system.

Establish the Network Connection

If the network includes a DHCP server, the DHCP server automatically assigns a dynamic IP address to the device. If the device does not find a DHCP server or does not receive a response within three to six seconds, the device assigns its own IP parameters using Zero Configuration Networking (Zeroconf), if configured.

Instead of using Zeroconf, you can manually assign an IP address to the device.

Note **These instructions use NDI ToolBox to perform the configuration. You can also configure the network using the API. For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide.***

Zeroconf Requirements

To use Zeroconf, the system must meet the following requirements:

- The host computer and devices must be connected to each other or on a local network with no DHCP server.
- The host computer must be configured for Zeroconf networking. This is the default for Window and macOS. On Linux, it may be enabled on a "per interface" basis.

When you power on a host computer that is using Zeroconf, the host computer assigns IP addresses to itself and devices on the 169.254.xxx.xxx subnet. After it assigns all IP addresses, you can connect to the device.

Static IP Addressing Requirements

You can use static IP Addressing to manually assign an IP address to the device. When you set a static IP address, you must provide the IP address for the device and host computer. Optionally, you can also provide the gateway and subnet mask for each.

If you set up the device for Static IP addressing, you must also set up your host computer network appropriately to enable the connection.

- On Windows, use the **Control Panel**.
- On Linux, use **Yast** or equivalent.
- On macOS, use **Setup**.

Connect to a System Using ToolBox

The **Connect to** menu in the NDI Configure ToolBox utility lists all available devices. If you set a custom host name for a device, the Connect to menu shows the custom name. Otherwise, the menu displays the device's serial number as the host name. For example P9-12345.

If direct connection to the video camera is enabled, the **Connect to** menu also displays any connected video cameras (VCUs). For information on this setting, see "Connecting to the Video Camera" on page 58.

You can also connect to a system using the API. See "Connect to a System Using the API" on page 20.

If your system includes a System Control Unit, you cannot connect to the Position Sensor and the System Control Unit at the same time.

Tip: To see the IP address for a Position Sensor, in the **Connect to** menu, hover over the name.

1. Open the NDI Configure ToolBox utility.
2. If necessary, select **File > Connect to > *<Device>*.**

For help connecting to a system, see "Troubleshooting Connections" on page 89.

Modify a Connection

After you connect to a device, you can change the network connection.

1. In NDI Configure, in the left pane, select **Settings and Options > Network**.
2. Select the **IP Method** drop-down and choose the appropriate setting.
   - To configure Static IP networking, select **Static**. Provide the IP address for the device. If necessary, provide the gateway and subnet mask.
   - If you enabled DHCP or Zeroconf, there is no additional information to provide.
3. Select **File > Save Parameters**.
4. To reset the system, select **File > Reset System**.

Troubleshoot Connections

If you cannot connect to the device from the host computer, see "Troubleshooting Connections" on page 89.

Connect to a System Using the API

To connect to a device using the API instead of ToolBox, you must connect to the system TCP listening port.

Example

```
telnet P9-12345.local 8765
ver 4
```

Where `P9-12345` is the device serial number/host name, `8765` is the port as defined in the `Param.Network.Host Port` parameter, and `ver 4` is the command to retrieve the version information for the device.

For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

4 Tutorial: Learning to Use the Polaris Vega System

This chapter describes the basic functionality of the Polaris Vega System using NDI ToolBox. For more detailed information on NDI ToolBox, see the *NDI ToolBox Online Help.*

The tutorial helps first-time users:

- connect to the system and track tools,
- observe error and information flags while tracking tools,
- track using a reference tool, and
- pivot a tool to determine the tool tip offset.

4.1 Tracking Tools

Before you try tracking tools, ensure that you have set up and connected the hardware, as described in “Connecting the Hardware” on page 16, and have installed NDI ToolBox, as described in “Installing NDI ToolBox” on page 18.

Connect to a System

1. Open the NDI Track ToolBox utility.
2. If NDI Track does not automatically connect to the system, perform the following actions:
   a) Select **File > Connect to.**
   b) Select the system from the list or select **New Connection.**
   c) Type the host name or IP address and select **OK**.

Track Tools

1. To load the tool definition files for the tools you want to track, click  .
2. In the dialog, navigate to the desired tool definition files.
3. Select one or more files.
4. Click **Open**.
5. To track an active wired tool, plug the tool into the System Control Unit.

   After the tool is plugged in or the tool definition file loads, the Polaris Vega System automatically attempts to track the tool.
6. Move the tool throughout the characterized measurement volume, making sure the markers on the tool face the Position Sensor.

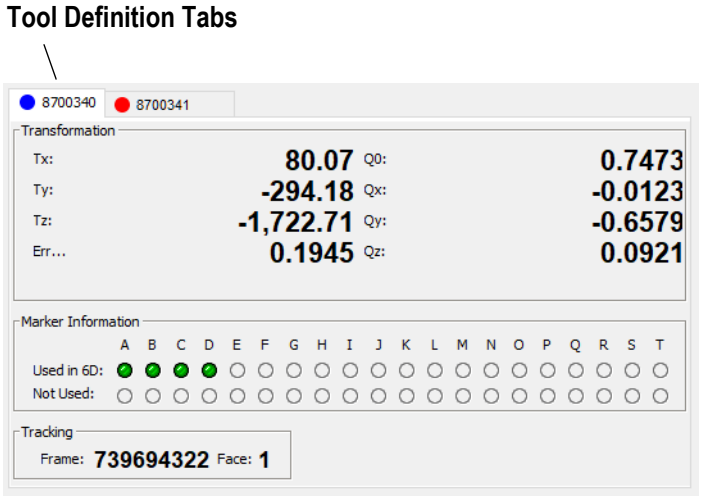
   As you move the tool, the symbol representing the tool in the graphical representation moves to reflect the tool’s position.

4.2 Triggering Information and Error Flags

As the Polaris Vega System tracks tools, it returns information about those tools to the host computer, such as whether the tool is out of volume, partially out of volume, or missing. You can use Toolbox to trigger some of the most common flags.

The bottom-right section of the tool tracking utility includes a tab for each loaded tool definition file. The tool tracking utility displays errors, warnings, and marker information in these tabs.

Figure 4-1 Tutorial: NDI ToolBox Tool Tracking Window



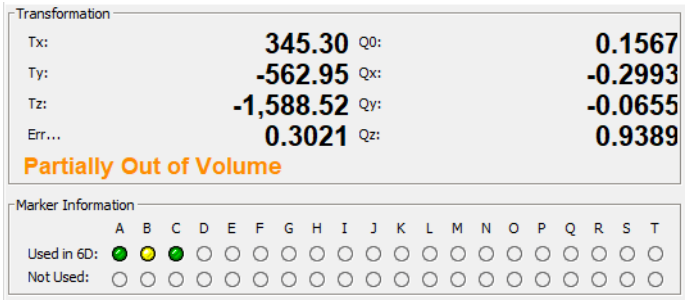
View the “Partially Out of Volume” and “Out of Volume” Flags

1. In the bottom-right section of the tool tracking utility, select a tab.
2. Start to move the tool to the edge of the characterized measurement volume.

   As some markers pass the edge of the volume, NDI ToolBox displays the message “Partially Out of Volume.”
3. Continue to move the tool past the edge of the characterized measurement volume.

   When all the markers are outside of the volume, NDI ToolBox displays the message “Out of Volume.”

Figure 4-2 Tutorial: "Partially Out of Volume" Flag

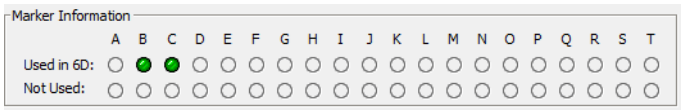


View the "Too Few Markers" Flag

1. In the bottom-right section of the tool tracking utility, select a tab.
2. Position the tool inside the characterized measurement volume, with the markers facing the Position Sensor.
3. Cover one or more markers, without touching them.

   The Position Sensor can no longer detect the covered markers.

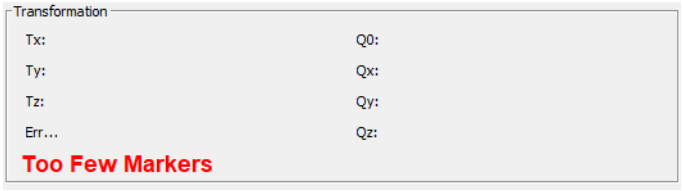
Figure 4-3 Tutorial: Detected Markers Indicator



4. Continue to cover more markers, without touching them.

   When the Position Sensor can no longer detect the minimum number of markers, NDI ToolBox displays the message "Too Few Markers" and does not report a transformation.

Figure 4-4 Tutorial: "Too Few Markers" Flag

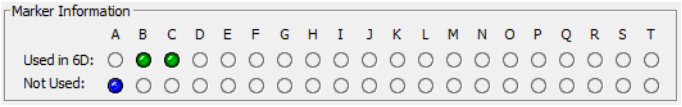


View the "Exceeded Maximum Marker Angle" Flag

1. In the bottom-right section of the tool tracking utility, select a tab.
2. Position the tool inside the characterized measurement volume, with the markers facing the Position Sensor.
3. Turn the tool gradually until the markers are no longer facing the Position Sensor.

   When a marker exceeds the maximum marker angle, NDI ToolBox displays a blue indicator in the Not Used section of the marker information.

Figure 4-5 Tutorial: "Exceeded Max Marker Angle" Indicator



4.3 Setting a Tool as Reference

When you set a tool as reference, the Position Sensor tracks all other tools with respect to the reference tool. For more information on reference tools, see "Reference Tools" on page 41.

Ensure you have loaded at least two tool definition files. For more information, see "Track Tools" on page 22.

1. In the bottom-right section of the tool tracking utility, select the tab for the tool you want to set as reference.
2. Right-click the tool tab, then select **Global Reference**.

   The reference tool appears as a square in the graphical display. Other tools are displayed inside a square that is the colour of the reference tool. The positions and orientations of other tools are now reported in the local coordinate system of the reference tool.

Note **The Polaris Vega System still calculates the tool transformations in the coordinate system of the Position Sensor. The NDI ToolBox software then calculates and reports the tool transformations with respect to the reference tool.**

4.4 Determining the Tool Tip Offset by Pivoting

You can determine the tool tip offset of a probe or pointer tool by performing a pivoting procedure or using a calibrator. After NDI ToolBox calculates the tool tip offset, it can report the position of the tip of the tool instead of the position of the origin of the tool. For more information on determining the tool tip offset and how to use a calibrator, see "Tool Tip Offset" on page 40.

You must have a divot to rest the tool tip in while you pivot the tool; to ensure that the tip does not move, the size and shape of the divot must match the tool tip. For example, a probe with a 1-mm ball tip requires a hemispherical divot with a 1-mm diameter.

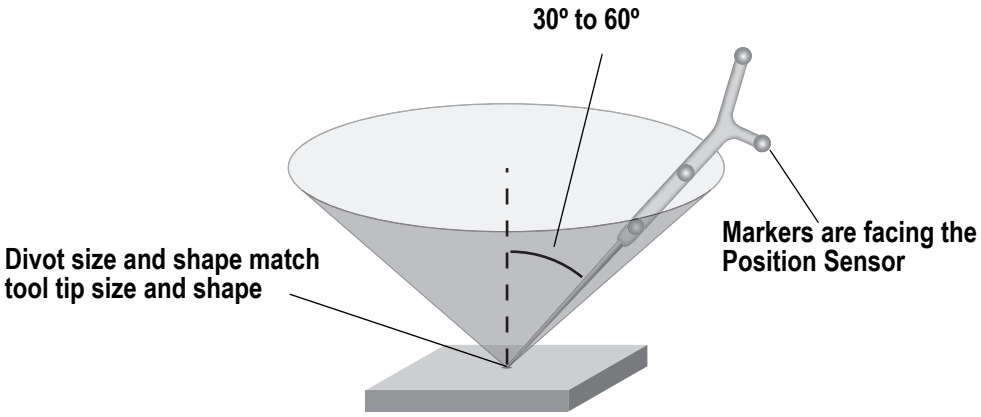
Ensure that you have loaded a tool definition file for a probe or pointer tool. For more information, see "Track Tools" on page 22.

1. In the bottom-right section of the tool tracking utility, select the tab for the tool you want to pivot.
2. To open the **Pivot** dialog, click  .
3. Select a start delay of about 5 seconds and a duration of about 20 seconds.
4. Place the tool tip in the divot.
5. Ensure that the tool is within the characterized measurement volume and remains within the volume throughout the pivoting procedure.
6. In the **Pivot tool** dialog, click **Start Collection**.
7. Pivot the tool in a cone shape, at an angle of 30º to 60º from the vertical.

a) Keep the tool tip stationary and ensure that there is a line of sight between the markers on the tool and the Position Sensor throughout the pivoting procedure.

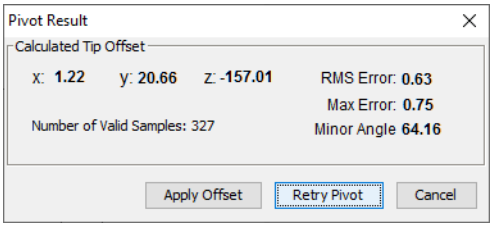
b) Pivot the tool slowly until the specified pivot duration time has elapsed.

Figure 4-6 Tutorial: Pivoting Technique



8. When the pivot is complete, the **Pivot Result** dialog appears. To report the position of the tip of the tool, click **Apply Offset**.

Figure 4-7 NDI ToolBox Software: Pivot Result Dialog



5 How the Polaris Vega System Works

The information in this chapter can help increase your technical understanding of the system, but is not required to use the system. To learn how to use the system, see “Tutorial: Learning to Use the Polaris Vega System” on page 22.

5.1 Information Returned by the Polaris Vega System



Do not use the Polaris Vega System for absolute measurements; the system is designed for relative measurements only. Treating measurements as absolute may result in an incorrect interpretation of results. Incorrect interpretations may result in personal injury.

When the Polaris Vega System is tracking tools, it returns information about those tools to the host computer. By default, the system returns:

- **the position of each tool’s origin**, given in mm, in the coordinate system of the Position Sensor. See “Global Coordinate System” on page 29.
- **the orientation of each tool**, given in quaternion format. The quaternion values are rounded off, so the returned values may not be normalized.
- **an error value** for each tool transformation. This RMS value, given in mm, is the result of the least squares minimization between the marker geometry in the tool definition file and the tool’s measured marker positions.
- **the status of each tool**, indicating whether the tool is out of volume, partially out of volume, or missing. It also indicates whether the port handle corresponding to each tool is enabled and initialized. For more information on port handles, see the *Polaris Vega and Lyra Application Programming Interface Guide*.
- **the frame number** for each tool transformation. The frame number is incremented by 1 at the frame frequency. The frame number returned with a transformation corresponds to the frame in which the data used to calculate that transformation was collected.
- **the system status**, which includes some of the system errors.

Note **Transformations with respect to a reference tool (see “Reference Tools” on page 41) and transformations for a probe with a tool tip offset (see “Tool Tip Offset” on page 40) are calculated using application software, such as NDI ToolBox.**

If requested, the system can also return:

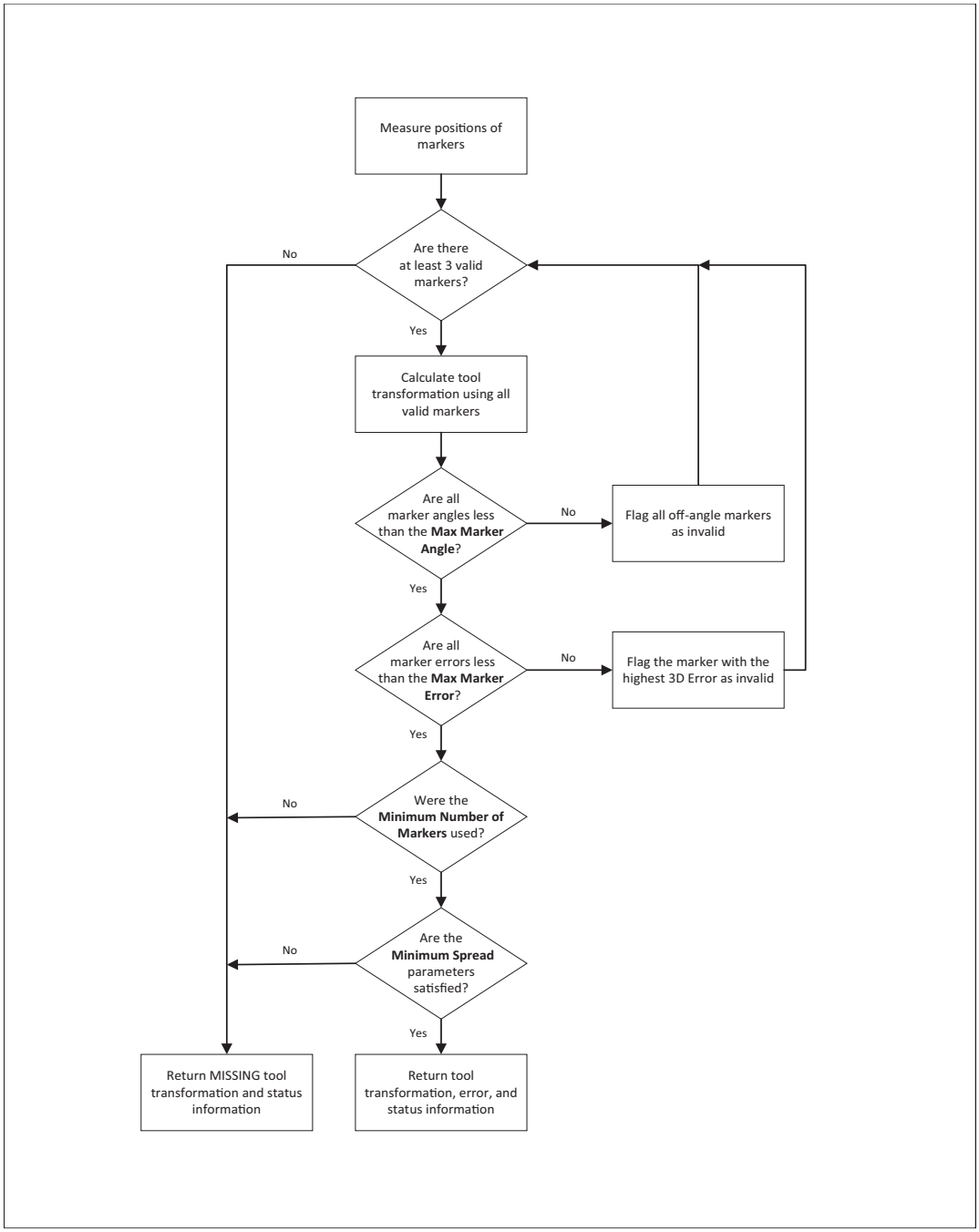
- timestamps for reported frames.
- status of buttons and switches on active wired tools.
- tracking errors and flags. Some tracking errors and flags are returned by default.
- marker status information, such as whether a particular marker was used to calculate a tool transformation.
- positions of stray markers (3D positions not associated with any tool).
- positions of individual markers on tools.

- the marker error value for individual markers on tools (the difference between the measured and expected location of the marker on the tool).
- transformations for tools that are outside of the characterized measurement volume.
- transformations for tools when the system has detected one of the following error conditions:
  - the bump sensor has been triggered.
  - the system is outside of operating temperature range.
  - the bump sensor battery power is low.
  - the temperature sensors are outside of functional range.
  - the input voltage is out of range.

The tool tracking utility of NDI ToolBox displays most of this returned information.

Note **For information on the API commands used to request tracking information from the Polaris Vega System, see the `BX`, `BX2`, and `TX` commands in the *Polaris Vega and Lyra Application Programming Interface Guide.***

Figure 5-1 Reporting Tool Transformation Conditions



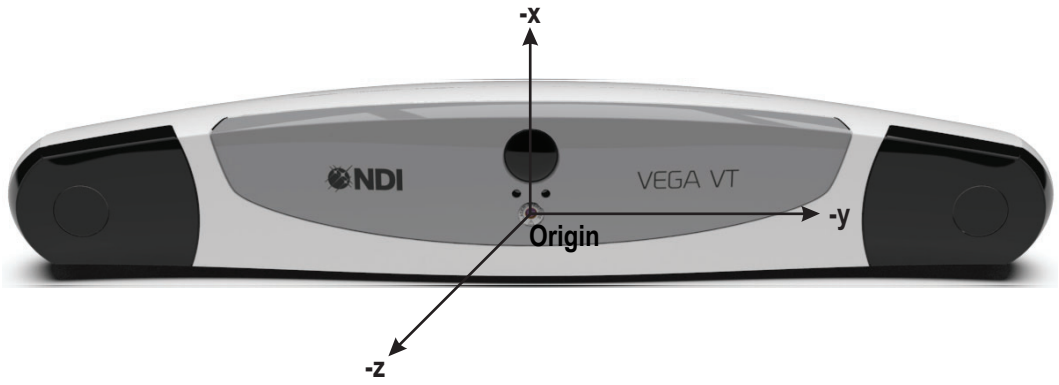
Note **The application must take appropriate action to detect the events listed above and determine whether they are detrimental to the application.**

5.2 Global Coordinate System

The Vega Position Sensor uses a coordinate system with an origin located at the Position Sensor and axes aligned as shown in Figure 5-2. This global coordinate system is defined during

manufacturing and cannot be changed. The origin of the coordinate system is not tied to a physical feature of the Position Sensor enclosure and can vary from system to system by up to 50 mm along all axes; two different Position Sensors may report different absolute transformations even when the physical setup is the same.

Figure 5-2 Global Coordinate System



Do not use the Polaris Vega System for absolute measurements; the system is designed for relative measurements only. Treating measurements as absolute may result in an incorrect interpretation of results. Incorrect interpretations may result in personal injury.

The Polaris Vega System reports the transformations of tools in the global coordinate system. However, if you are using a reference tool, software can calculate and report transformations in the local coordinate system of the reference tool. For more information on reference tools, see “Reference Tools” on page 41.

5.3 Field of View and Characterized Measurement Volume

The field of view is the total volume in which the Polaris Vega System can detect a marker, regardless of accuracy.

The characterized measurement volume is a subset of the field of view and is accuracy is within specified limits; therefore, the Position Sensor collects data within the characterized measurement volume. The Vega Position Sensor has two characterized measurement volumes available: the pyramid volume, illustrated in Figure 5-3, and the extended pyramid volume, illustrated in Figure 5-4.

The Position Sensor’s performance is determined using the calibration methodology described in Appendix A on page 101.

Figure 5-3 Pyramid Volume

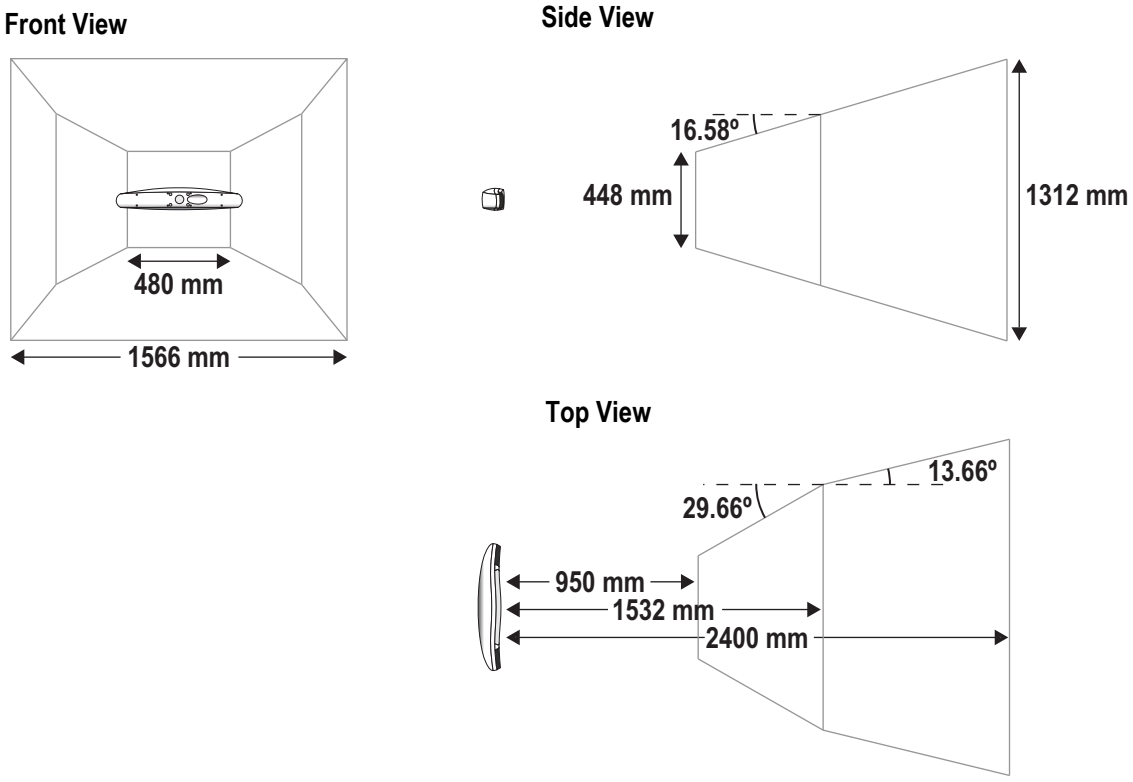
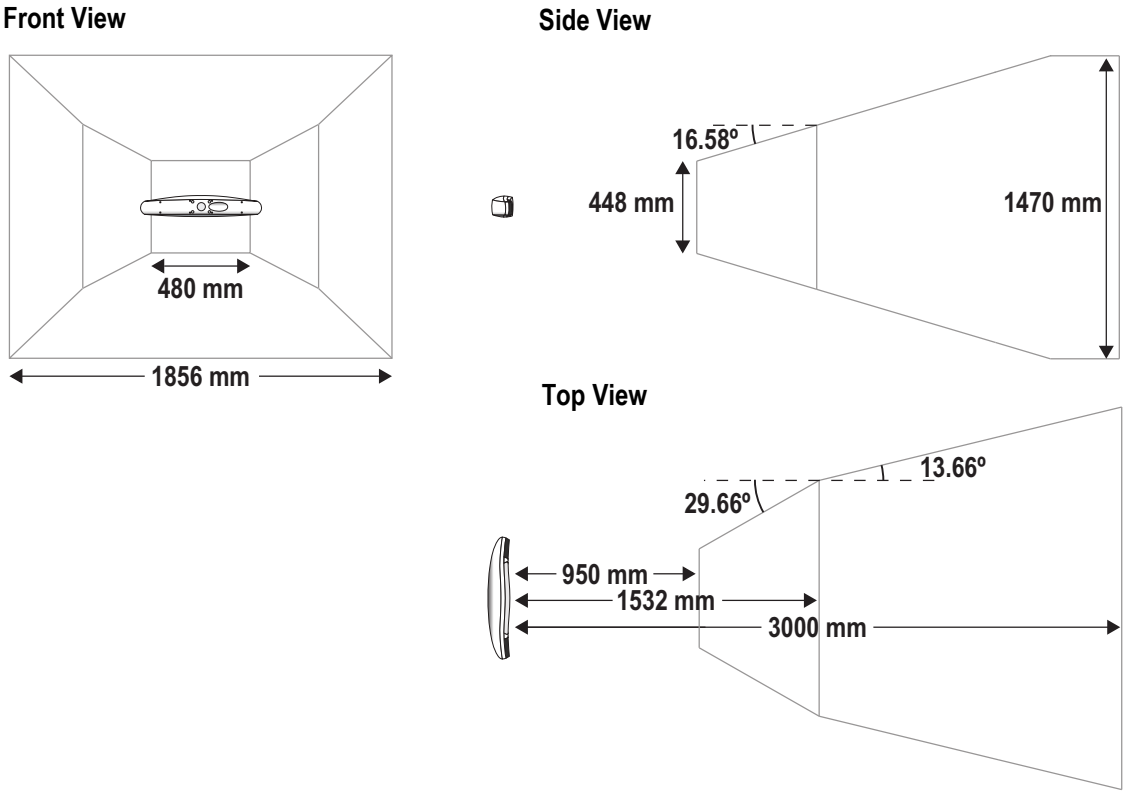


Figure 5-4 Extended Pyramid Volume



Out of Volume and Partially Out of Volume Flags

Out of Volume Flag The system flags a tool as out of volume if all of its markers are outside of the characterized measurement volume but the system can still detect the tool.

The API command `BX2` reports all measurements, regardless of whether the tools or markers are inside the characterized measurement volume. You must check the accompanying status code to determine whether the measurements are inside the volume, partially out of volume, or completely out of volume.

By default, the API commands `BX` and `TX` report tools or markers outside the characterized measurement volume as missing. You can request these measurements by using reply option 0x0800.

For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

Partially Out of Volume Flag The tool definition file specifies the minimum number of markers that the system must use in the calculation of a tool transformation. The system flags a tool as partially out of volume if at least one marker on the tool is inside the characterized measurement volume but fewer than the minimum number of markers are inside the characterized measurement volume. For more information, see the *Polaris Tool Design Guide*.

For example, consider a five-marker tool, with three markers inside the characterized measurement volume and two markers outside the volume. If the minimum number of markers is set to 3, the tool

is considered to be inside the volume. If the minimum number of markers is set to 4 or 5, the system flags the tool as partially out of volume.

5.4 Marker Detection and Tool Tracking

Detecting Markers

To detect passive markers, the Position Sensor's illuminators flood the field of view with IR. The passive markers are retro-reflective, directing the IR back to the Position Sensor.

To trigger active wireless tools, the Position Sensor emits a high-frequency infrared "chirp" signal just before exposure starts. The infrared receiver on the active wireless tool detects the chirp and activates the IREDs for the duration of the exposure time. For more information, see "Chirp Signal Specification" on page 39.

For active wired tools, the System Control Unit activates the markers on the tool.

For all tools, the Position Sensor collects IR for a period of time called the exposure time. This acts like an electronic shutter. Depending on your system configuration, the system may use variable or fixed exposure control. Variable exposure control automatically adjusts the exposure time using a process called dynamic range control to set the intensity of the brightest signal detected to a maximum value. The intensity of all other signals falls below this value.

The system distinguishes between potential marker data and background IR using a value called the trigger level. The trigger level is the minimum IR intensity considered to be valid marker data. The Position Sensor rejects background IR that falls below the trigger level. For systems that use variable exposure control, the trigger level generally increases with exposure time. For more information, see "Understanding Infrared Light Sensitivity" on page 75.

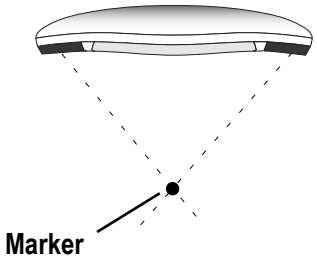
If the system cannot detect individual IR sources, or detects more IR sources than it can process, it reports an error.

Tracking Tools

When the Polaris Vega System first begins tracking a tool, it must acquire the tool. Similarly, if a tool goes missing, it must reacquire the tool. To acquire or reacquire a tool, the Polaris Vega System first measures the positions of all the visible markers it detects.

To determine the position of a marker, the Position Sensor calculates virtual lines, called the lines of sight, out from the measured center of the images of the marker on each sensor. In a perfect (error-free) system, a pair of lines of sight intersect exactly at the location of the marker in 3D-space, as shown in Figure 5-5. In a real system, the lines of sight do not intersect exactly but pass by each other at a very small distance. The shortest distance between the two lines of sight is the line separation. If the line separation at this point is less than a predefined limit, the Polaris Vega System considers the point to be a possible marker position. Otherwise, it discards the point.

Figure 5-5 Determining a Marker Position



At this point, the system has a set of un-associated 3D coordinates. Next, it associates the 3D positions with markers on the tool.

The system uses two different methods to associate the 3D positions with markers on the tools, depending on whether the tool is a unique geometry tool or standard geometry tool.

The unique geometry algorithm relies on the segment lengths (the distance between any pair of two 3D positions) and the segment angles (the angle between any pair of segments). The system compares the measured segments and segment angles to the “nominal” segments and segment angles as calculated from the tool definition. If the measured and nominal data are within predefined tolerances of each other, the measured data is a “match” and the system associates the corresponding markers with the tool.

Active wired tools can be defined as either unique geometry or standard geometry during tool characterization. Standard geometry tools match 3D positions to tool markers by activating each active marker in sequence.

For more information about unique geometry and standard geometry tools, see the *Polaris Tool Design Guide*.

If the system cannot match a 3D position to a tool, it reports it as a stray marker. Stray markers can be physically present markers not associated with any tool, can be caused by reflections, or can be phantom markers (mathematically valid solution for co-planar arrangements of real makers).

For more information, see “Stray Marker Reporting” on page 42 and “Phantom Markers” on page 43.

The Polaris Vega System acquires a tool if it can recognize three markers on the tool. The system reports a transformation only if all the tool tracking parameters are met. After it acquires the tool, the system uses a predictive algorithm.

Three-Marker Locking Option

By default, if a transformation is determined, but considered invalid because it does not meet the tool tracking parameters, the system discards the tool's transformation. The Position Sensor must re-identify the tool before it can calculate the transformation again.

If you enable the three-marker locking option during tool characterization, when a transformation is considered invalid, the system continues to internally track the tool. Because the system is still tracking the markers that are visible, it does not need to re-identify the tool when the tool tracking parameters are met.

For example, consider a four-marker tool with the three-marker locking option enabled and where the system can detect only three of the markers on the tool.

- If the minimum number of markers is set to three, the system continues to track the tool and report transformations.
- If the minimum number of markers is set to four, the system continues to track the tool but reports the tool as MISSING. When the system detects the fourth marker, it calculates and reports the transformation without having to re-identify the tool.

NDI recommends enabling three-marker locking, unless three of the markers are collinear. In this case, internally tracked transformation will be unstable about the collinear axis and the system may not properly identify the fourth marker when it becomes visible again.

For more details on three-marker locking, see the *Polaris Tool Design Guide*.

5.5 Tracking Frequency and Frame Frequency

The frame frequency is the rate at which the system takes images of the measurement volume. The tracking frequency is the rate at which the system reports transformations for each of the tracked tool types. The tracking frequency cannot exceed the frame frequency.

The default frame frequency is 60 Hz and the default tracking frequency is 20 Hz (1/3 the frame frequency). These defaults can be changed using the API or ToolBox.

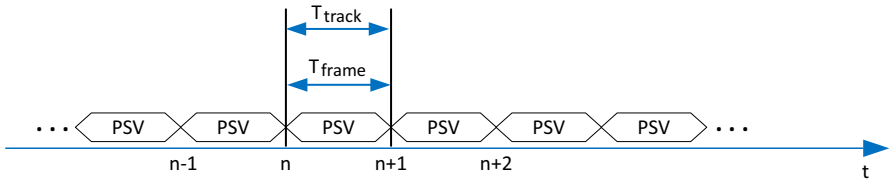
The number of tool types being tracked affects the tracking frequency. The tracking frequency decreases as more frames are needed to track all the tools. The number and type of tools have the following effects on the number of frames needed:

- If one type of tool is tracked, the maximum tracking frequency is the frame frequency. It is possible to slow the tracking frequency down to 1/2 or 1/3 the frame frequency with the addition of one or two "dummy” frames.
- If two types of tools are tracked, the maximum tracking frequency is 1/2 the frame frequency, with each tool tracked in alternating frames. It is possible to slow the tracking frequency to 1/3 the frame frequency with the addition of a "dummy” frame.
- If three tool types are tracked, the maximum tracking frequency is 1/3 the frame frequency, with each tool type tracked in its own frame.

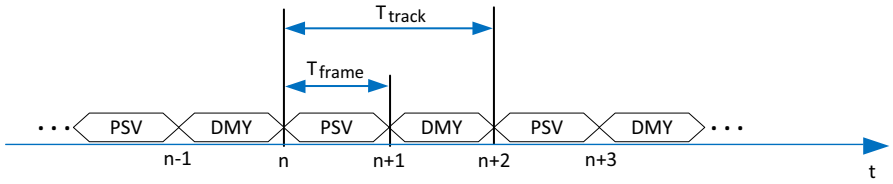
Figure 5-6 illustrates the relationship between the tracking frequency and frame frequency.

Figure 5-6 Tracking Frequency and Frame Frequency Relationship

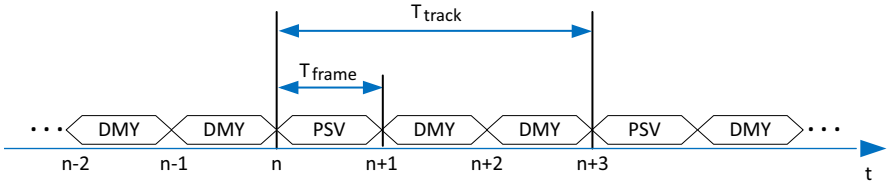
Example 1: Tracking Frequency = Frame Frequency



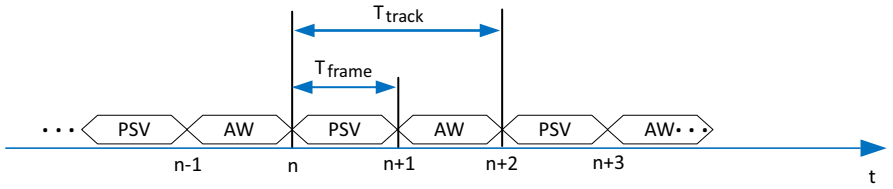
Example 2: Tracking Frequency = 1/2 Frame Frequency(one dummy frame added)



Example 3: Tracking Frequency = 1/3 Frame Frequency (two dummy frames added)



Example 4: Tracking Frequency = 1/2 Frame Frequency (different tool types tracked in alternating frames)



PSV: Passive Frame DMY: Dummy Frame AW: Active Wireless Frame

Tracking Frequency and Frame Frequency Measurements:

$$\text{Tracking Frequency} = \frac{1}{T_{track}}\text{, Frame Frequency} = \frac{1}{T_{frame}}.$$

5.6 Polaris Vega System Tools

A tool is a rigid structure with three or more fixed markers with no relative movement between them. You can enable up to 25 tools. However, having a large number of tools in view or a large number of extraneous markers in view may affect the speed of the system and its ability to return transformations.

Do not use markers without inspecting them for cleanliness and damage both before and during use. Reliance on data produced by unclean or damaged markers may lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.

Do not use a unique geometry tool whose design does not conform to the Polaris Vega System's unique geometry constraints. These unique geometry constraints ensure that the system can correctly identify a tool and can distinguish between multiple unique geometry tools. When two indistinguishable tools are used, the first tool that is detected will be tracked. If that tool moves out of the measurement volume, the second tool will be tracked. If this is repeated, the tracking data will appear to jump between the two tools. Reliance on data produced by an incorrectly identified tool or two indistinguishable tools can lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.

Passive Tools

Figure 1-5 shows an example of a passive tool. Passive tools have no active (light-emitting) components and do not require power from a cable or battery.

NDI offers two types of markers for passive tools:

Passive Spheres NDI Passive Sphere markers have a retro-reflective coating. The coating reflects IR light back to its source instead of scattering it. As such, the IR light from the Position Sensor illuminators reflects off the markers directly back to the sensors. NDI Passive Sphere markers snap-fit to the tool using NDI mounting posts, which are manufactured to firmly hold NDI spheres. For more information on passive sphere markers, see the *Polaris Tool Design Guide*.

You cannot re-sterilize passive sphere markers. NDI does not recommend using a passive sphere marker if it has been sterilized a second time, as multiple cycles of sterilization may adversely affect the marker's performance. Testing has shown that there is no significant degradation in the performance of these markers after one cycle of ETO sterilization. You cannot autoclave passive sphere markers.

Radix Lenses Radix Lenses are retro-reflective markers that can easily be wiped off if they become dirty. In contrast to the NDI Passive Sphere, you cannot use the Radix Lens directly as supplied. You must first incorporate it into a tool or mounting base (not provided by NDI). For more information, see the *Radix Lens Integration Guide*.

In either case, light from the illuminators around the sensors of the Position Sensor reflects back to the sensors, making the markers appear bright compared to the rest of the scene. To identify individual markers and associate them with a specific tool, the system uses geometrical properties like lengths and angles of the marker geometry derived from the tool definition file. For more information, see “Tool Definition File” on page 40.

Caution! **Do not handle passive markers with bare hands as this will leave residue from skin that affects the marker's reflectivity. Take care not to drop or scuff the markers, as this also affects the reflectivity of the markers.**

Active Wireless Tools

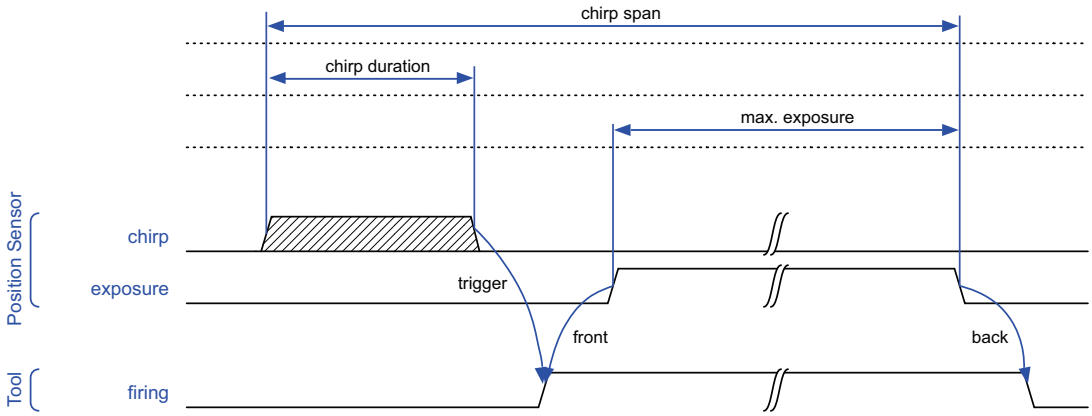
Active wireless tools have an IR receiver and IREDs. To track active wireless tools, the Position Sensor pulses (chirps) its illuminators in a way the tool’s IR receiver recognizes. When the receiver

detects the end of a valid chirp signal, the tool activates all of its IREDs, which the Position Sensor then sees.

These tools can be powered by a battery or an external source.

Figure 5-7 illustrates the interaction between the active wireless tool and the Vega Position Sensor.

Figure 5-7 Active Wireless Tool Timing Diagram



At time *chirp span* before the end of exposure, the Position Sensor starts chirping its illuminators for *chirp duration*.

When the tool detects the end of the chirp signal, it starts firing its IREDs for a time that is slightly longer than the maximum exposure time of the Position Sensor. This ensures minimal overlap on both ends of the exposure period where the IREDs are active before exposure starts and after exposure ends.

Table 5-1 Chirp Signal Specification

Parameter	Vega ST/VT	Vega XT	Comment
Chirp Frequency	107.3 kHz	143.0 kHz	
Chirp Duty Cycle	50%	50%	
Chirp Span	2400 µs	450 µs	Relative to end of exposure
Chirp Duration	345 µs	105 µs	
max. Exposure	2000 µs	300 µs	Relative to end of exposure
Front Overlap	>= 0	>= 0	Firing must start at or before start of exposure
Back Overlap	>= 0	>= 0	Firing must not stop before end of exposure
Irradiance	0.02 – 70 W/m^2	0.02 – 70 W/m^2	Signal strength of the chirp signal throughout the measurement volume
Wavelength	800 – 900 nm	800 – 900 nm	Wavelength of illuminator signal

Since there is no data communication between the Position Sensor and the tool (other than the chirp signal for synchronization), they act in the same manner as passive tools. This means:

- The host application must provide tool definition files that describe the tool geometry to the system.
- The tool geometry must follow the Unique Geometry Rules.
- Active wireless tools and passive tools are tracked in separate frames. Therefore, an active wireless tool can have the same tool geometry as a passive tool that is used at the same time.
- If active and passive tools are tracked at the same time, the track frequency is at most 1/2 the frame frequency (alternating tracking of active wireless and passive tools)

For customers interested in implementing active wireless tracking, NDI provides an evaluation kit that includes a reference design (schematic and source code). For more information, contact NDI Customer Support at https://support.ndigital.com.

Active markers are physically smaller than passive sphere markers and consist of an infrared light-emitting diode (IRED) mounted on a ceramic base. The ceramic base allows you to autoclave the markers.

Active Wired Tools

Active wired tools are connected to the System Control Unit and incorporate active markers. To track active wired tools, the System Control Unit fires the markers and the Position Sensor detects the IR emitted by the markers.

The tool definition file specifying the marker information for an active wired tool is stored in a serial read-only memory (SROM) device located on the tool. For example, in the connector. You can also specify the tool definition file for active wired tools using the API.

Active markers are physically smaller than passive sphere markers and consist of an IRED mounted on a ceramic base. The ceramic base allows you to autoclave the markers.

5.7 Tool Definition File

To track a tool, the client application must provide the tool definition file to the system. The tool definition file includes the geometry of the tool's markers, the tool's manufacturing data, tool face definitions (for a multi-faced tool), and the parameters and settings described in "Tool Tracking Parameters" on page 40. Without this information, the system cannot accurately interpret the data it collects.

The procedure used to create a tool definition file is called *tool characterization*. For more information on tool characterization, see the *Polaris Tool Design Guide*.

5.8 Tool Tracking Parameters

The tool definition file includes four tool tracking parameters:

- Maximum Marker Error
- Maximum Marker Angle
- Minimum Number of Markers
- Minimum Spread

The system uses these four parameters to determine which markers to use to calculate the tool transformation. The system also uses these parameters to reject outliers and to limit the chance of returning inaccurate tool transformation data.

For information on how the system uses the tool tracking parameters, see the *Polaris Tool Design Guide*.

5.9 Tool Tip Offset



Do not use a tool with a tip without first verifying the tip offset. Any application that uses a tool with a tip must provide a means to determine the location of the tip. Reliance on data produced by a tool with an inaccurate tip offset may lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.

The tool definition file defines the origin of a tool as part of the tool's local coordinate system. When the Position Sensor tracks a tool, it reports the transformations of the origin of the tool.

In certain circumstances, it is useful to track a point on the tool other than the tool's origin. For example, it is useful to track the location of the tip of a probe. You can define the tool's origin at the tip of the probe; however, if the tool is later bent, the origin is no longer located at the tip.

NDI recommends that you determine the tool tip offset of the tool prior to each use. The tool tip offset is the vector between the tip of the tool and the origin of the tool. To determine the location of the tool tip, application software can apply the tool transformations reported by the Polaris Vega System to the tool tip offset.

Note **The Polaris Vega System always tracks the origin of the tool. The application software, not the Polaris Vega System, calculates the location of the tool tip.**

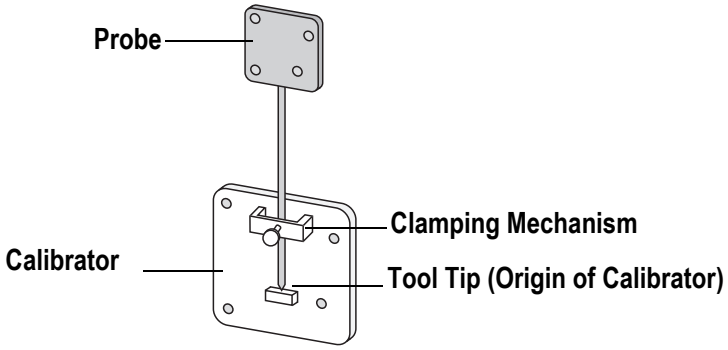
Determining the tool tip offset prior to each use ensures that the location of the tool tip is known as accurately as possible. You can determine the tool tip offset using a calibrator or by performing a pivoting procedure.

Determine the Offset Using a Calibrator

A calibrator is a rigid body that incorporates three or more markers and a clamping mechanism. The clamping mechanism allows you to clamp another tool (usually a probe) into place. Figure 5-8 shows an example of a calibrator.

To use a calibrator to determine the tool tip offset of a probe, clamp the probe in place on the calibrator. The point where the tool tip rests defines the origin of the calibrator. The Polaris Vega System can then measure the positions of the probe's origin and the calibrator's origin. The application software compares these measurements to determine the tool tip offset of the probe.

Figure 5-8 Sample Calibrator



Determine the Offset Using the Pivoting Procedure

You can also determine the tool tip offset using the pivot procedure in NDI ToolBox. During the pivoting procedure, the Polaris Vega System measures the positions of the markers while you pivot the tool. The software collects this data and uses it to determine the tool tip offset. For more information, see "Determining the Tool Tip Offset by Pivoting" on page 25 or the *NDI ToolBox Online Help*.

5.10 Reference Tools

A reference tool is a tool whose local coordinate system is used as the global coordinate system in which other tools are tracked. The Polaris Vega System tracks all the tools, including the reference tool, and reports the transformations in the coordinate system of the Position Sensor. For more information, see "Global Coordinate System" on page 29. Software, such as NDI ToolBox, then calculates and reports the positions and orientations of all other tools with respect to the position and orientation of the reference tool.

Note **Use a reference tool to ensure minimal drift in the measurements produced; specifically, drift caused by time, settling and/or temperature.**

The application software, not the Polaris Vega System, calculates the tool transformations with respect to the reference tool.

For example, in neurosurgery, the reference tool can be attached to the patient's head. The user then performs a registration procedure to define the reference tool's position relative to the patient's head. After that, if either the patient's head or the Position Sensor shifts, the measurements are not affected since they are reported with respect to the patient's head (the reference tool) and not with respect to the Position Sensor.

If the Vega Position Sensor cannot track the reference tool, the software cannot calculate the transformations of other tools with respect to the reference tool. For example, this may occur if the reference tool is occluded.

5.11 Stray Marker Reporting

A stray marker is a marker that is not part of a rigid body or tool. For example, by placing stray markers on a patient's chest, the markers may be used to gate or track the patient's breathing to time radiation therapy.

When you request stray marker data from the Polaris Vega System, the Polaris Vega System reports tool transformations, 3D data (position only, no orientation information) and out-of-volume information for markers that are not used in tool transformations, including phantom markers. The application software must eliminate phantom markers and verify that the stray markers are within the characterized measurement volume.

Be aware of the following potential hazards associated with using the stray marker reporting functionality:

- The system may report an external IR source as a stray marker. For example, an IR transmitter or incandescent light.
- Marker identification is not possible from frame to frame. Users must devise a method to keep track of which 3D position belongs to which marker.
- A stray marker does not have a marker normal, so there is no way to know if the marker orientation is exceeding a particular angle.
- There are no built-in checks to determine if the 3D result is a real marker or a phantom marker, generated by other IR sources or markers in view of the Position Sensor. The system tries to reject markers using the line separation qualifier, but if several markers are in a line parallel to the horizontal plane of the Position Sensor, phantom markers may still be generated that are within the line separation qualifier. For more information on phantom markers, see "Phantom Markers" on page 43.
- The Position Sensor cannot detect or compensate for partially occluded markers. The user may be able to detect the apparent shift if the marker position can be constrained in the application software. For example, the marker position must be constrained along a vector and its position relative to another marker is supposed to be fixed within some tolerance.



Do not rely on unqualified 3D results for stray markers. There are no built-in checks to determine if the 3D results for stray markers represent real markers, phantom markers or IR interference, so the host application must identify and qualify the reported 3D results for stray markers. Reliance on unqualified 3D data may lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.

To request stray marker data, use the `0x1000` reply option with the API command `BX` or `TX` or the `--3d=strays` option with the API command `BX2`. These reply options return out-of-volume information along with the 3D data. For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

Note **If the application is interested only in 3D data (stray markers), a tool definition file is not required. However, for each tool type you want the system to track, you must load at least one "dummy" tool to tell the system to add a frame for this tool type. You can add dummy tools via optional parameters in the API command `PHRQ`. For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide*.**

5.12 Phantom Markers

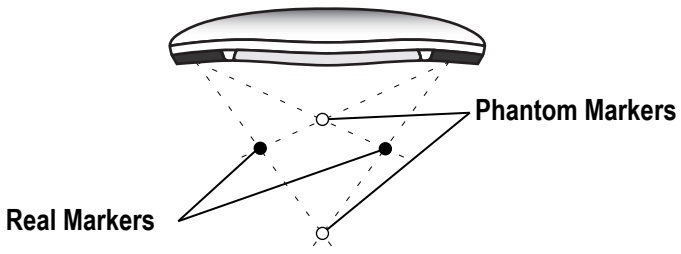
Phantom markers are the result of the calculation that the Polaris Vega System uses to determine the position of a source of IR. They appear and are reported as markers but are not actual markers.

To determine the position of a source of IR, the Position Sensor calculates a line between the source of IR and each sensor (displayed as dotted lines in Figure 5-9). Where the lines cross each other, the Polaris Vega System calculates the *line separation* (the distance between the lines). If the line separation at this point is within a predefined limit, the Polaris Vega System considers the point to be a possible marker position.

The system reports phantom markers when the imaginary lines calculated from the sensors intersect in more than one place with a line separation within a predefined limit. This generally occurs when two or more markers are in the same plane as the sensors. Figure 5-9 shows an example of two coplanar markers with four mathematical solutions. Two are the actual marker locations, and two are the phantom marker locations. In this example, one phantom marker is closer to the Position Sensor than the actual markers, and the other phantom marker is farther away from the Position Sensor than the actual markers, but this is not the only possible scenario.

The number of phantom markers increases with the number of coplanar markers. When there are *n* coplanar markers, there are up to *n* • (*n* - 1) phantom markers.

Figure 5-9 Phantom Markers



When you request stray marker data from the Polaris Vega System, the system reports data for both phantom markers and stray markers. If you do not request stray marker data from the Polaris Vega System, the system does not return any phantom marker data.

Note **The system cannot distinguish which solutions are phantom markers; you must eliminate the reported phantom markers using application software.**

5.13 System Spectral Response

The Polaris Vega System uses near infrared light for tracking. The system responds to wavelengths in the 800 nm to 1100 nm range. Light sources with spectral content in this wavelength range may cause interference with tracking. Examples include sunlight, halogen lamps, and other incandescent light bulbs. The system signals the presence of extraneous light sources by setting the “Interference” flag. Whether or not the interference impacts tracking depends on the actual scene, but it is generally advisable to remove the source of interference before using the system in critical applications. The easiest way to identify the location and source of the interference is to use the NDI Capture ToolBox Utility to capture images of the scene with the interference present.

Note **To prevent interference with the system, it is important to reduce environmental IR. Some operating room lights may emit IR.**

5.14 Data Transmission Rate

The host computer update rate (the rate at which the host computer receives data) depends on the following factors:

Network Utilization If the system is operated on a busy shared network, the transmission may slow down due to bandwidth limitations. If the system is operated on a separate network, this issue is not significant.

API command reply length The more data the system must return with every transformation, the slower the host update rate. The amount of data returned with each transformation increases as you track more tools and select more options. The slower update rate is not significant, unless a large amount of data, such as image data, is being transferred.

Application speed The host transmission rate can vary according to how often the application asks for data and how often the graphical user interface (in particular, graphics) must be updated.

6 Vega System Features

This chapter provides details about additional features of the Polaris Vega System.

6.1 Video Camera - Vega VT Systems Only

Polaris Vega VT Position Sensors include an integrated video camera located centrally between the two image sensors, with a small vertical offset, as shown in Figure 2-2. The video camera provides a live, colour video stream of the Vega measurement volume.

Important notes:

- The video camera is not used for tracking tools.
- Streaming from the video camera is via port 554.
- Control of the video camera is either via the Position Sensor (proxying) or through a direct connection to the video camera. See “Connecting to the Video Camera” on page 58.

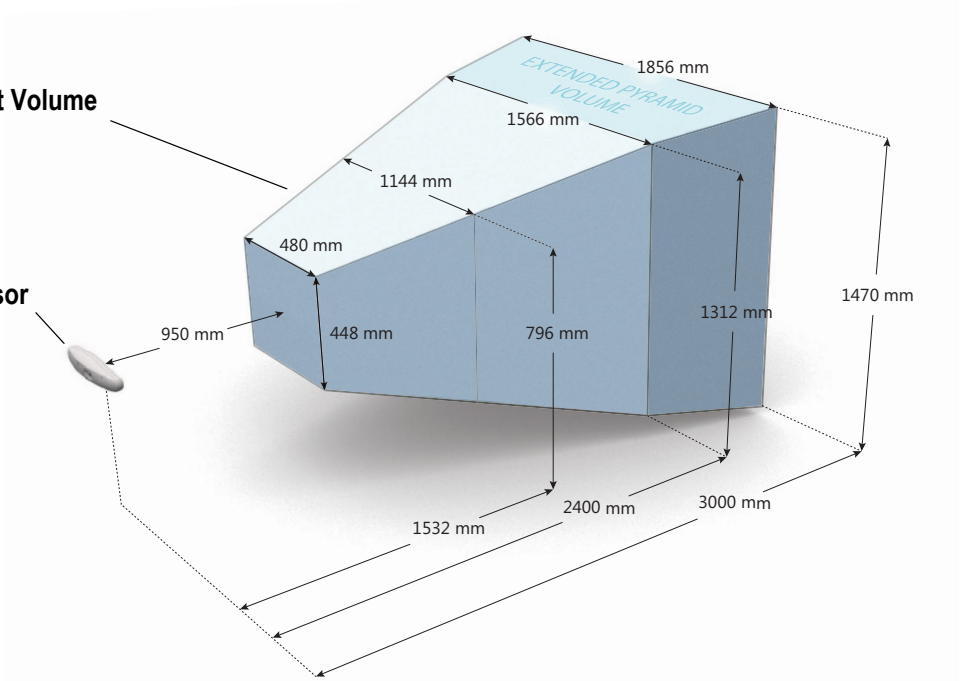
For augmented reality overlays, the Position Sensor’s tool tracking data stream can be projected onto the live video stream. For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

For more information about configuring the video camera and connecting to the streaming output, see “Video Camera” on page 53.

6.2 Extended Pyramid Volume

The optional extended pyramid volume provides a larger characterized measurement volume. Figure 6-1 shows the extended pyramid volume as the back section of the measurement volume, 2400 mm to 3000 mm from the Position Sensor.

Figure 6-1 Polaris Vega Extended Pyramid Volume



For the accuracy specification for the extended pyramid volume, see “Polaris Vega System Performance” on page 101.

6.3 Positioning Laser

The positioning laser, if included, is located in the Position Sensor and indicates the general centre of the characterized measurement volume. This feature allows you to properly position the Position Sensor, or position objects in the measurement volume, relative to the measurement volume. An aperture on the front of the Position Sensor emits the laser beam and directs it along the *z*-axis of the Position Sensor’s global coordinate system. For more information, see “Global Coordinate System” on page 29.

Note **The laser spot diverges with increasing distance from the Position Sensor.**

Activate the Laser

Do not look directly into the laser-emitting aperture. The Class 2 laser module on the Position Sensor emits radiation that is visible and may be harmful to the human eye. Direct viewing of the laser diode emission at close range may cause eye damage.

Ensure that people with restricted movement or reflexes (for example, patients undergoing medical procedures) do not look directly into the laser-emitting aperture. Patients undergoing medical procedures may be restricted in the availability of adverse-effects reflexes (turning away eyes and/or head, closing eyes) due to pharmaceutical influences and/or mechanical restraints. The Class 2 laser module on the Position Sensor emits radiation that is visible and may be harmful to the human eye. Direct viewing of the laser diode emission at close range may cause eye damage.

Do not use controls or adjustments or perform procedures other than specified in this guide. Doing so may result in hazardous light exposure.

1. To activate the laser, perform one of the following actions:
   - Use the ToolBox Configure utility. The **Laser** option is available under the **Position Sensor Settings and Options** node. The laser automatically turns off after 35 seconds.
   - Use the API command `SET` to set the value of the user parameter `Param.Laser.Laser Status`. A value of **1** turns the laser on, and a value of **0** turns the laser off. The laser automatically turns off after 35 seconds.
   - Connect a switch to the external laser trigger connector. For more information, see "External Laser Trigger Switch" on page 48.

For more information on user parameters and API commands, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

Laser Specifications and Standards

The positioning laser is a Class 2 laser with the following characteristics:

Table 6-1 Laser Characteristics

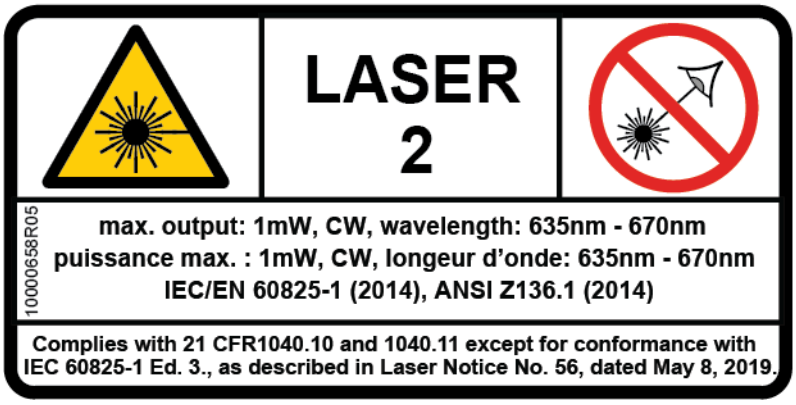
Wavelength	635 nm to 670 nm
Maximum Output	1 mW
Divergence (Half Angle)	0.7 mRad

The Polaris Vega System containing a positioning laser conforms to the following standards:

- ANSI Z136.1 (2014)
- IEC 60825-1 (2014) and IEC 60825-1 (2007))
- FDA/CDRH 21 CFR 1040.10 and 1040.11 except for deviations pursuant to Laser Notice No. 56, dated May 08, 2019

The label shown in Figure 6-2 is located on the back of the Position Sensor and lists the laser specifications and safety information.

Figure 6-2 Position Sensor Laser Label



Note **If the Polaris Vega System is incorporated into another product, the laser safety information must be included in the product manual.**

External Laser Trigger Switch

You can use the Laser Activation Port Connector to connect a switch to trigger the laser. When the switch is closed, the signal is pulled internally to ground and the laser is activated. The Polaris Vega has an on-board 1 kΩ resistor to limit the current. When the switch is open or is not connected, the signal is pulled internally to +3.3 V.

The port is a four-conductor, 3.5-mm audio jack. Only the first two conductors are used, as follows:

Conductor	Signal
1 (tip)	Laser switch contact input
2	Laser switch contact input
3	Not used
4	Not used

If you intend to design and integrate an external laser activation switch, to provide the maximum protection against electrostatic discharge (ESD), consider the following:

- Design the handle and select a switch that minimizes the possibility of energy flowing into the laser switch circuits in the Position Sensor.
- Electrically isolate the body of the switch from the handle and provide a good grounding path from the handle to earth ground. This directs the ESD energy to flow through the handle and to ground, instead of passing through the laser switch wiring into the Position Sensor.

6.4 Gamma Radiation Hardening

The Vega ST model is available in a radiation robust variant that is suited for use in radiation oncology environments. Hardening of the Vega ST Position Sensor against gamma and neutron radiation is achieved by a shielded processor and modified memory management.



Do not use the Polaris Vega System in an x-ray, proton radiation, or MRI environment without first determining the performance, including accuracy, of the Position Sensor in these environments. NDI has not validated the Position Sensor in these environments. Failure to do so may result in personal injury.

6.5 Serial Interface

The Vega ST Position Sensor is available with an optional LEMO connector to support serial communications between the Position Sensor and host computer.

For more information, see “Serial Communications” on page 63.

6.6 Bump Sensor and Internal Battery

All Vega Position Sensors contain a bump sensor that detects when the Position Sensor has suffered an impact. Although each instance is different, NDI expects that a representative trigger threshold is equivalent to a 50 mm to 400 mm (depending on orientation) drop onto a vinyl tiled concrete surface.

When the bump sensor detects a bump:

- The error LED on the Position Sensor is on, indicating that a potentially minor recoverable fault has been detected. The on state persists until the bump is cleared.
- The “bump detected” bit in the `Info.Status.Alerts` user parameter is set. This bit persists until the bump is cleared.
- The “bump detected” bit in the `Info.Status.New Alerts` user parameter and the “diagnostic pending” bit in the TX and BX responses are set to indicate a new alert. When the `Info.Status.New Alerts` parameter is read, the “bump detected” bit is cleared.
- The API command `BX2` reports transformations and includes a System Alert component with data to indicate the cause of the alert.
- By default, the API commands `TX` and `BX` do not report transformations if a bump is detected. Use the reply option `0x0800` to override this behaviour.
- The `Info.Status.Bump Detected` and `Param.Bump Detector.Bumped` user parameters are set to **1**. These user parameter values persist until the bump is cleared.

Note **For more information on user parameters and API commands, see the *Polaris Vega and Lyra Application Programming Interface Guide*.**

If the bump sensor detects a bump, NDI recommends that you perform an accuracy assessment procedure with the NDI Accuracy Assessment Kit (AAK) to ensure that the Position Sensor is still calibrated. For information on the accuracy assessment procedure and AAK, contact NDI or visit the NDI Customer Support site at https://support.ndigital.com.

If the bump sensor is triggered, the Position Sensor may no longer be covered under warranty, as the impact required to trigger the bump sensor is greater than that expected to occur through proper use and handling of the Position Sensor.

Clear the Bump Sensor

1. To clear a bump, perform one of the following actions:
   - Use the Configure utility of NDI ToolBox. For instructions, see the *NDI ToolBox Online Help*.
   - Use the API command `SET` to set the value of the user parameter `Param.Bump Detector.Clear` to **1**.
     This clears all bumps detected up to this point. The system automatically resets this user parameter to **0**. For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

Internal Battery

The Position Sensor contains an internal battery that powers the system's real-time clock and powers the bump sensor when the system is powered off. When the system is powered on, the bump sensor circuit draws its power from the external power supply.

The battery reaches full charge when the Position Sensor is powered on for more than 12 hours. If the bump sensor is not triggered, a fully-charged battery will power the bump sensor and real-time clock for approximately two years. If the bump sensor is triggered, the battery will power the sensor and clock for approximately two months.

When the battery is drained where it is unable to power the bump detector and real-time clock, the bump detector no longer functions, the real-time clock loses its setting, and the system reports a low-battery status.

If the battery can no longer power the bump sensor, the system reports a bump every time it is disconnected from the external power. This persists until the battery is recharged and the bump is cleared.

After the battery is recharged, NDI recommends that you perform an accuracy assessment with the NDI Accuracy Assessment Kit (AAK) to ensure that the Position Sensor is still calibrated. If the system passes the AAK threshold, you can clear the bump alert.

Reset the Real-Time Clock

1. Perform one of the following actions:
   - If you use NDI ToolBox version 6 or later, connect to the system. ToolBox automatically adjusts the clock and adds an entry to the system’s event log indicating the clock was adjusted.
   - If you use an earlier version of NDI ToolBox, use the ToolBox Configure utility to enter to correct date and time. The **Clock** option is available under the **Position Sensor Settings and Options** node.
   - Use the API command `SET` to set the value of the user parameter `Param.Clock.Date Time`.

For more information on user parameters and API commands, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

6.7 System Watchdog

A system watchdog monitors the Polaris Vega Position Sensor. If the system watchdog detects a hardware or firmware malfunction, it writes an entry to the event log and forces the Position Sensor to restart. If the system watchdog initiates repeated restarts, you should investigate to determine the cause.

If the system is used in an environment that has a high level of gamma and neutron radiation, radiation-induced bit-flips (soft-errors) can cause the system to temporarily malfunction. A restart ensures the system is returned to a known “good” state. If the system is used in gamma and neutron radiation environments, see “Gamma Radiation Hardening” on page 49.

After restarting, the host application must reconfigure the Polaris Vega System, such as loading tool definition files, before tracking can resume.

6.8 Multi Firmware

The multi firmware feature is a configuration option available for purchase from NDI that allows the system to house more than one combined firmware revision. When you enable the multi firmware feature, you can specify which combined firmware revision the system uses on its next reset or power up.

You can install new combined firmware revisions using NDI ToolBox and can use ToolBox or API commands to specify which combined firmware revision the system will use.

For more information on NDI ToolBox, see the *NDI ToolBox Online Help*. For more information on using API commands, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

6.9 Data Averaging

Data averaging is available for all Polaris Vega Systems in combination with the API command `BX2`. Data averaging allows the user to trade data reporting frequency for a reduction in position jitter (noise).

Data averaging is most useful with the higher frame frequency available on Vega XT.

Data averaging is disabled by default, but you can enable and configure it through user parameters. For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

6.10 Log Files and User-Defined Parameters

The System Control Unit and Position Sensor each contain a log file. The system records events to the log file, such as bump detector events, firmware updates, and changes in system status. On the Position Sensor, each entry includes the date and time, using UTC. If the System Control Unit is connected to a Position Sensor when the log entry is made, the log entries in the System Control Unit are also labelled with the date and time.

You can read and write to the log file using the API commands `GETLOG` and `SYSLOG` or by using the NDI ToolBox Configure utility; however, record only major events to the log file. For example, you may want to record an incoming inspection result in the system log.

Record minor events for the Position Sensor or System Control Unit in user-defined strings instead. For example, a system maintenance or cleaning schedule. The system does not use the user-defined parameters.

You can read and update the user-defined parameters in the ToolBox Configure utility or by using the API commands `GET` and `SET` to read or set the value of the user parameter `Param.User.String`*X*, where *X* ranges from 0 to 4.

For more information on user parameters and API commands, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

7 Video Camera

You can use the “live view” video camera integrated into the Vega VT Position Sensor to capture a video stream of the Vega measurement volume. The video camera streams data using the RTSP protocol; to display the stream from the video camera, use a third-party media player that supports this protocol.

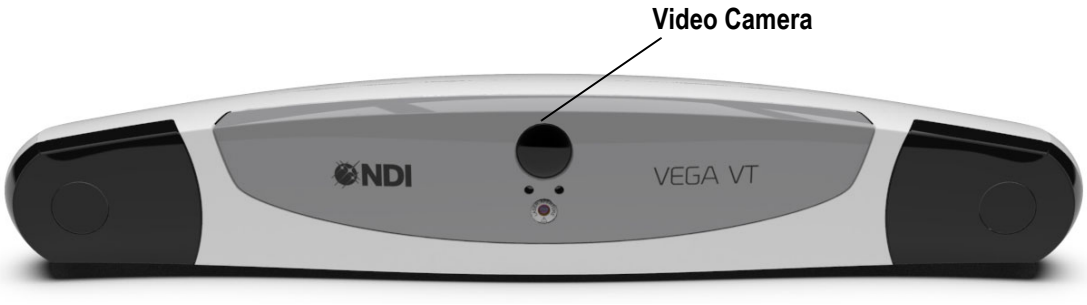
The video camera is fixed focus, with the point of focus being 2.5 m from the centre of the Position Sensor. The focal length of the camera is 4.35 mm. Table 7-1 details the field of view for each supported resolution.

The camera provides video output in standard video formats. The lowest available resolution is 1024 x 768 at 47fps (maximum) and the maximum resolution is 2048 x 1536 at 20fps (maximum).

Table 7-1 Video Camera Field of View

Resolution	Horizontal Field of View	Vertical Field of View
2048 x 1536, 1024 x 768	54.8°	42.5°
1920 x 1088	51.8°	30.6°

Figure 7-1 Vega Video Camera - Front View



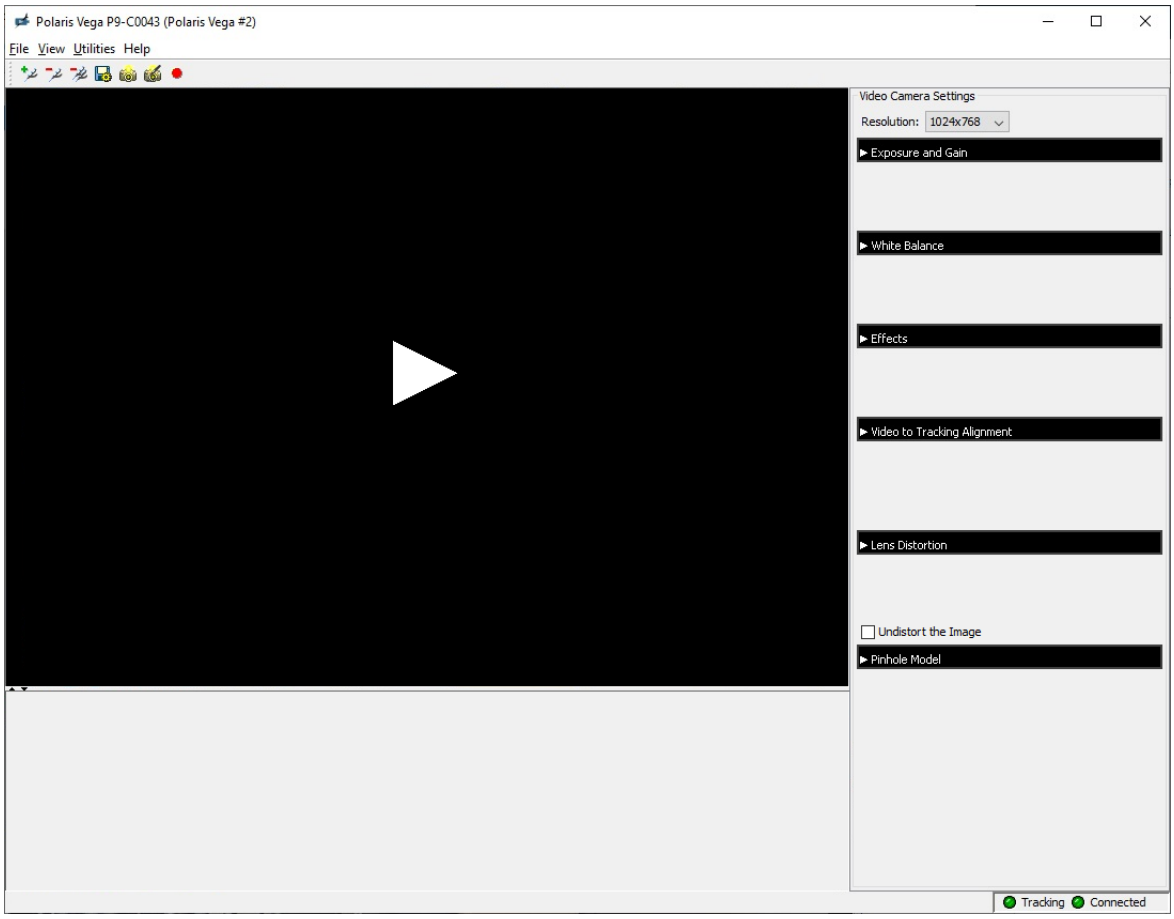
7.1 View the Video Stream

Before you can view the video stream, ensure that you have set up and connected the hardware, as described in “Connecting the Hardware” on page 16, and have installed NDI ToolBox, as described in “Installing NDI ToolBox” on page 18. During the installation, select the option to install GStreamer.

Figure 7-2 shows the main window of the Video Client utility when streaming is paused. Any tools that are being tracked are displayed under the video stream display area. The toolbar above the video stream display area allows you to perform different functions, such as taking a snapshot of the

video stream and recording the video stream to a file. For more information on the Video Client utility toolbar, see the *NDI ToolBox Online Help*.

Figure 7-2 ToolBox Video Client Utility - Main Window



1. Open the NDI ToolBox Video Client utility.
2. To connect to the Position Sensor that contains the video camera you wish to stream from, select **File> Connect to**, then select the Vega Position Sensor of interest from the list.

   The Video Client utility displays the stream from the video camera.
3. Perform one or more of the following actions:
   - To adjust aspects of the video stream, use the **Video Camera Settings** panel on the right side of the utility. For more information on video camera settings, see “Common Adjustments” on page 54.
   - To pause streaming, hover the mouse over the video stream display area and click the **Pause** button that is displayed.
   - To resume streaming, hover over the same area and press the **Play** button that is displayed.

7.2 Common Adjustments

This section describes some common adjustments you might use to optimize the appearance of the video camera output in your environment.

Adjust the Resolution

You can change the resolution when not streaming. Increasing the resolution decreases the number of frames per second.

1. Select the **Resolution** drop-down and select one of the following resolutions:
   - 1024 x 768 at 14 to 47 fps
   - 1920 x 1088 at 11 to 30 fps
   - 2048 x 1536 at 10 to 20 fps

Adjust the Frame Frequency

To reduce flicker caused by lighting, adjust the frame frequency within the range specified by the resolution. For example, if the light source is running at 50 Hz and the exposure time is short (as it usually would be in a bright scene), you may notice the image flicker at a constant rate. Adjusting the frame frequency from 30 Hz to 25 Hz will mitigate this issue because 50 is an even multiple of 25.

Lowering the frame frequency also allows for longer exposure times, which can produce better images of static scenes - they are typically less noisy than those that rely on the gain.

1. Perform one of the following actions:
   - Use the NDI ToolBox Configure utility.
   - Use the API to set the value for the `Param.Frame Rate` video camera user parameter.

For more information, see “Configure the Video Camera Using the ToolBox Configure Utility” on page 57 or “Configure the Video Camera Using the API” on page 57.

Adjust the Image Brightness

1. Start streaming from the video camera so you can see the effects of your adjustments.
2. Expand the **Exposure and Gain** menu.
3. Enable **Automatic Exposure and Gain**.
4. Adjust the **Brightness Target** slider to achieve optimal brightness.
5. If necessary, select a different **Metering** mode to match your environment. Perform one of the following actions:
   - To set the metering for general scenes, select **Center Weighted**.
   - To set the metering for backlit scenes, select **Segmented**.
   - To set the metering for spotlighted scenes, such as an operating room with a particularly bright area, select **Spotlight**.
   - To set the metering using a zone in the image, select **Weighted ROI**. To view the zone, select the **Display ROI** option. To select a different zone, click **Define ROI**, then click and drag a rectangle over the desired zone in the video stream display area.
6. To prevent motion blur, adjust the **Maximum Exposure** slider to the highest setting.

Adjust the Saturation

You can improve high contrast, highly-saturated images by increasing the gamma correction value to reduce the amount of saturation in the image.

1. Start streaming from the video camera so you can see the effects of your adjustments.
2. Expand the **Effects** menu.
3. In the **Gamma Mode** drop-down, select **User Defined** or **User Defined - Enhanced**.
4. Use the **Gamma** slider to increase the gamma correction value.

Adjust the White Balance

The White Balance settings adjust the colour gains and offset to achieve a realistic colour video. When **Automatic White Balance** is enabled, the algorithm adjusts the red and blue gains to try to maintain an image with equal amounts of red, green, and blue.

1. Start streaming from the video camera so you can see the effects of your adjustments.
2. Expand the **White Balance** menu.
3. Select the **Automatic** option.
4. Adjust the **Blue Green Factor** and **Red Green Factor** sliders to achieve optimal colour balance.

Adjust the Streaming Latency

There are several steps that can be taken to reduce the latency of the video stream.

On the computer running the streaming client,

- Install a high quality, high-speed (gigabit) network interface card.
- Install a high-speed video card.
- Increase the CPU power.
- Increase the priority of the application that is processing the video. In Windows, for example, this is done in the Task Manager.
- Select a lower image resolution. A resolution of 1024x768 offers the lowest latency.
- Increase noise filtering to reduce the amount of transmitted image data (at the expense of fine detail).

The network setup can also affect the latency of the video stream. The lowest latency configuration is a direct connection between the Vega Position Sensor and the client computer. Connecting the Position Sensor to the client computer via network switch or hub may increase latency, depending on the amount of traffic on the network.

Note **Integrating the Vega Position Sensor into a 10Base-T network is not recommended.**

Performance of the video camera is highly dependent on network throughput. Avoid devices, such as Ethernet to USB converters, that might reduce this throughput.

7.3 Configuring the Video Camera

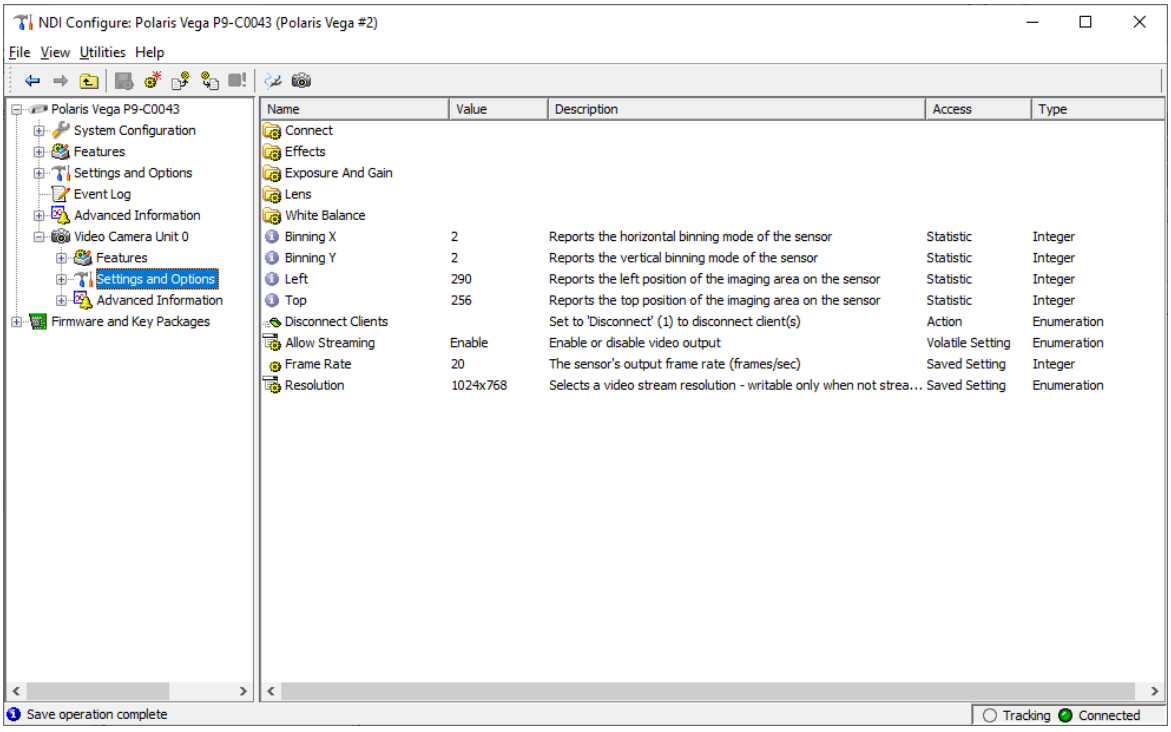
You can configure the video camera using the NDI ToolBox Video Client utility, NDI Toolbox Configure utility, or the API.

Configure the Video Camera Using the ToolBox Configure Utility

The NDI ToolBox Configure utility displays system information and controls the setting of video camera parameters and options.

1. Open the NDI ToolBox Configure utility.
2. To connect to the Position Sensor that contains the video camera you wish to work with, select **File> Connect to**, then select the Vega Position Sensor of interest from the list.
3. In the panel on the left of the Configure utility, expand the **Polaris Vega** node, then the **Video Camera** node**.**
4. Select **Settings and Options**.
5. In the panel on the right, use the various settings to adjust the configuration of the video camera for your environment. See Figure 7-3.

Figure 7-3 Video Camera Settings in the ToolBox Configure Utility



Configure the Video Camera Using the API

The method for setting and getting parameters for the video camera depends on whether the communication with the video camera is performed through the Position Sensor (proxying) or via a direct connection to the video camera. For information on changing this setting, see “Connecting to the Video Camera” on page 58.

If the connection is through the Position Sensor, prefix all parameter names with "VCU-0.". If a direct connection is used, this prefix can be omitted.

Table 7-2 Video Camera API Commands

Action	API Command
Set a parameter via the Position Sensor	`SET VCU-0.<parameter name>`
Get a parameter via the Position Sensor	`GET VCU-0.<parameter name>`
Retrieve all video camera parameters via the Position Sensor, including details	`GETINFO VCU-0.*`
Default all video camera parameters via the Position Sensor	`DFLT VCU-0.*`

Note **If you have adjusted video camera parameters, you can use the INIT command to reset the parameters to their saved values. This will also close any currently streaming video connections.**

For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

7.4 Connecting to the Video Camera

You can connect to the video camera through the Position Sensor or directly to the video camera. When you connect through the Position Sensor, only one connected application can change parameter values for the video camera. To change the parameter values for the video camera with a different application, you must enable the direct connection to the video camera and use that connection for setting video camera parameter values.

- When Direct Connection is **Disabled**, connection to the video camera is through the Position Sensor via port 8765. This is the default setting.
- When Direct Connection is **Enabled**, the connection is directly to the video camera, via port 8766.
- Streaming is always via port 554.

Note **The instructions in this guide assume the connection to the video camera is through the Position Sensor. If you connect directly to the video camera instead, you must adjust your actions accordingly.**

Change the Direct Connection Setting Using the ToolBox Configure Utility

For information on changing the value of the parameter `Param.Video Camera.Direct Connection` via the API, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

1. In the ToolBox Configure utility, connect to the desired Position Sensor.
2. In the panel on the left of the Configure utility, expand the **Polaris Vega** node, then select **Settings and Options**.

3. Click the **Video Camera** folder.
4. In the panel on the right, change the value for **Direct Connection** to **Enabled** or **Disabled**.
5. Save the change, restart the system, then reconnect to the video camera.

When you configure the video camera for a direct connection, the **Connect to** menu in the NDI ToolBox utilities lists it as a separate device, identified by its serial number and the suffix “VCU” for “Video Camera Unit.”

7.5 Using Different Streaming Clients

The video camera streams data using the RTSP protocol; to display the stream from the video camera, use a third-party media player that supports this protocol.

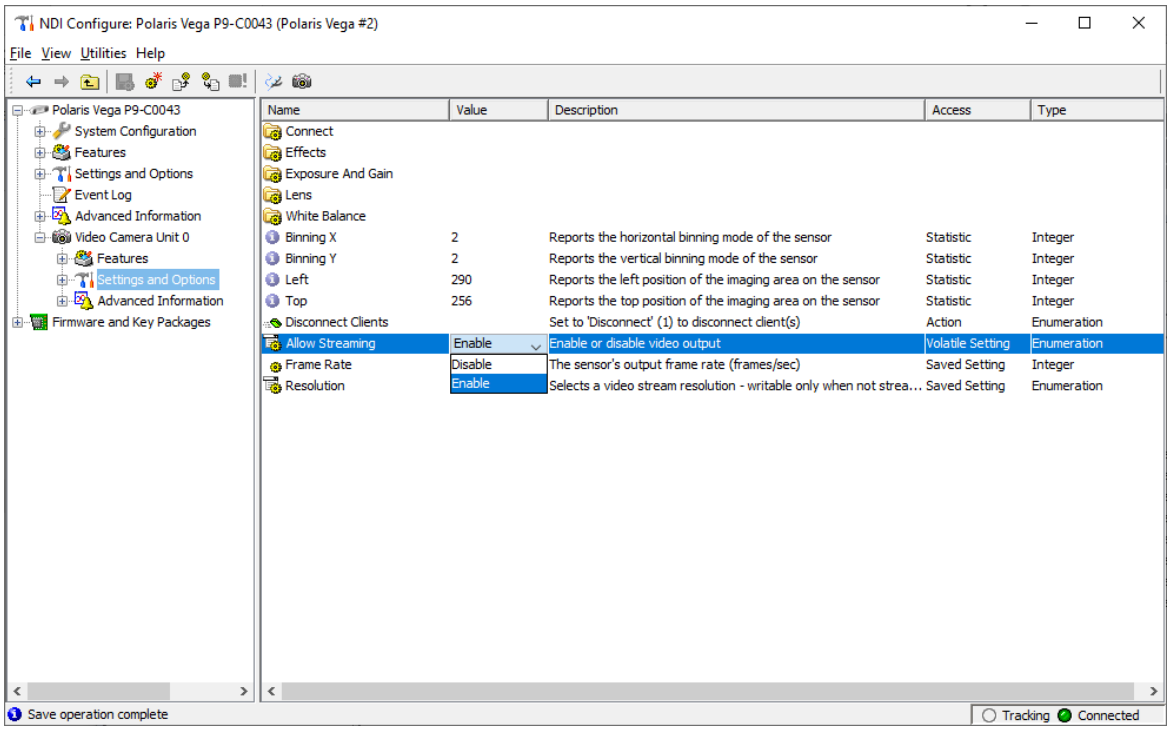
Note **The Vega video camera can support one streaming connection at a time.**

Enable Streaming

The ToolBox Video Client utility automatically enables streaming from the video camera. To prevent unauthorized access to video data from third parties, you must enable streaming when using any other client. Without streaming enabled, any connection attempt to port 554 will fail.

1. Start the NDI ToolBox Configure utility.
2. In the panel on the left of the Configure utility, expand the **Polaris Vega** node, then the **Video Camera** node**.**
3. Select **Settings and Options**.
4. Set the **Allow Streaming** option to **Enabled**. See Figure 7-4.

Figure 7-4 Enable Video Streaming with ToolBox



Find the IP Address for the Video Stream URL

To establish a connection between the video camera and the media player, you must provide a URL in the format **rtsp://hostname.local:554/video** or **rtsp://*<IP Address>*:554/video**. If you do not know the IP address, you can find it using the NDI ToolBox Configure utility.

1. In the NDI ToolBox Configure utility, navigate to **File> Connect to**.
2. Hover over the Vega Position Sensor in the list and make note of the IP address that is displayed.

Note **To allow video streaming to the computer running the media player, you may need to configure the computer's firewall to allow UDP traffic on ports 30000-30500 and TCP/UDP traffic on port 554. Contact your network administrator for details.**

Stream with GStreamer

GStreamer is a framework for creating streaming media applications that is freely available and recommended by NDI because it does not introduce significant latency.

Note **The Vega video camera was tested by NDI with GStreamer version 1.12.1 with the libav wrapper installed.**

GStreamer provides a library that can be used to process the video coming from the Vega video camera. To use the library from the command line, you must use the gst-launch-1.0.exe application and construct command line arguments that can communicate with the video camera (rtspsrc),

decode and decompress the video data (rtph264depay, avdec_h264) and display it on the screen (autovideosink). Each processing element is delimited by an exclamation point (!) and can be configured using its available properties (e.g. latency is a property of rtspsrc).

The line below is the structure to use to display live video on your screen:

```
gst-launch-1.0.exe rtspsrc location=rtsp://[IP Address/hostname.local or IP
address of Position Sensor]:554/video latency=1 protocols=1 drop-on-latency=true
do-retransmission=false udp-reconnect=false ! rtph264depay ! avdec_h264 ! queue !
autovideosink sync=false
```

Following is specific example of the command:

```
gst-launch-1.0.exe rtspsrc location=rtsp://192.168.1.186:554/video latency=1
protocols=1 drop-on-latency=true do-retransmission=false udp-reconnect=false !
rtph264depay! avdec_h264 ! queue ! autovideosink sync=false
```

Note **This example uses the IP address of the Position Sensor (192.168.1.186:554). You can use the host name instead. For example, P9-C0043.local:554.**

Stream with VLC

VLC is a simple, freely-available media player. However, it internally buffers the video data before displaying it on the screen, which adds significant latency to the video display. NDI recommends GStreamer or the NDI ToolBox Video Client utility over VLC.

Ensure that you have downloaded and installed VLC and that you have the IP address for the Video stream. For more information, see “Find the IP Address for the Video Stream URL” on page 60

1. In the VLC main window, select **Media> Open Network Stream**.
2. In the **Open Media** dialog, on the **Network** tab, type the URL. For example, **rtsp://192.168.1.186:554/video**
3. Click **Play**.
4. If you encounter errors when trying to stream to the VLC media player, you may need to configure VLC as follows.

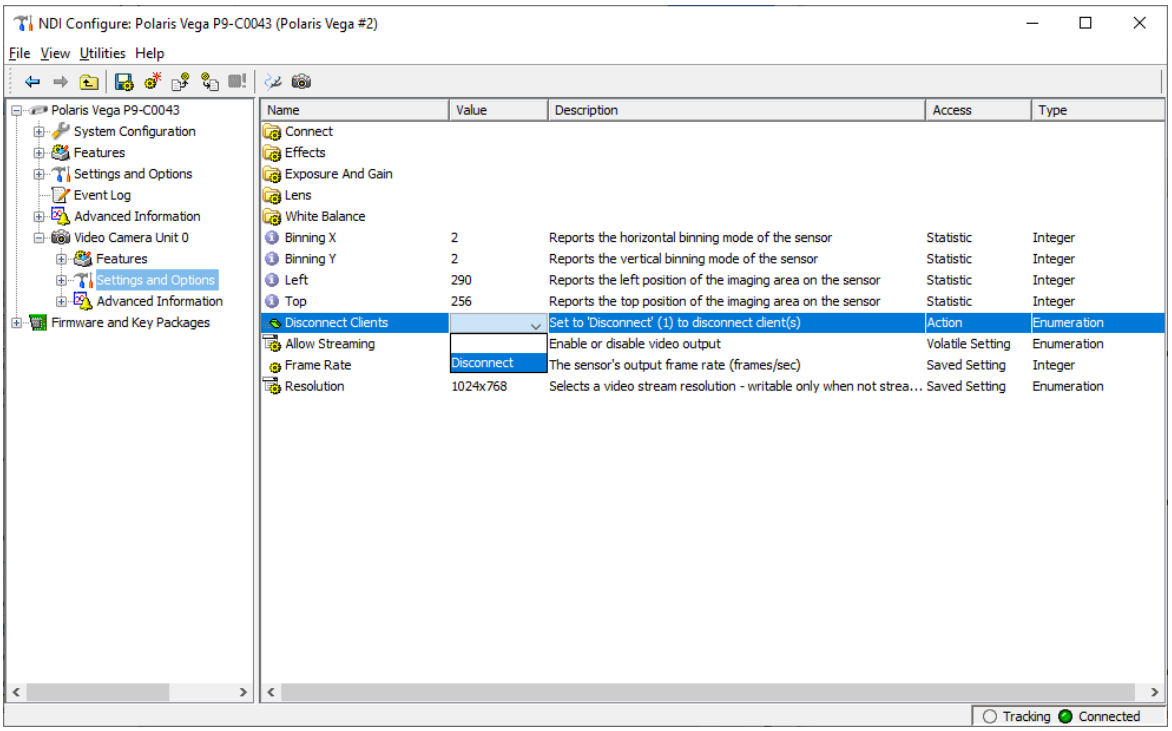
1. In the VLC main window, select **Tools> Preferences** then click **Input/Codecs**.
2. At the bottom of the screen, select the **HTTP** option.

Disconnect Streaming Clients

To force a streaming client to disconnect from the video stream, use the **Disconnect Clients** setting.

1. Start the NDI ToolBox Configure utility.
2. In the panel on the left of the Configure utility, expand the **Polaris Vega** node, then the **Video Camera** node**.**
3. Select **Settings and Options**.
4. Set the **Disconnect Clients** option to **Disconnect**. See Figure 7-5.

Figure 7-5 Toolbox Setting: Disconnect Clients

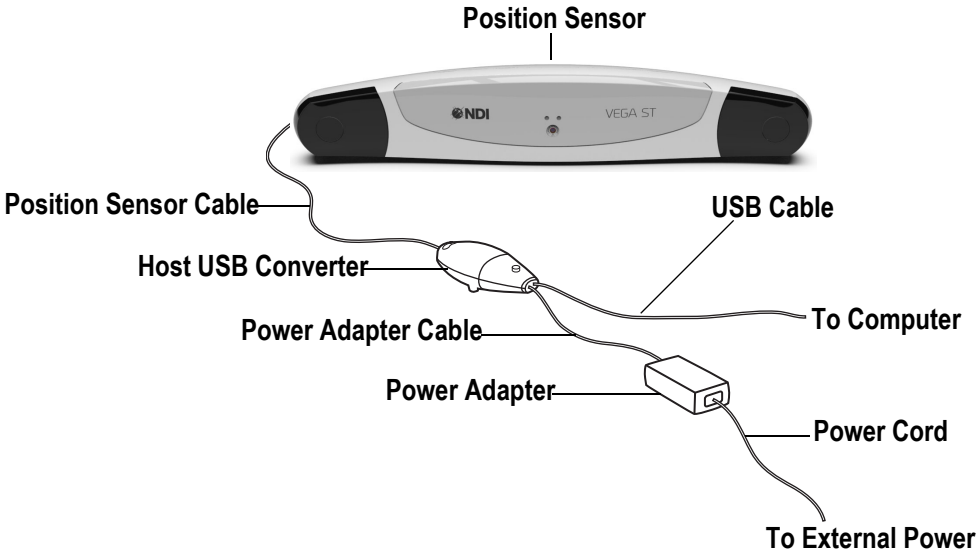


8 Serial Communications

The Vega ST Position Sensor is available with an optional LEMO connector to support serial communications between the Position Sensor and host computer. NDI recommends this option only for customers migrating from Polaris Spectra to Polaris Vega, not for new designs.

If you have questions regarding communication interfaces. contact NDI. See “Contact Information” on page viii.

Figure 8-1 Serial Connection



This chapter provides the information and instructions required to set up the Polaris Vega ST System for serial communications.

LEMO Connector

The LEMO connector is a 14-pin connector that provides power to the Position Sensor and allows communications to and from the Position Sensor. The LEMO connector is located on the back of the Position Sensor. Figure 8-2 shows the pin details, and Table 8-1 shows the signals on each pin.

If you make a custom cable, use LEMO part number FGA.1B.314.CYCD62Z or equivalent LEMO connector to mate to the connector on the Position Sensor.

Cable shield ground connection must be maintained to the shell of the LEMO connector.

Any unused contacts can be left floating.

Figure 8-2 Position Sensor Connector Layout

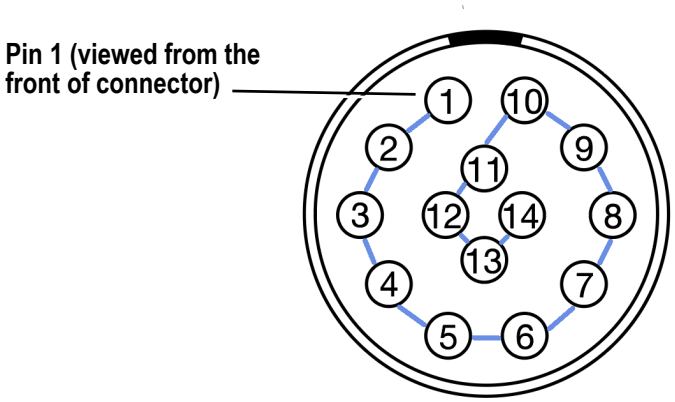


Table 8-1 Position Sensor Connector - Signals

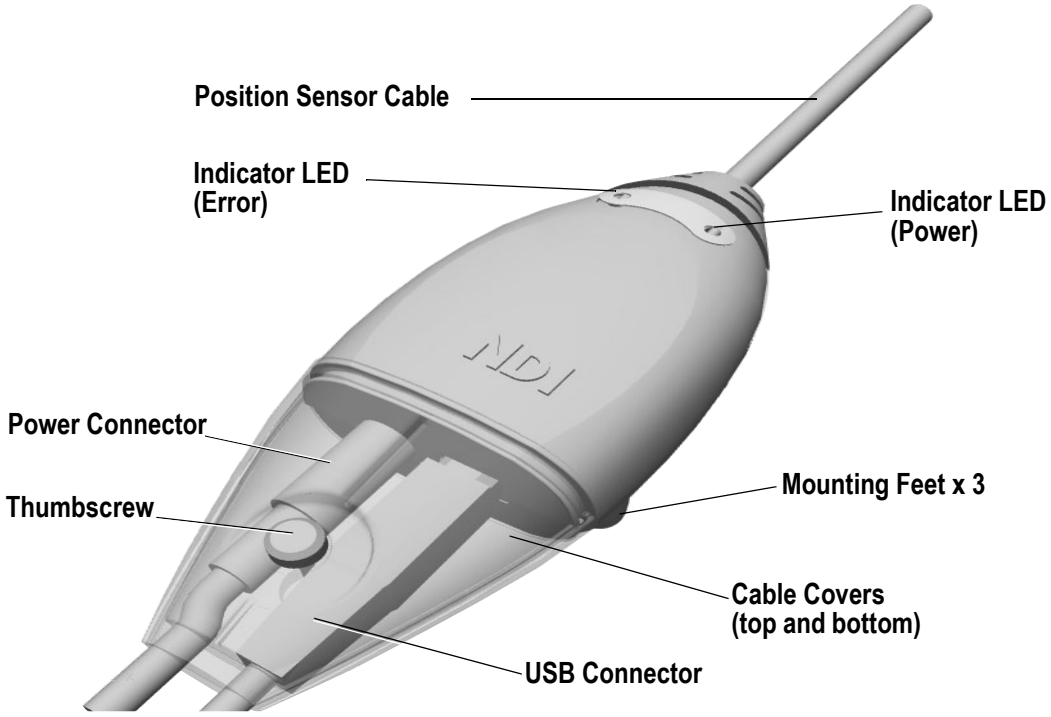
Pin	Signal
1	Power
2	Power
3	Rx +
4	Rx -
5	Ground
6	Ground
7	Ground
8	Tx +
9	Tx -
10	Power
11	RTS +
12	RTS -
13	CTS +
14	CTS -

Host USB Converter

The Host USB Converter provides the interface between the Position Sensor, the power adapter, and the host computer:

- The attached Position Sensor cable connects the Host USB Converter to the Position Sensor.
- The power adapter and USB cable plug into the Host USB Converter.
- Modem status bits are provided. See “Serial Port Emulation” on page 66.

Figure 8-3 Host USB Converter



Cable Covers: Two covers (top and bottom) secured together by a thumbscrew keep the USB and power adapter cables in place.

Position Sensor Cable: Provides the data connection and power between the Position Sensor and Host USB Converter. This cable is permanently connected to the Host USB Converter and is available in lengths up to 15 m, depending on the configuration of your system.

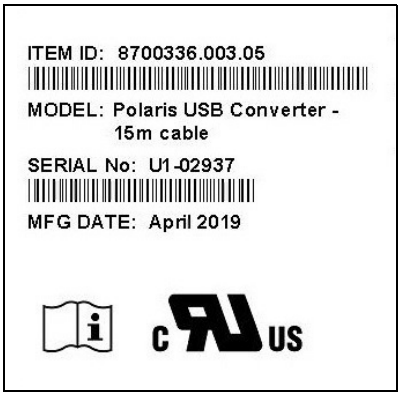
Power Indicator: Lights green when power is supplied to the Host USB Converter.

Error Indicator: Flashes briefly when the Host USB Converter is first powered on, then lights amber if the Host USB Converter detects a fault.

Mounting Feet: Three feet incorporate an internal thread (M6 x 1 mm pitch x 4 mm depth) to attach the Host USB Converter to an external structure. For example, a cart. For free-standing use, rubber inserts are located in the mounting feet.

The serial number label on the bottom of the Host USB Converter shows the item ID, model, serial number, and manufacture date of the Host USB Converter.

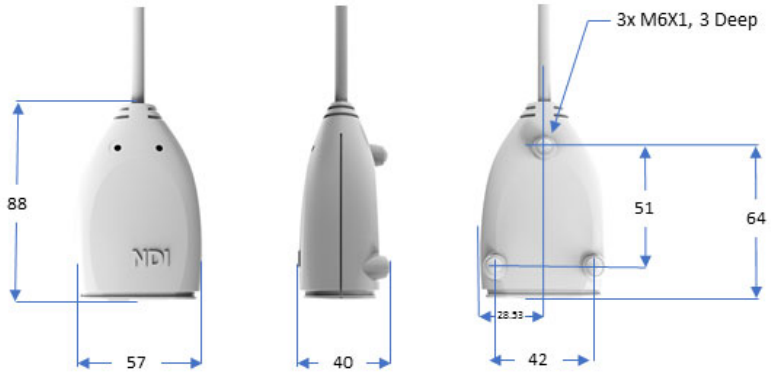
Figure 8-4 Host USB Converter Serial Number Label



The Host USB Converter can be free-standing or can be mounted via M6 x 1 mm pitch x 3 mm deep threaded holes located in the three mounting feet. Figure 8-5 shows the Host USB Converter dimensions and mounting arrangement.

Note **Before you design a custom enclosure or any attachments for the Host USB Converter, contact NDI for assistance. See "Contact Information" on page viii.**

Figure 8-5 Host USB Converter Mounting Details



All dimensions in mm

Figure 8-5 is provided for information purposes only and dimensions are subject to change without notice.

If the Host USB Converter is located within the patient vicinity where it may be touched by the patient, the operating ambient temperature range is +10°C to +30°C. If the Host USB Converter is located outside this patient vicinity, the operating ambient temperature range permitted for the system is +10°C to +40°C.

Serial Port Emulation

Modem Status Bits The Host USB Converter emulates a standard PC serial port, allowing the host application to communicate with the Polaris Vega System as if it was a standard RS-422 device. The following modem status bits are used:

- When the Host USB Converter senses that a Position Sensor is connected and power must also be connected to the Host USB converter), the Data Set Ready (DSR) is set.
- If the Host USB Converter error indicator shows a fault condition, the Ring Indicator (RI) is.

Communication Rate If your operating system cannot set the serial port rate directly to 1.2 MBd, an aliased 19 200 Bd rate is provided to enable you to run at the higher speed. Do not set the Polaris Vega System to 19 200 Bd; if you set the system to 19 200 Bd, it cannot communicate with the host computer, because setting the host application to 19 200 Bd results in the aliased rate of 1.2 MBd. You can set the baud rate of the Polaris Vega System using the API command `COMM`. For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

Power Adapter

The system is powered by an NDI-supplied power adapter: Advanced Power Systems, part number APS49EMG-24021-7/Hitron, part number HEMG49-S240210-7 - applicable to IEC 60601-1 2nd Edition and 3rd Edition.

The power adapter connects to the external power supply and provides DC power to the Position Sensor via the Host USB Converter.

Alternatively, a customer-provided power adapter that meets the following criteria can be used:

- medical grade, double insulated (required if end-user will be in a medical environment)
- 24 V DC, 1.2 A output (40 W typical)
- Switchcraft part number 760 plug (or equivalent)

Note **The Underwriters Laboratories Inc. (UL) recognition of the Polaris Vega System includes the Vega Position Sensor, Host USB Converter, and the power adapter as follows:**

IEC 60601-1 2nd Edition and 3rd Edition - APS APS49EMG-24021-7/Hitron HEMG49-S240210-7 power adapter.

This investigation does not include the use of any other power adapter with the Polaris Vega System. Any such configuration will require further investigation. If the Polaris Vega System is used in a medical application, the final end-use configuration must be independently investigated to the IEC 60601 family of standards and all applicable national differences.

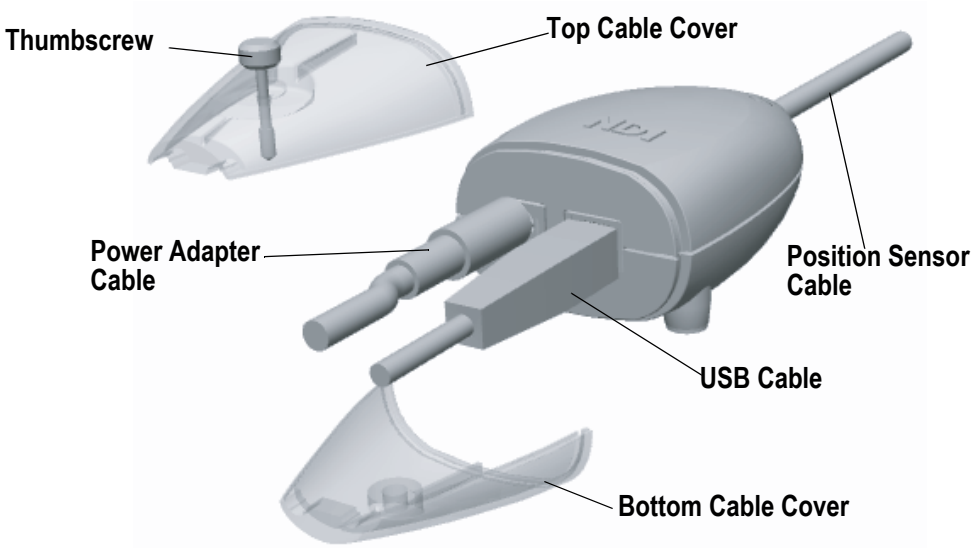
If a non-NDI supplied power adapter is used, it should be chosen to suit the particular use and the resulting system configuration must be verified for electrical safety by an approved test laboratory. For more information, contact NDI. See "Contact Information" on page viii.

8.1 Setting up for Serial Communications

1. Connect the Position Sensor cable (attached to the Host USB Converter) to the LEMO connector on the back of the Position Sensor.
   - Ensure that you align the red marking on the Position Sensor connector with the red marking on the cable connector and that the double keys of the connectors are aligned.
   - Make sure the Position Sensor cable is securely connected to the Position Sensor. A loose connection may result in partial functionality or unpredictable system behaviour.

2. Connect the USB cable B plug and power adapter cable to the Host USB Converter, as shown in Figure 8-6.
3. Place the top and bottom cable covers on the Host USB Converter and secure the covers with the thumbscrew.

Figure 8-6 Host USB Converter Connecting the Cables



4. Plug the USB cable A plug into the host computer.
5. Make sure all the cables are connected firmly, and placed where they will not be stressed, stepped on, or bent.
6. Plug the power cable into the power adapter.
7. Plug the power cable into the external power supply and turn on the power.

Note **The Position Sensor initially communicates to the host computer at 9600 Bd. If you are using the NDI ToolBox application, it increases the baud rate as high as possible. If you are using your own application, you can set the baud rate using the API command COMM. For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide*.**

8.2 Installing Host USB Drivers

If you are using the Host USB Converter for serial communications between the Position Sensor and host computer, you must install USB drivers on the host computer.

Follow one of the operating-system specific procedures below to install the drivers:

- “Install on Windows” on page 69
- “Install on Linux” on page 70
- “Install on macOS” on page 70

Install on Windows

About the Drivers

The Windows driver model consists of two parts: low-level USB and high-level virtual serial port. It appears as a Windows serial port, COMx (where x is enumerated), and emulates a standard PC serial port such that the Host USB Converter may be communicated with as a standard RS-422 device.

When you first connect the NDI Host USB Converter to the host computer, you must install drivers for the Host USB Converter. There are two sets of drivers for the Host USB Converter:

- The first set of drivers enable the Host USB converter to work with the host computer.
- The second set of drivers set up the USB port where the Host USB Converter is connected to emulate a serial port.

Driver Location

If you have already installed NDI ToolBox, the USB drivers were installed as part of that installation. By default, the drivers are located at C:\Program Files\Northern Digital Inc\ToolBox\USB Driver.

You can also download the drivers from the NDI Customer Support site at https://support.ndigital.com.

Installation

Note **When you first connect the Host USB Converter to the host computer, an error message may appear over the task bar. Ignore the message and complete the following steps to install the USB drivers.**

1. Start the Device Manager. In Windows 10, click the **Search** icon in the Windows task bar and begin typing **Device Manager**. When it appears in the list of matches to your search term, select the **Device Manager (Control Panel)** item.
2. Under **Other Devices**, right-click **NDI Host USB Converter**.
3. Select **Update Driver Software**, then select **Browse my computer for driver software**.
4. Perform one of the following actions:
    - Navigate to where you saved the downloaded driver.
    - Navigate to **Program Files\Northern Digital Inc\ToolBox\USB Driver**.
5. Click **Next**.
6. In the Windows security dialog, select **Install this driver software anyway**.
7. After the first set of drivers are installed, browse to **Other Devices** and right-click **USB Serial Port**.
8. Repeat steps 3 to 6.

Driver installation is complete and the system will now appear to be connected through a virtual COM port.

Host USB converters are interchangeable on the same USB port, without having to re-install drivers or losing the COM port previously established.

Reassign the Windows COM Number

If necessary, after the Host USB Converter drivers have been installed, you can reassign the Windows COM number.

1. Start the **Device Manager**.
2. Navigate to **Ports (COM & LPT)**.
3. Right-click the Host USB Converter.
4. Click **Properties**.
5. Click the **Port Settings** tab.
6. Click the **Advanced Settings** button.
7. In the dialog, select the **Com Port Number** dropdown list.
8. Reassign the COM port as necessary.
9. Click **OK**.

If you unplug the Host USB Converter from the USB port, Windows remembers the mapping. When reconnect the Host USB Converter using the same USB port, the COM port mapping is re-established.

Install on Linux

On Linux kernel versions 2.6 and later, USB serial devices appear as driver files /dev/ttyUSB*x*, where *x* is the port number. These drivers emulate a standard tty serial port and allow applications to communicate through the USB device as if it were an RS-422 port.

Kernel versions 2.6.32 and later automatically recognize the NDI devices.

Note **When connected, the Host USB Converter appears as "/dev/ttyUSB*x*", where *x* is the port number.**

To access the Polaris Vega from a non-root user account, you must add the user account to the "lock" group. You may also need to add the user account to the following groups:

- uucp
- tty
- dialout

You can do this only as root user with the command "usermod –G <group> <account name>." For the changes to take effect, log out and log in again.

Install on macOS

This file is also placed in the ToolBox installation folder during ToolBox installation.

Note **To manage NDI software on a Mac platform, you will need administrator account privileges.**

1. Perform one of the following actions:
    - Navigate to where you saved the downloaded driver.
    - Navigate to the Toolbox folder.
2. Double-click **FTDIUSBSerialDriver (NDI).pkg**.
3. Follow the on-screen instructions.

Note **The driver supplied with NDI ToolBox is necessary for the creation of a VCP software interface to an NDI device with a USB interface. The driver is a macOS kernel extension provided by FTDI and configured by NDI to support NDI devices.**

Modified FTDI USB to Serial Driver

If you have other devices, such as a USB to serial port converter, that also use a FTDI VCP driver, if you install the driver from NDI, you may:

- change the version of the driver previously installed
- possibly disable access to your other devices

The NDI configured version of the driver is based on v2.2.7 of the original FTDI VCP driver. If this version of the driver does not recognize your device, access to your device is disabled. For assistance, contact NDI. See “Contact Information” on page viii.

8.3 Connecting to a System Using Serial Communications

1. Perform one of the following actions:
    - On Windows, select **File > Connect to > (COMx).**
    - On Linux, select **File > Connect to > (/dev/ttyUSBx).**
    - On Mac, connect using /dev/cu.usbserial-xxxxxxxx or /dev/tty.usbserial-xxxxxxxx.

9 Maintaining the Polaris Vega System

User maintenance of the Polaris Vega System is limited to the following procedures:

- Cleaning the Position Sensor
- Checking the calibration of the Position Sensor
- Updating the firmware

The System Control Unit does not require any special cleaning considerations. Follow standard procedures to clean the System Control Unit.

Note **Do not open any component of the Polaris Vega System. Doing so will void the warranty.**



Maintenance Warnings

Before performing any maintenance on the Polaris Vega System, read the following warnings:

1. All user maintenance must be done by appropriately trained personnel and while equipment is not in use. Individual components of the Polaris Vega System contain no user-serviceable parts. Maintenance by untrained personnel may present an electric shock hazard.
2. Do not use the Position Sensor without inspecting it for cleanliness and damage before use. The Position Sensor should also be monitored during use. Reliance on data provided by an unclean or damaged Position Sensor may lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.
3. Do not immerse any part of the Polaris Vega System or allow fluid to enter the equipment. If fluids enter any part of the system, they may damage it and present a risk of personal injury.
4. Do not sterilize the Polaris Vega Position Sensor as this may cause irreversible damage to its components. Reliance on data provided by a damaged Position Sensor may lead to inaccurate conclusions. Inaccurate conclusions may result in personal injury.

9.1 Cleaning the Position Sensor

Inspect the Position Sensor regularly for cleanliness. Clean the Position Sensor, particularly the illuminator filters and lenses, only when necessary. You must determine the frequency of cleaning. This may include “in-use” cleaning.

Caution! **Use only 70% isopropanol solution and a soft lint-free cloth to remove handling smudges from the enclosure and illuminator covers. Other fluids may cause damage to the illuminator filters. Do not use any paper products for cleaning. Paper products may cause scratches on the illuminator filters.**

1. To remove dust from each illuminator filter and lens, gently pull a photographic lens duster brush across the surface in one direction only.
2. Continue cleaning the remainder of the Position Sensor, taking care not to wipe debris from the Position Sensor case onto the illuminator filters or lenses. Avoid prolonged contact between the isopropanol solution and the Position Sensor.

9.2 Cleaning the Video Camera

1. To remove any dust and particles from the window, use a photographic lens duster brush or an air puffing lens cleaner. Do not use pressurized air.
2. To remove fingerprints or other marks from the window, wipe gently in a circular motion with a clean, dry, microfibre lens cleaning cloth.
3. If necessary, slightly dampen a lens cleaning cloth with 70% isopropanol solution or a standard lens cleaning solution and wipe the window. The cloth should only be slightly damp, not wet.

Note **When wiping the window with a cloth or duster brush, use a circular motion and sweep up and away from the window to draw dust and debris out of the enclosure. Do not wipe dust, debris, or cleaning solution towards the edges of the window. An accumulation around the edges of the window can impact the quality of the image from the video camera. An air puffing lens cleaner is useful for removing dust and debris from the edges of the window.**

9.3 Checking the Calibration of the Polaris Vega System

All measurement systems require calibration to be confident that the system is performing within specification. Over time, measurement systems drift as mechanical and electrical components are exposed to different conditions; this is the primary reason that measurement systems decalibrate. Measurement systems may decalibrate from component degradation due to environmental exposure, age, mechanical vibration, or shock. Polaris Vega Systems are designed to minimize measurement drift, but it is impossible to eliminate decalibration completely.

You should perform a periodic calibration check on the Position Sensor. The frequency of the calibration check depends on the specific application and environment you use the Position Sensor in.

If the Position Sensor begins to lose calibration, it may lose the ability to track some tools before others. The various constraints used by the Polaris Vega System make certain tool designs more sensitive to loss of calibration than others. For example, if a tool has several similar segment lengths, similar angles between segments, or segment lengths similar to those of another tool, an out-of-calibration Position Sensor may not measure segment lengths as accurately and may not be able to determine which markers belong to which tool. In this case, the system reports the tools as missing. For more information on segment lengths and angles, see “Marker Detection and Tool Tracking” on page 33.

You can use NDI’s Accuracy Assessment Kit (AAK) to help determine whether a Position Sensor is performing acceptably for the user’s application. However, to ensure that all calibrations are conducted in accordance with procedures established specifically for the Polaris Vega Position Sensor, return the Position Sensor to NDI for all calibration procedures. For instructions on returning equipment to NDI, see “Return Procedure” on page 93. Appendix A on page 101 describes the methodology NDI uses to calibrate the Position Sensor.

Note **An impact can affect the calibration. A bump sensor detects when the Position Sensor has suffered an impact. For more information, see “Bump Sensor and Internal Battery” on page 49.**

9.4 Updating the Firmware

Each Polaris Vega configuration has a unique part number, and each configuration includes a specific combined firmware reversion. When you order a Polaris Vega, NDI programs the system with the combined firmware revision that is part of the configuration for the part number you ordered.

Note **If you return the system to NDI for repair, NDI reprograms the system with the original combined firmware revision. For more information, see "Return Procedure" on page 93.**

The Polaris Vega System's firmware is stored in flash memory devices in the Position Sensor and System Control Unit (if included). You must update the firmware on the System Control Unit separately from the Position Sensor. NDI recommends that the Position Sensor, System Control Unit, and Video Camera Unit are at the same version. You can download the latest firmware from the NDI Customer Support site at https://support.ndigital.com.

Use the Configure Utility in NDI ToolBox to update the Polaris Vega System's firmware. NDI ToolBox also includes command line functionality to embed a ToolBox task, such as upgrading firmware, into your application software. For more information, see the *NDI ToolBox Online Help*.

Note **If the LEDs flash during a firmware update, do not reset the system. Resetting the system during a firmware update could cause a non-recoverable error, requiring you to return the affected component to NDI to restore the firmware.**

If your system includes the multi firmware feature, you can specify which combined firmware revision the system uses on its next reset or power up. For more information, see "Multi Firmware" on page 51.

Note **A safe boot loader within the Position Sensor and System Control Unit (if included) provides a fallback if the main control firmware becomes unusable. For example, a communication fault or power fault could cause a field firmware upgrade to fail. In these cases, the Position Sensor and System Control Unit (if included), can still start up by running the safe boot loader. This provides minimal support to allow you to retry the control firmware upgrade.**

10 Understanding Infrared Light Sensitivity

10.1 Infrared Light Sensitivity Levels

The IR light sensitivity level determines how sensitive the Polaris Vega System is to IR light. Sensitivity levels rely on three aspects:

- The *exposure time* is the time in which the Vega Position Sensor collects IR light.
- The *trigger threshold* is the minimum pixel intensity required for the system to detect a marker. To detect a marker, one or more pixels in the marker image must have intensities above this threshold.
- *Background IR light* is IR light the Vega Position Sensor detects that is not reflected or emitted from a marker. Background IR light can be direct, such as light bulbs or sunlight, or indirect, such as reflections off shiny surfaces or draping. In particular, IR light in the 800 nm to 1100 nm range can interfere with the Polaris Vega System's ability to track tools. For example, some operating room lights emit IR light that the Position Sensor detects as background IR.

Changing the sensitivity level does not change the exposure time; therefore, the maximum IR light value does not change. However, changing the sensitivity level changes the trigger threshold and can help the Position Sensor detect IR light from tool markers while ignoring background IR light that has significantly lower intensity than the marker intensity.

There are seven sensitivity levels. Level 1 is the least sensitive, where the Position Sensor is least affected by background IR light because the trigger threshold is high. With each increasing sensitivity level, the system becomes more sensitive to IR light. At higher sensitivity levels, the Position Sensor can detect fainter IR light levels but will include IR light from background sources, making the Position Sensor more susceptible to background IR interference.

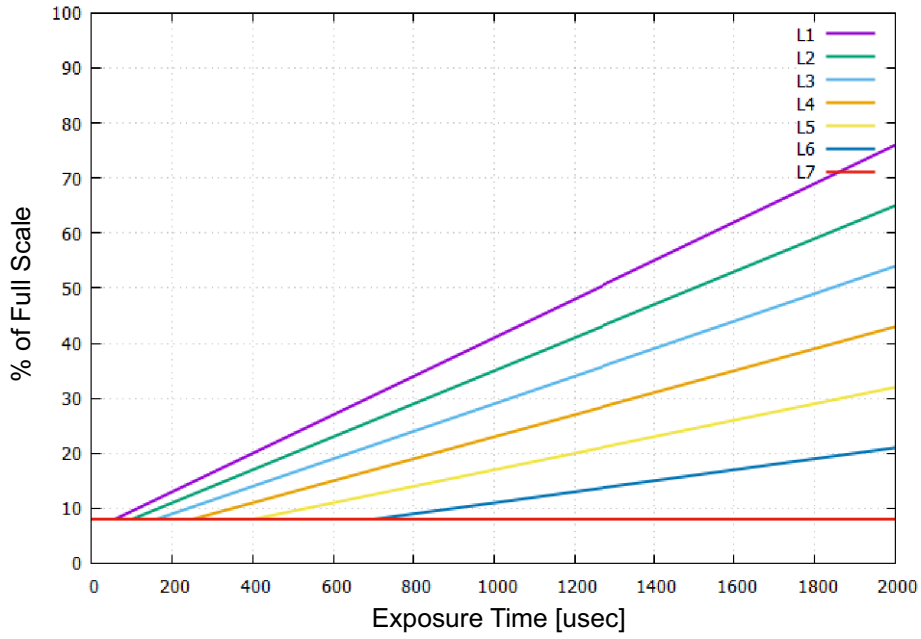
Note **If you have experience with Polaris Spectra systems, be aware that the meaning of the levels has changed. For Vega systems, a higher level means higher sensitivity; for Spectra systems, a higher level means lower sensitivity.**

Vega ST and VT Sensitivity Levels

The Vega ST and Vega VT Position Sensors use variable exposure time, where the Position Sensors continuously adjust the exposure time so that the intensities of the marker images are close to a predefined target value.

Vega ST and Vega VT Position Sensors implement the trigger threshold as a piecewise-linear function of exposure time. Figure 10-1 shows the relationship between trigger threshold and exposure time for each sensitivity level.

Figure 10-1 Vega ST and VT Sensitivity Levels



Vega XT Sensitivity Levels

Vega XT uses a fixed exposure time. Because of this, the trigger thresholds are constant for each sensitivity level. Table 10-1 shows the relationship between trigger threshold and exposure time for each sensitivity level.

Table 10-1 Vega XT Sensitivity Levels

Sensitivity Level	% of Full Scale
L1	55
L2	50
L3	45
L4	40
L5	35
L6	30
L7	25

10.2 Determining When to Change the Sensitivity Level

Unless the system is experiencing interference from background IR light or you are having trouble tracking tools with low IR intensity, use the default sensitivity level 4. Setting the level too high may cause an increase in background interference, which may prevent the system from performing optimally, and setting the level too low may cause tools to stop tracking.

If you experience a large amount of background interference that impacts tracking, first identify and try to remove the cause of the interference. If you cannot remove the interference from the scene, then use the NDI Capture ToolBox Utility to assess whether decreasing the sensitivity level will help eliminate more background IR light.

- Generally, changing the sensitivity level to a less sensitive level helps only in situations where the intensity of the interference is lower than that of the markers. If the intensity of the interference is similar to or higher in intensity than the markers, changing sensitivity levels cannot resolve the problem.
- Changing the sensitivity level may reduce tracking problems only when the tools are in a different physical location from the background IR. The sensitivity level cannot reduce tracking problems when the tools are embedded in background IR.

If you are having trouble tracking tools with low IR intensity, try increasing the sensitivity level. This allows the Position Sensor to detect fainter IR light and can be helpful for tools that are far from the Position Sensor.

Check for Background IR Light

1. If a tool is tracking intermittently or not tracking at all, perform one of the following actions:
   a) Use the API command `BX`, `BX2` or `TX` to check the **IR interference** flag in the port status or tool information reply components. If the IR interference flag is intermittently or constantly on for any of the tools, background IR may be present.
   b) In the NDI Track ToolBox utility, check for the **Interference** flag.
   c) In the NDI Capture ToolBox utility, take images of the scene and identify sources of IR interference, such as reflections.

Note **The system may be tracking a tool even when the tool's IR interference flag is on. You should still increase the sensitivity level, since the behaviour is dependent on the setup. For example, moving the tool to another part of the measurement volume may prevent the Position Sensor from tracking it properly.**

Change the Sensitivity Level

If you cannot eliminate the source of the background IR light, start with a low sensitivity level and increase the level until the tools track reliably.

1. To change the IR sensitivity, perform one of the following actions:
   - Use the Configure utility of NDI ToolBox to select a sensitivity level and to save the sensitivity level as the default setting in the Position Sensor memory.
   - Use the API command `SET` to change the value of the user parameter `Param.Tracking.Sensitivity`. To save the changed value as the default setting in the Position Sensor memory, use the API command `SAVE`.

For more information on API commands, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

11 Approvals

11.1 Electrical Safety and Electromagnetic Compatibility

The Polaris Vega System, consisting of a Position Sensor (Model P9), is listed in the "*Declaration of Conformity*" available for download from the NDI Customer Support site at https://support.ndigital.com.

Note **The System Control Unit has been designed for compliance; however, NDI has not certified the electrical safety or electromagnetic compatibility of the System Control Unit.**

The electrical safety and electromagnetic compatibility of active wired and active wireless tools available from NDI have not been assessed. These tools are intended to demonstrate the base technology and are not meant to be certified for use as a medical device accessory.

11.2 Optical Radiation Safety

Position Sensor Illuminators

The Position Sensor has been assessed against the standards listed in the *Declaration of Conformity* available for download from the NDI Customer Support site at https://support.ndigital.com and found not to pose a potential hazard to the eye under any foreseeable viewing condition.

Note **The Polaris Vega System emits IR light that may interfere with IR-controlled devices, such as operating room tables. It is recommended that you test the Polaris Vega System if you intend to use it in an environment where other IR-controlled devices are in use.**

The Polaris Vega System conforms to the IEC 62471:2006 and EN 62471:2008.

Positioning Laser



Do not use controls or adjustments or perform procedures other than specified in this guide. Doing so may result in hazardous light exposure.

The positioning laser is a Class 2 laser, with a wavelength of 635 nm to 670 nm and a maximum output of 1 mW. The Polaris Vega System containing a positioning laser conforms to the following standards:

- ANSI Z136.1 (2014)
- IEC 60825-1 (2014) and IEC 60825-1 (2007)
- FDA/CDRH 21 CFR 1040.10 and 1040.11 except for deviations pursuant to Laser Notice No. 56, dated May 08, 2019

Note **For information on laser operation, see "Positioning Laser" on page 46.**

11.3 IEC 60601-1 recommendations for the Polaris Vega System

The Polaris Vega System has been evaluated as a potential component of a Medical Electrical System intended for use in health care facilities outside the patient environment. The Polaris Vega System was evaluated using an IEC 60950 certified power supply. It should be possible to certify the system for use within the patient environment if the power supply of the end-use configuration is certified to IEC 60601-1. When integrated into an end-use configuration, consider the following recommendations:

- The maximum investigated branch circuit rating is 20 A.
- The investigated Pollution Degree is 2.
- The following tests shall be performed by the system integrator in the end product application:

a) Marking Legibility after installation

b) Touch Current (Leakage Test)

- Two (2) MOOP (Means of Operator Protection) are provided from AC input to output of the Power Injector, and one (1) MOOP is provided from AC input to Protective Earth (per Power Injector Certification).
- Complete evaluation of risk management requirements must be conducted by the system integrator in the end product investigation.
- NDI has not considered Essential Performance.
- The end-product evaluation shall ensure that the requirements related to Accompanying Documents, Clause 7.9 of IEC 60601-1 v3.1 are met.
- Suitability of the power source shall be verified by the respective certification body.
- The system was tested together with a personal computer. Since, in final application the product may be used with any certified personal computer, a leakage current test shall be conducted to verify suitability of the final system configuration. End product investigation shall consider compliance with enclosure leakage current test with the personal computer's protective earth disconnected.
- The optional external switch intended to be connected to the laser activation port in any end-use application must be separated from live parts by at least 2MOOPs (Means of Operator Protection).
- The product is not suitable for use in the presence of a flammable anaesthetics mixture with air or oxygen or with nitrous oxide.

12 Classifications

Table 12-1 Classifications

Classification	Description
Electric shock protection	Class I - no applied parts
Degree of protection from electric shock	Not classified
Degree of protection against ingress of solids and liquids	IP20 (Protection against solid objects over 12 mm. No protection against liquids.)
Flammable atmosphere	Not suitable for use in the presence of a flammable anaesthetic mixture with air, oxygen or nitrous oxide
Mode of operation	Continuous when supplied by external power
Method of sterilization or disinfection	Not suitable for sterilization
Laser classification	Class 2
IRED illuminators risk group	Exempt

13 Technical Specifications

13.1 Operating Environmental Conditions

The Polaris Vega System, including the Position Sensor and cables, has been tested to function in the conditions listed in Table 13-1.

Table 13-1 Operating Environmental Conditions

Environmental Condition	Value
Atmospheric Pressure	70 kPa to 106 kPa
Relative Humidity	30% to 75%
Temperature	+10°C to +35°C (see warning below)

The Position Sensor requires a warm-up time every time it is powered on. The warm-up time is typically 30 seconds; if the Position Sensor is stored at low temperatures, the warm-up time may be longer. The power LED flashes while the Position Sensor warms up; once the LED is steady, the system is ready for use.

13.2 Transportation and Storage Environmental Conditions

The Polaris Vega System, including the Position Sensor and cables, has been tested to be stored and transported in the conditions listed in Table 13-2.



Do not transport or store the Position Sensor outside the recommended storage conditions, as this may cause the system to go out of calibration. Reliance on data provided by an out of calibration Position Sensor may lead to inaccurate conclusions and may cause personal injury. A calibration procedure must be performed before using the Position Sensor after it has been transported or stored outside the recommended storage temperature range.

Caution! **To ship the Polaris Vega System, repack it in the original containers with all protective packaging. The provided packaging is designed to prevent damage to the equipment..**

Table 13-2 Transportation and Storage Environmental Conditions

Specification	Value
Atmospheric Pressure	50 kPa to 106 kPa
Relative Humidity	10% to 90% non-condensing
Temperature	-10°C to +50°C

13.3 Technical Specifications

For information about the technical specifications of the System Control Unit, see the data sheet for your System Control Unit. To obtain the data sheet, contact NDI.

Table 13-3 Position Sensor Technical Specifications

Specification	Value
Dimensions	591 mm x 103 mm x 106 mm
Weight	1.7 kg +/- 0.1 kg[1]
Mounting	Via four M4 x 0.7 mm pitch x 10 mm deep threaded holes, rear mount
Power Requirement	Compatible with IEEE 802.3at Type 2[2]

[1] The weight of the Position Sensor when equipped with the radiation hardening feature is 2.15 kg +/- 0.15kg.

[2] For IEC 60601-1 testing, the power injector used was Power Dsine (Microsemi Corp.), PD-9001GR/AC (PD-6561G300).

13.4 Mean Time Between Failures

The mean time between failures (MTBF) for the Polaris Vega Position Sensor is calculated at >100,000 hours. The MTBF drops by half for every 10°C increase in environmental temperature. Therefore, care must be taken to use the Polaris Vega Position Sensor within the recommended operating environmental conditions defined in this guide.

13.5 Shelf Life

The Position Sensor contains an internal battery that powers the system's real-time clock and powers the bump sensor when the system is powered off. If the bump sensor is not triggered, a fully-charged battery will power the bump sensor and real-time clock for approximately two years. If the bump sensor is triggered, the battery will power the sensor and clock for approximately two months.

If the battery is drained, keep the Position Sensor powered on for approximately 12 hours to recharge the battery. If the system continues to indicate the battery is low, contact NDI. See "Contact Information" on page viii.

For more information, see "Bump Sensor and Internal Battery" on page 49.

14 Electromagnetic Compatibility

The Polaris Vega System requires special precautions regarding EMC. It must be installed and put into service in accordance with the EMC information detailed here. Failure to do so may result in personal injury.

Radio frequency communications equipment, including portable and mobile devices, may affect the Polaris Vega System and result in personal injury.

Do not use the Polaris Vega System either adjacent to, or stacked with, other equipment as this may result in increased electromagnetic emissions and/or decreased electromagnetic immunity. If the Polaris Vega System is used adjacent to, or stacked with other systems, check that both systems are operating normally. Failure to do so may result in personal injury.

14.1 Cables and Accessories

Ethernet cables must be at least Cat 5e shielded to maintain compliance to the emissions and immunity requirements of IEC 60601-1-2:2014.

Use of accessories and cables other than those specified or provided by NDI could result in increased electromagnetic emissions or decreased electromagnetic immunity of this equipment and result in improper operation.

14.2 Guidance and Manufacturer's Declaration: Electromagnetic Emissions

The Polaris Vega System is intended for use in the electromagnetic environment specified below. The customer or the user of the Polaris Vega System should ensure that it is used in such an environment

Table 14-1 Manufacturer's Declaration for Electromagnetic Emissions

Emissions Test	Compliance	Electromagnetic Environment Guidance
RF emissions CISPR11	Group 1	The Polaris Vega System uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF emissions CISPR11	Class A	The Polaris Vega System is suitable for use in all establishments other than domestic and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
Harmonic emissions IEC61000 3-2	Class A	
Voltage fluctuations/ flicker emissions IEC61000-3-3	Complies	

14.3 Guidance and Manufacturer's Declaration: Electromagnetic Immunity

The Polaris Vega System is intended for use in the electromagnetic environment specified below. The customer or the user of the Polaris Vega System should ensure that it is used in such an environment.

Table 14-2 Electromagnetic Immunity

Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment - Guidance
Electrostatic discharge (ESD) IEC 61000-4-2	±8 kV contact ±2, ±4, ±8, ±15 kV air	±8 kV contact ±2, ±4, ±8, ±15 kV air	Floors should be wood, concrete, or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Electrical fast transient/burst IEC 61000-4-4	±2 kV for power supply lines. ±1 kV for input/ output lines	±2 kV for power supply lines. ±1 kV for input/ output lines	External power quality should be that of a typical commercial or hospital environment.
Surge IEC 61000-4-5	±1 kV line(s) to line(s)	±1 kV differential mode	External power quality should be that of a typical commercial or hospital environment.
	±2 kV line(s) to earth	±2 kV common mode	
Voltage dips (3), short interruptions (1), and voltage variations on power supply input lines (1) IEC 61000-4-11	0% U_T for 0.5 cycles (0º, 45º, 90º, 135º, 180º, 225º, 270º, 315º)	0% U_T for 0.5 cycles (0º, 45º, 90º, 135º, 180º, 225º, 270º, 315º)	External power quality should be that of a typical commercial or hospital environment. If the user of the Polaris Vega System requires continued operation during external power interruptions, it is recommended that the Polaris Vega System be powered from an uninterruptible power supply or a battery. U_T is the external power voltage prior to application of the test level.
	0% U_T for 1 cycle (0º)	0% U_T for 1 cycle (0º)	
	70% U_T (30% dip in U_T) for 25/30 cycles(0º)	70% U_T (30% dip in U_T) for 25/30 cycles(0º)	
	0% U_T for 250/300 cycles(0º)	0% U_T for 250/300 cycles(0º)	
	49 Hz & 61 Hz	49 Hz & 61 Hz	
Power frequency (50/60 Hz) magnetic field IEC 61000-4-8	30 A/m	30 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.

Table 14-3 Electromagnetic Immunity—Not Life Supporting

Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment - Guidance
Conducted RF IEC 61000-4-6	3 Vrms 150 kHz to 80 MHz	3 Vrms	Portable and mobile RF communications equipment should be used no closer to any part of the Polaris Vega System, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter. Recommended separation distance: $d = 1.2\sqrt{P}$ See Table 14-4.
Radiated RF IEC 61000-4-3	3V/m, 80 - 2700 MHz, 1 kHz, 80% AM modulation	3 V/m	$d = 1.2\sqrt{P}$ 80 MHz to 800 MHz $d = 2.3\sqrt{P}$ 800 MHz to 2.5 GHz See Table 14-4. Where 'P' is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and 'd' is the recommended separation distance in metres. Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey[a], should be less than the compliance level in each frequency range[b]. Interference may occur in the vicinity of equipment marked with the following symbol: 

Note **At 80 MHz and 800 MHz, the higher frequency range applies.**

These guidelines may not apply to all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.

a - Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast, cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location where the Polaris Vega System is used exceeds the applicable RF compliance level above, observe the Polaris Vega System to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as re-orienting or relocating the Polaris Vega System.

b - Over the frequency range of 150 kHz to 80 MHz, field strengths should be less than 3 V/m.

14.4 Recommended Separation Distances

The Polaris Vega System is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the Polaris Vega System can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the Polaris Vega System, as recommended below, according to the maximum output power of the communications equipment.

Table 14-4 Recommended Separation Distances between Portable and Mobile RF Communications Equipment and the Vega System

Rated maximum output power of transmitter (watts)	Separation distance according to frequency of transmitter (metres)		
	150 kHz to 80 MHz $d = 1.2\sqrt{P}$	80 MHz to 800 MHz $d = 1.2\sqrt{P}$	800 MHz to 2.5 GHz $d = 2.3\sqrt{P}$
0.01	0.12	0.12	0.23
0.1	0.38	0.38	0.73
1	1.2	1.2	2.3
10	3.8	3.8	7.3
100	12	12	23

For transmitters rated at a maximum output power not listed above, the recommended separation distance (d) in metres (m) can be estimated using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.

Note **At 80 MHz and 800 MHz, the higher frequency range applies.**

These guidelines may not apply to all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.

14.5 Radio Frequency Emissions

FCC

This device complies with Part 15 of the FCC rules. Operation is subject to the following two conditions:

1. This device may not cause harmful interference and,
2. This device must accept any interference received, including interference that may cause undesired operation.

Note **This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:**
— Reorient or relocate the receiving antenna.
— Increase the separation between the equipment and receiver.
— Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
— Consult the dealer or an experienced radio/TV technician for help.

Changes or modifications not expressly approved by Northern Digital Inc. could void the user's authority to operate the equipment.

CE Mark

CISPR 11: Class A

Industry Canada

Industry Canada Compliance Statement: This device complies with CAN ICES-003(A).

Avis de Conformité à la réglementation d'Industrie Canada: Cet appareil est conforme à la norme CAN NMB-003(A).

15 Troubleshooting

For more troubleshooting information, regularly check the NDI Customer Support site at https://support.ndigital.com. If you cannot find the answer to your question here or on the support site, contact NDI. See “Contact Information” on page viii.

15.1 Troubleshooting Connections

The device is visible in the Connect To menu but when I try to connect, I get the error message “Failed to Connect”

Problem: The device may not be configured properly for static IP addressing.

Solution: Configure the host computer and device for static IP addressing as follows:

1. In the NDI Configure ToolBox utility, select **File > Connect To**.
2. Hover over the device you are trying to connect to.
3. Make a note of the IP address in the tool tip.
4. Change the host network interface to be on the same subnet as the device.
5. In the NDI Configure application, attempt to connect to the device again.

If the problem persists, it is likely due to an issue with your hardware configuration. Contact your network administrator.

When I navigate to the Connect To menu, I do not see the device I am trying to connect to

Problem: The device may not be powered on or connected to the network.

Solution: Ensure the devices and PoE midspan injector are powered on and connected to the network.

If the problem persists, it is likely due to an issue your network configuration. Contact your network administrator.

15.2 Troubleshooting Common Problems

Most problems that may occur with the Polaris Vega System fall into one of the following categories:

- A system hardware failure (for example, a faulty Position Sensor or cable)
- A tool error (for example, dirty markers)
- Environmental conditions (for example, incidental IR light)
- User error (for example, obscuring the optical path)

The Error LED on the Position Sensor indicates most faults. Use Table 1-2 to help determine your next steps.

You can diagnose the fault using the API command `GET` to read the `Info.Status.Alerts` user parameter or by observing the error message in the Configure utility of NDI ToolBox. For more information on the `Info.Status.Alerts` user parameter, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

The tool is inside the measurement volume, but the software reports that the tool is partially out of volume

This may mean that fewer than the **minimum number of markers** specified in the tool definition file are inside the characterized measurement volume, but at least one marker on the tool is outside the characterized measurement volume.

For example, if a four-marker tool with its minimum number of markers parameter set to 4 has three markers inside the characterized measurement volume and one marker outside the characterized measurement volume, the Polaris Vega System continues to track the tool, but the accuracy is unknown and the system reports the tool as partially out of volume.

The system doesn't track tools at the back of the characterized measurement volume

If the system tracks at the front of the measurement volume but not at the back, the Position Sensor may be damaged or may be calculating high line separation values. For more information on line separation, see "Marker Detection and Tool Tracking" on page 33.

You can check the line separation values using the Marker Line Separation component of the API command `BX2`. For more information, see the *Polaris Vega and Lyra Application Programming Interface Guide*.

Other IR-based devices are not working properly

Using the Polaris Vega System in the same room as other IR-based devices may cause the other devices to malfunction. The Position Sensor's illuminators flood the surrounding area with IR light, which could saturate other IR receiving devices, preventing them from properly receiving other IR signals.

You may be able to synchronize other devices with the Polaris Vega System so that IR signals from the other devices are not emitted at the same time as the illuminators emit IR. For more information, contact NDI Technical Support or visit the NDI Customer Support site at https://support.ndigital.com.

The Polaris Vega System is tracking some tools but not others

As the Position Sensor begins to lose calibration, it may lose the ability to track some tools before others. The various constraints used by the Polaris Vega System make certain tool designs more sensitive to loss of calibration than others.

For example, if a tool has several similar segment lengths, similar angles between segments, or segment lengths similar to those of another tool, an out-of-calibration Position Sensor may not be able to determine which markers belong to which tool and reports the tools as missing.

For more information on segment lengths and angles, see "Marker Detection and Tool Tracking" on page 33. For more information on calibration, see "Checking the Calibration of the Polaris Vega System" on page 73.

Reflections or other IR sources are present

Reflections and other IR sources may cause markers to become “lost” in the background IR light. Eliminate or minimize these reflections and extra IR sources as much as possible. Reflections occur when IR light from the illuminators reflects off surfaces such as:

- surgical drapes: position drapes in a way to minimize reflections
- reflective surfaces: minimize reflective surfaces in the environment
- tools: design and manufacture tools in non-reflective materials. For more information, see the *Polaris Tool Design Guide*.

When positioning the system, consider other sources of IR light, such as operating room lights, to minimize interference from such devices.

Markers are partially blocked from the view of the Position Sensor

The Polaris Vega System requires a clear line of sight to the markers. Anything that interferes with the line of sight can reduce the measurement accuracy. The magnitude of the errors caused by partial occlusion of the markers depends on the number of markers, the geometry of the tool, and the severity of the occlusion. Errors caused by partial occlusion can have the same magnitude for active and passive tools. However, passive markers are partially occluded more frequently because they are larger than active markers. When designing a new tool, it is important to consider the effect of partial occlusion on its accuracy. For more information on tool design, see the *Polaris Tool Design Guide*.

The Position Sensor reports a tool as missing

The Position Sensor may report a tool as missing if:

- it has been rotated so that too few markers are visible to the Position Sensor
- the tool is no longer in the field of view
- the tool is damaged (for example, it is bent)
- the condition of the makers has deteriorated (for example, the markers are scuffed or occluded with foreign matter).

The Position Sensor seems too warm

The Position Sensor is warm to the touch during normal use. If the temperature goes out of range, NDI ToolBox shows an error message.

The Polaris Vega System does not have full functionality or is behaving intermittently

Check the connection between the Ethernet cable and the Position Sensor. A loose connection may result in partial functionality or unpredictable system behaviour.

The virtual serial port cannot be opened

If you cannot open the virtual serial port, check the system connections, the Host USB Converter, and the host computer USB port.

15.3 Troubleshooting LEDs

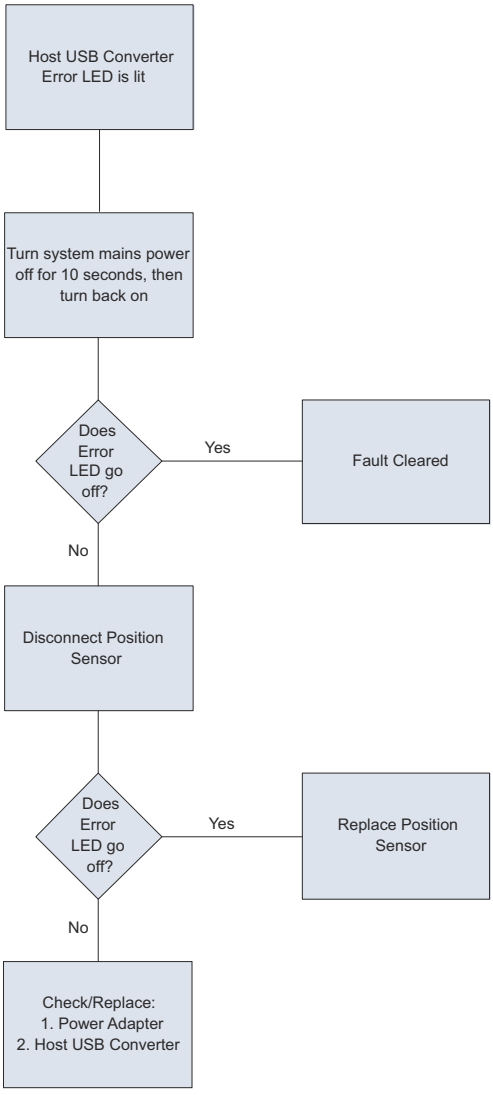
For more information on the system LEDs, see “Status LEDs and Audio Codes” on page 7.

Host USB Converter

- **Power** : Lights green when power is being supplied to the Host USB Converter.
- **Error** : Lights amber when the Host USB Converter has detected a fault.

If the Host USB Converter Error LED is lit, follow the flow chart below to help diagnose the fault.

Figure 15-1 Host USB Converter: Error LED Fault Chart



16 Return Procedure, Warranty, and Disposing of Equipment

16.1 Return Procedure

If you must return equipment to NDI for repairs, contact NDI to request an RMA number and shipping instructions. You should ship any materials you are returning to NDI in their original packaging.

You are responsible for the shipping costs of returning equipment to NDI. NDI will inspect the equipment to determine whether the required repair is covered under warranty and can provide you with a quote of repair costs either before or after repairs are made. If the equipment is covered under warranty, NDI pays the return shipping costs. If the equipment is not covered under warranty, you must pay the return shipping costs.

All Polaris Vegas sent to NDI for repair are reprogrammed with their original combined firmware revision (the combined firmware revision that corresponds to the part number on the label).

16.2 Warranty

Unless otherwise agreed to in writing by NDI, the warranty is as follows, and applies only to the original purchaser.

This warranty is void if you open the case of any system component.

Hardware

NDI warrants to the Buyer that NDI's hardware product(s) will be free from defects in material and workmanship only for a period twelve (12) months from the date such product(s) is/are shipped from NDI to the Buyer.

Software

NDI's software product(s) is/are licensed and provided "as is, where is" without warranty of any kind. NDI makes no warranties, express or implied, that the functions contained in the software product(s) will meet the Buyer's requirements or that the operation of the programs contained therein will be error free.

General Provisions Applicable to Warranty

NDI's obligations under this warranty shall be limited to repairing or replacing (at NDI's option) the product(s), EXW (Incoterms 2000) NDI's plant (Waterloo, Ontario, Canada). Any original parts removed and/or replaced during any repair process shall become the property of NDI. This warranty shall apply only to the original Buyer [being that person or legal entity which has contracted directly with NDI for the supply of the product(s)]. Repair work shall be warranted on the same terms as stated herein except such warranty shall be for a period of sixty (60) days or for the remainder of the unexpired warranty period, whichever is longer. In respect of any product(s) supplied hereunder which are manufactured by others, NDI gives no warranty whatsoever, and the warranty given by such manufacturers, if any, shall apply.

The obligations of NDI set forth in this warranty are conditional upon proper transportation, shipping, handling, storage, installation, use, maintenance and compliance with any applicable recommendations of NDI. Without limiting the generality of the foregoing, this warranty shall not apply to defects or damage resulting from: fire; misuse; abuse; accident; neglect; improper installation; improper care and/or maintenance; lack of care and/or maintenance; customer supplied software interfacing; modification or repair which is not authorized by NDI; power fluctuations; operation of hardware product(s) outside of environmental specifications; improper site preparation and maintenance; permitting any substance whatsoever to contaminate or otherwise interfere with optics; and any other cause beyond the control of NDI. The obligations set forth in this warranty are further conditional upon the Buyer promptly notifying NDI of any defect and, if required, promptly making the product(s) available for correction. NDI shall be given reasonable opportunity to investigate all claims and no product(s) shall be returned to NDI without NDI first providing the Buyer with a return material authorization number and shipping instructions. All product(s) returned to NDI shall be packaged in the containers originally used by NDI to ship the product(s) to the Buyer.

NDI, for itself, its agents, contractors, employees, providers, and for any parent or subsidiary of NDI, expressly disclaims all warranties, express or implied, including, without limitation, of merchantability or fitness for a particular purpose.

The foregoing warranty is the entire warranty of NDI. NDI neither assumes nor authorizes any person, purporting to act on its behalf, to modify or to change this warranty, or any other warranty or liability concerning the product(s).

16.3 Disposing of Equipment

To ensure environmentally responsible disposal after the equipment is decommissioned, please contact NDI. See “Contact Information” on page viii.

17 Abbreviations and Acronyms

Acronym or Abbreviation	Definition
AAK	Accuracy Assessment Kit
CAPI	Combined Application Programming Interface
DSR	Data Set Ready
EMC	Electromagnetic Compatibility
ESD	Electrostatic Discharge
FCC	Federal Communications Commission
I/O	Input/Output
IR	Infrared Light
IRED	Infrared Light-Emitting Diode
LED	Light-Emitting Diode
mDNS	Multicast Domain Name System
MOOP	Means of Operator Protection
MRI	Magnetic Resonance Imaging
MTBF	Mean Time Between Failures
NDI	Northern Digital Inc.
PoE	Power Over Ethernet
PTP	Precision Time Protocol
RF	Radio Frequency
RI	Ring Indicator
RMA	Return Materials Authorization
RMS	Root Mean Square
UDP	User Datagram Protocol
UL	Underwriters Laboratories Inc.
VCU	Video Camera Unit

18 Equipment Symbols

Table 18-1 Equipment Symbols

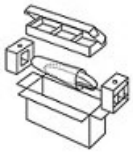
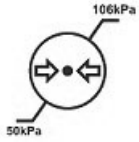
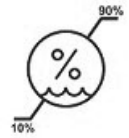
Symbol	Meaning	System Components
	Consult accompanying documents.	Position Sensor
	Laser Warning (To avoid personal injury, consult accompanying documents.)	Position Sensor
	Laser Aperture Caution (To avoid personal injury, consult accompanying documents.)	Position Sensor
	On (power: connection to the external power supply)	Position Sensor
	Error	Position Sensor
	Ethernet	Position Sensor
	Unused communication port	Position Sensor
	Keep away from rain	Packaging
	Fragile	Packaging
	Retain packaging	Packaging
	Acceptable pressure during shipping	Packaging

Table 18-1 Equipment Symbols (Continued)

Symbol	Meaning	System Components
	Acceptable humidity during shipping	Packaging
	Acceptable temperature during shipping	Packaging
	Cardboard recycling indicator (Chinese)	Packaging
	Cardboard recycling indicator (German)	Packaging
	Paper recycling indicator (German)	Packaging
	Cut packaging here	Packaging
	Do not cut packaging here	Packaging

19 Glossary

3D RMS Error

The RMS error is the square root of the sum of the squares of the measurement errors. This can be approximated by the square root of the mean square added to the standard deviation squared of the errors.

Absolute Measurements

Absolute measurements are measurements taken directly in the Position Sensor's coordinate system, that is, measurements reported by the system. NDI recommends not to use the system for absolute measurements.

Calibration

Calibration is the process of establishing, under specified conditions, the relationship between values produced by the Polaris Vega System and corresponding known values established by a device that is traceable to a national standard.

Characterized Measurement Volume

The characterized measurement volume is the volume within the detection region where accuracy is within specified limits. NDI cannot guarantee measurement accuracy outside this region.

Faces

Tool faces are separate rigid bodies that make up a tool.

Field of View

The field of view is the total volume in which the Polaris Vega System can track a marker, regardless of accuracy.

Firmware

Firmware is a computer program stored in an NDI hardware device and controls the Polaris Vega System.

Frame

A frame contains the measured positions of the markers in the field of view at a particular point in time.

Frame Frequency

The frame frequency is the rate at which the system takes images of the measurement volume.

Global Coordinate System

The global coordinate system is the Polaris Vega coordinate system. The Polaris Vega System uses the global coordinate system as a frame of reference against which tool transformations are reported. By default, the global coordinate system's origin is set at the Position Sensor.

Illuminator

The illuminator is an array of IR light-emitting diodes that surround the sensor lenses on the Position Sensor. These flood the area in front of the Position Sensor with IR light, which is reflected back to the Position Sensor by the passive markers.

Latency

Latency is the time between when the Position Sensor collects data and when that data is available to the host application. The Vega XT Position Sensor offers the lowest latency. Latency is measured in milliseconds.

Line Separation

To determine the position of an IR source, the Position Sensor calculates a line between the source of IR and each sensor. The line separation is the distance between these two lines where they cross.

Local Coordinate System

A local coordinate system is a coordinate system assigned to a specific tool or rigid body.

Maximum Marker Error

Maximum marker error applies to individual markers. It specifies, in the tool definition file, the maximum allowable difference between the measured and expected location of a marker on a tool or rigid body. Previously, this was referred to as the Maximum 3D Error.

Maximum Marker Angle

Maximum marker angle is used to determine if a marker will be used in the calculation of a rigid body or tool. If the marker is determined to be farther off-angle to the Position Sensor than the maximum marker angle, this data is not used to determine the rigid body or tool.

Passive Marker

A passive marker is a retro-reflective passive sphere or Radix Lens that reflects IR light emitted by the Position Sensor.

Pivoting

Pivoting is a procedure (of rotating a tool about its tip) used to determine the tool tip offset.

Position Sensor

The Position Sensor is the component of the Polaris Vega System that provides a source of IR light for passive markers, collects marker position data from both active and passive markers, calculates tool transformations, and sends the results to the host computer.

Quaternion

A quaternion is a compact representation of rotations, or correspondingly, orientations in 3D space (rather than having to use orthogonal matrices).

Reference Tool

A reference tool is a tool or rigid body whose local coordinate system is used as a frame of reference in which other tools are reported or measured.

Relative Measurements

Relative measurements, or referenced measurements, require the use of at least two tools, one of which is the reference. For relative measurements, all tool transformations are transformed into the coordinate system of the reference tool. When the system is used in this way, movements of the Position Sensor itself are cancelled out. NDI recommends using the system in this way.

Rigid Body

A rigid body is an object on which three or more markers are fixed relative to one another.

Tool Definition File

A tool definition file stores information about a tool or rigid body. This includes information such as the placement of the tool's markers, the location of its origin, and its manufacturing data. Tool definition files use the .rom extension.

Tool Tip Offset

The tool tip offset is the vector between the tip of the tool and the tool origin.

Tracking Frequency

The tracking frequency is the rate at which the system reports transformations for all the tools being tracked. The tracking frequency cannot exceed the frame frequency.

Transformation

A transformation is a combination of translation and rotation values that describe a change of the tool or rigid body in position and orientation.

Unique Geometry Tool

Unique geometry tools incorporate markers positioned in such a way that, when detected in the measurement volume, the tool can be uniquely identified from other tools.

Appendix A Polaris Vega System Calibration Performance and Methodology

Standard industry practice dictates that all measurement and testing instruments should be periodically calibrated to ensure they are operating within tolerances acceptable to the user and/or the user's customers.

The user must establish a calibration procedure and interval that is appropriate for the accuracy requirements of their application.

The Position Sensor is a highly specialized instrument developed exclusively by NDI. For all calibration procedures, return the Position Sensor to NDI. This practice ensures that all calibrations are conducted in accordance with procedures established specifically for the Vega Position Sensor.

Note **The NDI Accuracy Assessment Kit (AAK) can be used in the field as an aid to determine whether a Position Sensor is performing acceptably for the user's application.**

If, at any time, a concern should arise that the Position Sensor is not measuring accurately, it should be returned to NDI.

Note **The calibration procedure at NDI applies to single markers and cannot be directly applied to an application that uses tools with several markers.**

A.1 Polaris Vega System Performance

The Polaris Vega System performance is determined by a statistical analysis of the 3D Euclidean distance error between the reported position of an NDI marker and its true position, based on measurements taken throughout the entire Polaris Vega System's measurement volume. Acceptance criteria for the Polaris Vega System's performance are based on the RMS values of the accuracy and repeatability. Performance specifications are detailed in Table A-1.

This criteria is based on a statistically representative set of positions distributed uniformly throughout the measurement volume, using 30 samples per position at 20°C.

Table A-1 Polaris Vega System Performance

	Vega ST/VT		Vega XT	
	Pyramid Volume	Extended Pyramid Volume	Pyramid Volume	Extended Pyramid Volume
Volumetric Specification				
3D RMS Accuracy[1]	0.12 mm	0.15 mm	0.12 mm	0.15 mm
3D RMS Repeatability[2]	0.06 mm	0.08 mm	0.06 mm	0.08 mm
Infield Accuracy Test				
AAK Result (RMS)	0.25 mm	0.25 mm	0.25 mm	0.25 mm

[1] The 3D RMS volumetric accuracy error is determined as the best-fit error of a statistically representative set of measured positions distributed uniformly throughout the measurement volume when compared to the nominal positions, as reported by a NIST-traceable CMM.

For each measured position the mean of 30 samples is used. The data is collected with the system at thermal equilibrium, at an ambient temperature of 20.0 ± 0.5 °C.

[2] The 3D RMS repeatability error is determined as from the same data set as the 3D RMS volumetric accuracy error. For each measured position the mean position is calculated.

The local repeatability error is then calculated as the standard deviation of the distance of each of the 30 samples, relative to the mean position.

The volumetric 3D RMS repeatability error is the RMS error of all local repeatability errors.

A.2 Calibration Method

The following method is used to calculate the Polaris Vega System's accuracy and repeatability:

An NDI marker is accurately moved to each of *n* locations (X_i, Y_i, Z_i) spread throughout the measurement volume. The mechanism that moves the marker is assumed to have an accuracy that is at least 10 times better than the measured accuracy of the Polaris Vega System. This assumption allows the errors in the marker positioning to be ignored.

At each of the *n* locations, the Polaris Vega System takes *m* readings of the marker's 3D position (x_{ij}, y_{ij}, z_{ij}).

The **accuracy of the Polaris Vega System** is calculated as the RMS variation of the mean of *m* readings about the true 3D location calculated across all *n* locations throughout the measurement volume.

The **repeatability of the Polaris Vega System** is calculated as the RMS variation of the *m* readings about the average of the 3D readings at each location *n*. This RMS variation is calculated across all *n* locations throughout the measurement volume.

3D average measurements:

$$x_{i_{average}} = \frac{\sum_{j=1}^{m} x_{i,j}}{m}, \quad y_{i_{average}} = \frac{\sum_{j=1}^{m} y_{i,j}}{m}, \quad z_{i_{average}} = \frac{\sum_{j=1}^{m} z_{i,j}}{m}$$

3D RMS accuracy =

$$\sqrt{\frac{\sum_{i=1}^{n}\left[(x_{i_{average}} - X_i)^2 + (y_{i_{average}} - Y_i)^2 + (z_{i_{average}} - Z_i)^2\right]}{n}}$$

3D RMS repeatability =

$$\sqrt{\frac{\sum_{i=1}^{n}\left[\sum_{j=1}^{m}\left[(x_{i,j} - x_{i_{average}})^2 + (y_{i,j} - y_{i_{average}})^2 + (z_{i,j} - z_{i_{average}})^2\right]\right]}{n \cdot m}}$$

Appendix B Deprecated Features

B.1 Password Protection

The current password protection feature is deprecated. NDI is developing a more comprehensive version of this feature to be released with a future firmware revision.

If you choose to enable the current password protection feature, it provides the following security against changes to the system configuration:

- You must enter the correct password before you can save user parameter values, update the firmware, or install, disable, or enable a keyed feature.
- If you do not enter the correct password, you can change user parameter values but cannot save them. Upon system reset or initialization, the parameters return to their previous values.
- If the system is reset or initialized after you enter the password, you must re-enter the password before you can make changes to the system configuration.

To enter the password, use NDI ToolBox or the API command `SET Config.Password=<password>`, where *<password>* is the correct password. Obtain the password from NDI.

In the ToolBox user interface and API, this feature is referred to as “Password”.

B.2 250 Hz Tracking Frequency (MR250)

Previously, the Vega ST Position Sensor supported the option of tracking at 250 Hz with lower latency. This option has been discontinued and is no longer available. The replacement option is the Vega XT. For more information about the Vega XT, see “Vega XT Position Sensor” on page 11.

For historical reference, system capabilities associated with the 250 Hz tracking frequency are described in Table 19-1.

Table 19-1 System Capabilities at 250 Hz

Feature / Capability	
Volumetric Accuracy, Pyramid	0.25 mm
Volumetric Accuracy, Extended Pyramid (Optional)	0.30 mm
Tracking Frequency	250 Hz
Latency (when tracking two tools)	<8 ms
Supported Tool Types	• Passive Spheres • Radix Lenses
Communication Interface	• Ethernet
Other Configuration Options	• Positioning Laser • Extended Pyramid Volume